



CONFIDENTIAL

Nomadix Nexus



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Ruckus SmartZone

Download this guide 

Introduction

Scope and Purpose

Thank you for purchasing the Trusted WiFi solution. This document is a hardware configuration guide describing how to setup a Ruckus SmartZone WLAN Controller for a Trusted WiFi Passpoint service.

- For more information on how to setup an end-to-end Trusted WiFi Passpoint service, please refer to the Trusted WiFi Passpoint Administration Guide.
- For complete information on how to setup a Ruckus SmartZone WLAN Controller, please refer to the vendor's original documentation.

Documentation Conventions

{{snippet.Documentation Conventions}}

Passpoint Overview

Passpoint – also known as Hotspot 2.0 – is an industry-wide next generation approach to public internet access driven by the Wi-Fi Alliance that brings the following benefits:

- Frictionless onboarding and roaming, thanks to a one-time registration followed by automatic access to interconnected hotspots
- More secure and private Wi-Fi connections, compared with general visitor networks

Passpoint is based on the IEEE 802.11u standard, which is a set of protocols enabling cellular-like roaming. Following the initial enrollment, frequent users such as visitors, guests or employees bypass repeated logins, forms and passwords, as their mobile devices automatically join the Wi-Fi subscriber service when they return to a venue or roam between inter-linked Passpoint enabled hotspots and providers, while being better protected against potential cyber threats.

If a device supports 802.11u and is enrolled to a service, it automatically communicates with the Wi-Fi infrastructure via the access points to discover the network SSID and connects securely to it by presenting its access credentials. Upon successful authentication, the device is provisioned with Passpoint standards-based management objects - known as Per-Provider Subscription Management Objects (PPS-MO).

GlobalReach - an ASSA ABLOY company - has been involved with Passpoint since its inception and even contributed to the creation and initial pilot testing of the standard. As a result, it is one of the few trusted worldwide experts on this topic, with a proven platform backed by real-world operational experiences at scale.

The best user experience is to offer Passpoint through a customer/brand mobile app integration, as it further simplifies the onboarding process, while incentivizing app downloads and customer loyalty, leading to further engagement and monetization opportunities. To this effect, Trusted WiFi offers a Software Development Kit (SDK) for easy app integration.

Seamless Onboarding and Roaming

Best user experience via brand mobile loyalty app integration:

- One-time setup in 3 "taps" via a mobile app
- Followed by automatic connection for returning visitors and roaming within any of the participating venues
- No manual SSID selection, no login/password, no reset after a given period
- No impact from MAC address randomization
- Incentive to download the app, leading to visitor engagement opportunities
- Non-compatible devices can still use the legacy onboarding process

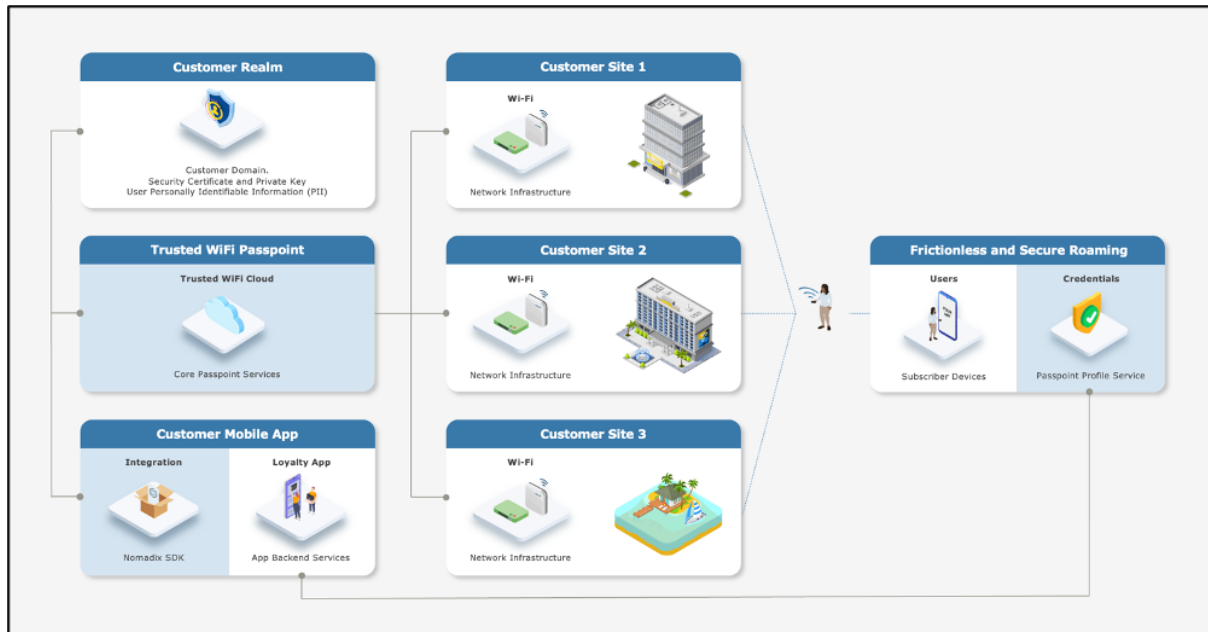
End-to-End Service Components

Implementing an end-to-end Trusted WiFi Passpoint service requires a combination of the following software and hardware components:

- **Trusted WiFi Passpoint:** the core services are offered and managed centrally via the Trusted WiFi Passpoint Module (hosted and operated by GlobalReach). Following an initial setup performed by the GlobalReach Operations team, Managed Service Providers (MSPs) can then add sites to customer realms and monitor the solution.
- **Customer Realm:** the service requires connectivity to a realm including domain, security certificate and private key, as well as the database holding the users' Personally Identifiable Information (PII). It is the responsibility of the Identity Provider – typically the customer/brand – to make this part available.
- **Mobile App:** the best user experience is to offer Passpoint through a customer/brand mobile app integration. Trusted WiFi offers a **Software Development Kit (SDK)** for easy implementation. It is the responsibility of the app owner – typically the customer/brand – to perform the integration.
- **Local Networks:** the local networks must be compliant with Passpoint (Hotspot 2.0). – Most recent Wi-Fi access points and controllers from major vendors support Passpoint today, however older models or products from more exotic manufacturers might not do. The configuration of the local infrastructure is typically handled by the Managed Service Providers (MSPs).
- **User Devices:** the subscribers' mobile devices (belonging to visitors, guests, employees, etc.) must be compliant with Passpoint (Hotspot 2.0) – All recent iOS and Android-based smartphones or tablets support Passpoint today, however older models or products running less popular operating systems might not do. Laptops compatibility is also more erratic. It is therefore essential to maintain a traditional onboarding service in parallel with Passpoint to handle non-compatible devices.

High-Level Topology

The diagram below illustrates the high-level topology for the end-to-end Trusted WiFi Passpoint service:



Glossary

The following is a glossary of the most common terms used regarding this solution.

Term	Abbreviation	Description
Deployment	-	Enabling of a Trusted WiFi product module for a property via the Trusted WiFi interface.
Module	-	Product or service purchased from Trusted WiFi that is managed through its own sub-section via the Trusted WiFi interface.
License	-	Legal agreement that grants users the right to use specific software, outlining terms and conditions for its usage, distribution, and potential modifications, while protecting the intellectual property of the software developer. GlobalReach licenses comprise of two different types: one-off licenses - typically to initially enable a software module. recurring licenses - typically including software updates and technical support, or more in the case of OpEx consumption commercial models based on price per month/quarter/year. Trusted WiFi is sold as a combination of one-off licenses to activate the service and recurring licenses based on a price per Wi-Fi access point per month.
Managed Service Provider	MSP	The third-party company that remotely manages and monitors a client's IT infrastructure and end-user systems, offering services like network and infrastructure management, security, and 24/7 technical support.
Organization	Org	A company account in Trusted WiFi.
Operator	-	An organization type account in Trusted WiFi that is used by MSPs to manage a property's Wi-Fi network.
Customer	-	An organization type account in Trusted WiFi typically used for customers/brands that allows grouping to view all properties belonging to the same company even if managed by several different MSPs.
Linked Organization	Linked Org	A link creating a relationship between an operator and a customer account, allowing a customer to view a property while allowing an operator to manage it.
User	-	An individual accessing a product, service or system. In the context of Trusted WiFi, a user is setup to access products and deployments information at Admin/Editor/Viewer levels, according to their relevant role permissions. In the context of a given product or service, a user is another term referring to a subscriber: the person at the end of the chain interacting with that product or service – usually through a device.
Property	-	Trusted WiFi concept representing an Individual site or location where products are deployed.
Linked Property	-	Site or location shared between operator and customer accounts.

Passpoint Software Development Kit	Passpoint SDK	The service that sits within the customer's mobile app and that is connected to the Trusted WiFi RADIUS infrastructure allowing Passpoint profiles to be created for a given Passpoint realm.
Passpoint Realm	-	The customer specific domain that is used to provision Passpoint profiles that are approved for connection to any associated network.
Passpoint Profile	-	The security certified profile that sits on a subscriber's device. If installed correctly it allows seamless authentication to the secure Passpoint SSID.
Secure Passpoint Service Set Identifier	Secure Passpoint SSID	The Wi-Fi network that is associated to the Passpoint realm configured to allow subscriber devices with a valid Passpoint profile to seamlessly connect to the Passpoint network at a property.
Subscriber	-	An individual person using a service.
Subscriber Device	-	The equipment – typically a smartphone, tablet or computer – the subscriber is using to connect to the service.
Collision-Resistant Unique Identifier	CUID	A unique identifier designed to be collision-resistant – meaning engineered to minimize the likelihood of generating duplicate IDs even in distributed systems – and more efficient in terms of space and database indexing performance due to its sequential nature. In the context of Passpoint, a CUID is delivered to a subscriber device when it requests a Passpoint profile.
Customer Loyalty Mobile Application	Mobile App	The iOS and/or Android digital application used by businesses to engage and reward their customers through loyalty programs. In the context of Passpoint, the best user experience is to offer Passpoint through a customer/brand mobile app integration using the Trusted WiFi SDK.

RUCKUS SMARTZONE CONFIGURATION

Supported Versions

Version 3.62 is the oldest version that supports Passpoint. Ruckus recommends running version 5.22 on the controller and either 3.62 or 5.12 on the APs. This will ensure support for both older and newer APs.

Prerequisites

It is assumed the following prerequisites are met before configuring a Ruckus SmartZone WLAN Controller for a Trusted WiFi Passpoint service:

- A supported Ruckus SmartZone running version 5.22 or greater, activated and licensed.
- A Trusted WiFi account with operator permissions.
- A core Trusted WiFi Passpoint service configured and tested.
- A property in Trusted WiFi with deployed Gateway and Passpoint modules.

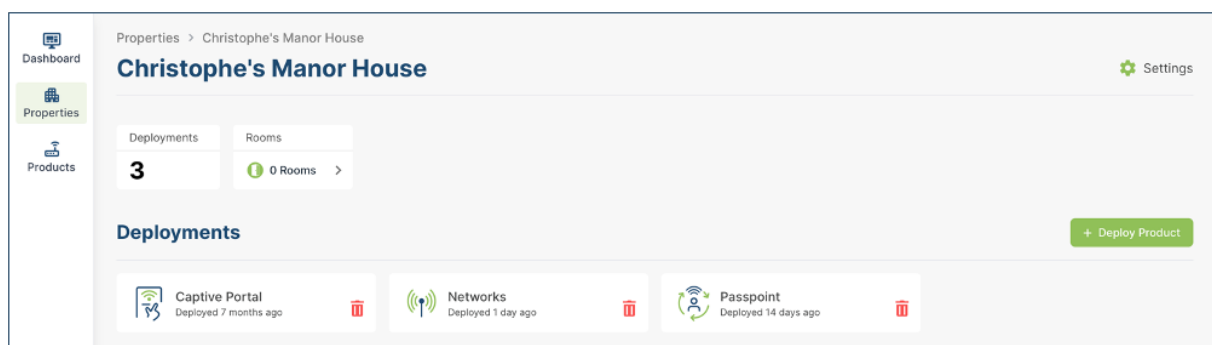
- A deployed and configured Wi-Fi network.

Warning

The Ruckus configuration is focusing on Passpoint v1.0. All properties must share the same NAI realm, RADIUS IP / port settings and SSID name for the Passpoint service (CustomerPasspoint). Each property requires a separate RADIUS secret and NAS identifier, both generated when a configuration is activated within the Trusted WiFi management platform.

Core Passpoint Settings

- Log into Trusted WiFi, then click on the **Properties** icon in the left menu to view the properties list.
- Select or search for the property you wish to work on. This will open its deployments page:



- Click on the **Passpoint** tile and then click on the **Configuration** option in the left menu to display the Passpoint settings summary as per the example below:

Configuration	
NAI Realm	customer.cloud.global
Friendly Name	customer.cloud.global
Primary RADIUS IP	116.214.120.1
Secondary RADIUS IP	116.214.121.1
RADIUS Authentication Port	1812
RADIUS Accounting Port	1813
NAS Identifier	1ft386a9
RADIUS Secret	5dc7b90b296789h08a6ef764b1e3d

- Take note of the details for your respective installation as they will be required at a later step.

Ruckus SmartZone Configuration

RADIUS

- Proceed to **Security > Authentication > Proxy (SZ Authenticator) > Create** to display the page below:

Create Authentication Service

* Name:

Friendly Name:

Description:

Service Protocol: ☒ RADIUS ☐ Active Directory ☐ LDAP

RADIUS Service Options

Encryption: ☐ OFF ☐ TLS

RFC 5580 Out of Band Location Delivery: ☐ OFF ☐ Enable for Ruckus AP Only

Primary Server

* IP Address/FQDN:

* Port:

* Shared Secret:

* Confirm Secret:

Secondary Server

Backup RADIUS: ☒ ON ☐ OFF Enable Secondary Server ☐ OFF Automatic Fallback Disable

* IP Address/FQDN:

* Port:

* Shared Secret:

* Confirm Secret:

- Complete the fields as per the property RADIUS settings and click **OK**.
 FieldDescriptionNameEnter the name of the Server.Friendly NameEnter the Friendly Name. It should match the SSID name to ensure Passpoint connection is seamless.DescriptionEnter an optional description.Service ProtocolSelect the RADIUS radio button.Primary ServerIP addressEnter the <Primary RADIUS IP> from Trusted WiFi.PortEnter the < RADIUS Authentication Port> from Trusted WiFi.Shared SecretEnter the < RADIUS Shared Secret> from Trusted WiFi.Confirm SecretRe-enter the <RADIUS Shared Secret> from Trusted WiFi.Secondary ServerBack RADIUSEnable the Secondary Server toggle.IP addressEnter the <Secondary RADIUS IP> from Trusted WiFi.PortEnter the < RADIUS Authentication Port> from Trusted WiFi.Shared SecretEnter the < RADIUS Shared Secret> from Trusted WiFi.Confirm SecretRe-enter the <RADIUS Shared Secret> from Trusted WiFi.

Wi-Fi Operator

- Click on **Services> Hotspot & Portals > Hotspot 2.0** to display the page below:

Navigation: Monitor | Network | Security | **Services** | Administration | search menu

Breadcrumbs: Services > Hotspots & Portals > Hotspot 2.0

Sub-headers: Guest Access | Hotspot (WISPr) | **Hotspot 2.0** | Web Auth | UA Blacklist | Portal Detection & Suppression | WeChat

Wi-Fi Operator

Organization: D System, D NMX

Actions: + Create | Configure | Clone | Delete

Name	Manage By	Description	Last Modified On	Last Modified By
No data 1				

Identity Provider

Actions: + Create | Configure | Clone | Delete

Name	Manage By	Online Signup Service	Description	Last Modified On	Last Modified By
No data 1					

- Under the **Wi-Fi Operator** heading, click **+Create** to display the page below:

Create Hotspot 2.0 Wi-Fi Operator Profile

Name:

Description:

Domain Names: + Add X Cancel Delete

Domain Name

Signup Security: ☐ OFF Support Anonymous Authentication (OSEN)

Certificate: No data available + Add X Cancel Delete

Friendly Names: + Add X Cancel Delete

Language	Name
English	

Advice of Charge: + Create Configure Delete

Type	NAI Realm	Plan Information

Operator Icon: Browse Clear + Add X Cancel Delete

Language	Icon	File Name
English		

Terms Conditions: File Name:

Time Stamp:

OK Cancel

- Complete the fields as per the description below and click **OK** to save the Operator configuration:
 - FieldDescriptionNameEnter the desired Name.
 - DescriptionEnter an optional description.
 - Domain NameEnter the first Domain Name as per the Trusted WiFi settings and click +Add to save.
 - LanguageSelect the appropriate language from the pull-down list and add the SSID as the Friendly Name. Click on the +Add to save.

Identity Provider

- This configuration enables authentication and accounting. The table below details the items for the Service Provider setup (listed as Identity Provider in the Ruckus configuration).

Option	Required	Data Source	Description	Name	Y	Installer's own	Name of Identity Provider	Description	Y	Trusted WiFi settings	List of Network Access Identifiers (NAI) realms corresponding to the Service Providers or other entities whose networks or services are accessible via this AP.	Authentication Realm	Y	Installer's own	RADIUS realm for "No Match" and "Unspecified"
- Under the **Identity Providers** heading on the same screen, click **+Create** to display the page below:

Create Hotspot 2.0 Identity Provider

[Network Identifier](#) → [Online Signup & Provisioning](#) → [Authentication](#) → [Accounting](#) → [Review](#)

* Name:
 Description:
 PLMNs: * MCC * MNC [+ Add](#) [X Cancel](#) [Delete](#)
 MCC MNC
 * Realms: * Name: [+ Add](#) [X Cancel](#) [Delete](#)
 * Encoding: RFC-4282
 EAP Methods:
 #1 #2 #3 #4
 EAP Method: N/A
 Name Encoding EAP Methods
 Home OIs: * Name * Length 5 Hex * Organization ID [+ Add](#) [X Cancel](#) [Delete](#)
 Name Length Organization ID
[Next](#) [Cancel](#)

- Complete the fields as per the description below and click **OK** to save the Operator configuration:
 - FieldDescriptionNameEnter the desired Name for the RADIUS service.
 - DescriptionEnter an optional description.
 - RealmsConfigure the first Realm information (NAI in Settings).Enter the Name of the Realm.Enter Encoding values RFC-4282.Enter the EAP method as EAP-TTLS.Click the +Add to save.
- Click the **Next** button to display the page below:

Create Hotspot 2.0 Identity Provider

[Network Identifier](#) → [Online Signup & Provisioning](#) → [Authentication](#) → [Accounting](#) → [Review](#)

☐ OFF Enable Online Signup & Provisioning

[Back](#) [Next](#) [Cancel](#)

Warning

Since this configuration is designed for Passpoint v1.0, no configuration is required and therefore the user should not Enable Online Signup & Provisioning.

- Click the **Next** button to display the page below:

Create Hotspot 2.0 Identity Provider

Network Identifier → Online Signup & Provisioning → **Authentication** → Accounting → Review

Authentication Services for Access WLAN

[+ Create](#) [Configure](#) [Delete](#)

Realm	Protocol	Auth Service	Dynamic VLAN ID
No Match	NA	NA-Disabled	N/A
Unspecified	NA	NA-Disabled	N/A

Note: If device onboarding was done with credential type 'remote', then map your 'realm' value to its respective authentication service PLUS define 'Unspecified' realm & map it to corresponding authentication service to properly handle legacy (non-Hotspot 2.0) devices.

[Back](#) [Next](#) [Cancel](#)

- Click on **No Match Realm** and then click on the **Configure** button to display the page below:

Edit Realm Based Authentication Service: No Match

* Realm: No Match

* Service: [NA] NA-Disabled [+](#) [✎](#)

Dynamic VLAN ID:

[OK](#) [Cancel](#)

- Select the correct RADIUS service for this realm from the drop-down list and click **OK** to save.
- Repeat the previous steps for the **Unspecified Realm** and click **OK** to save.
- Click **Next** to display the page below:

Create Hotspot 2.0 Identity Provider

Network Identifier → Online Signup & Provisioning → Authentication → **Accounting** → Review

☐ **Enable Accounting**

Accounting Services for Access WLAN

[+ Create](#) [Configure](#) [Delete](#)

Realm	Protocol	Accounting Service
No Match	NA	NA-Disabled
Unspecified	NA	NA-Disabled

Note: A realm to service mapping define the accounting service for each of the realm specified in this table. When the accounting service for a particular realm is 'NA', then accounting is disabled.

[Back](#) [Next](#) [Cancel](#)

- Select the **Accounting** tab to configure the accounting ports and toggle the **Enable Accounting On**, which will allow the default port to be used for this RADIUS service as per the Trusted WiFi settings.
- Click **Next** to display the page below:

Create Hotspot 2.0 Identity Provider

Network Identifier → Online Signup & Provisioning → Authentication → Accounting → Review

Name: Demo

Description:

PLMNs: MCC MNC

Realms:

Name	Encoding	EAP Methods
customer.cloud	RFC-4282	#1: N/A #2: N/A #3: N/A #4: N/A

Home OIs:

Name	Length	Organization ID

☐ OFF Enable Online Signup & Provisioning

Authentication Services for Access WLAN

Realm	Protocol	Auth Service	Dynamic VLAN ID
No Match	RADIUS	Passpoint	N/A
Unspecified	RADIUS	- Passpoint	N/A

Note: If device onboarding was done with credential type 'remote', then map your 'realm' value to its respective authentication service PLUS define 'Unspecified' realm & map it to corresponding authentication service to properly handle legacy (non-Hotspot 2.0) devices.

☒ ON Enable Accounting

Back OK Cancel

- Once satisfied with the configuration, click **OK** to save.

Passpoint WLAN Profile

- From the menu, go to the **Networks > Wireless > Wireless LANs** section to display the page below:

Monitor Network Security Services Administration

Network > Wireless > Wireless LANs

VIEW MODE: List Group

Wireless LANs

Organization: D System, D MMX

Name	Alerts	SSID	Auth Method	Encryption Method	Clients	Traffic	VLAN	Application Re
No data								

- Select the appropriate Zone from the side menu and then click on Create. Fill in the screen as follows

The screenshot displays a configuration page with four main sections:

- General Options:** Includes fields for Name, SSID, HESSID, Description, Zone (set to Home-Secure), and WLAN Group (set to default). A blue bracket labeled "Installer" groups these fields.
- Authentication Options:** Includes Authentication Type (with radio buttons for Standard usage, Hotspot (WISPr), Guest Access, Web Authentication, Hotspot 2.0 Access, Hotspot 2.0 Onboarding, and WeChat) and Method (with radio buttons for Open, 802.1X EAP, MAC Address, and 802.1X EAP & MAC). An arrow labeled "Installer" points to the Hotspot 2.0 Access radio button.
- Encryption Options:** Includes Method (with radio buttons for WPA2, WPA3, WPA2/WPA3-Mixed, OWE, WPA-Mixed, WEP-64 (40 bits), WEP-128 (104 bits), and None), Algorithm (with radio buttons for AES, AUTO, and AES-GCMP-256), 802.11r Fast Roaming (set to OFF), and 802.11w MFP (with radio buttons for Disabled, Capable, and Required).
- Data Plane Options:** Includes Access Network (set to OFF) and Tunnel WLAN traffic through Ruckus GRE.

- Complete the fields, as per the description below:

Field	Description
Name	Enter the name of the Service.
SSID	Enter the name of the SSID.
HESSID	Not required.
Description	Enter an optional description.
WLAN Group	Default, or can be changed to the installer's group if required.
- Authentication options
 Authentication Type: Check the Hotspot 2.0 Access radio button.
 Method: Check the 802.1x EAP radio button.
- Encryption Options
 Method: Check the WPA2 radio button.
 Algorithm: Check the AES radio button.
 802.11r: Default is off, but the installer can decide to enable it if required at the property.
- Scroll down to the **Hotspot 2.0 Profile** section:

Hotspot 2.0 Profile

Hotspot 2.0 Profile: NMX Passpoint +

Authentication Service: ☐ OFF RFC 5580 Location Delivery Support: Requires that Authentication Service be set to 'Use the controller as a proxy'

Accounting Service: Send interim update every 5 Minutes (0-1440)

Wireless Client Isolation

Client Isolation: ☐ OFF Isolate client traffic from all hosts on the same VLAN/subnet

Isolation Whitelist: Gateway Only (Automatic) +

(The whitelist requires entries for the subnet gateway and other allowed hosts.)
(UE with static IP will not be isolated when autowhitelist/hybrid whitelist with autowhitelist is enabled)

RADIUS Options

Firewall Options

Advanced Options

[?] BSS Priority: ☒ High ☐ Low

Wi-Fi Calling: ☐ OFF

Client Fingerprinting: ☒ ON

[?] Access VLAN: VLAN ID 833

☐ OFF Enable VLAN Pooling

If DHCP/NAT is enabled on an AP, the VLANs configured should be aligned with the VLANs in the DHCP Profile(s). Clients will have connectivity issues if the client resolves a VLAN other than those in the DHCP profile(s).

OK Cancel

- Click on the + to create a new Hotspot 2.0 WLAN profile and display the page below:

Create Hotspot 2.0 WLAN Profile

Name: Demo

Description:

Operator: Demo Operator +

Identity Providers: Identity Provider No data available + Add X Cancel Delete Create

Identity Provider	Online Signup Service	
Passpoint	Disabled	☒

You can configure single SSID and Onboarding SSID when you add an identity provider that has Online Signup & Provisioning enabled

Advanced Options

OK Cancel

- Complete the fields, as per the description below:
 - FieldDescriptionNameEnter the WLAN profile name.
 - DescriptionEnter an optional description.
 - OperatorSelect the operator from the drop-down list.
 - Identity ProviderSelect the identity providers from the pull-down list (which were previously configured) and select Add.
- Click **OK** to save the configuration.
- Under the **Advanced Options** section, ensure you configure the VLAN you have setup.
- Click **OK** to save this WLAN.

Hardware Configuration Guide for Ubiquiti UniFI Network Server

Download this guide 

INTRODUCTION

Scope and Purpose

Thank you for purchasing the Trusted WiFi solution. This document is a hardware configuration guide describing how to setup Ubiquiti UniFI Network Server software for a Trusted WiFi Passpoint service.

- For more information on how to setup an end-to-end Trusted WiFi Passpoint service, please refer to the Trusted WiFi Passpoint Administration Guide.
- For complete information on how to setup Ubiquiti UniFI Network Server software, please refer to the vendor's original documentation.

Documentation Conventions

{{snippet.Documentation Conventions}}

Passpoint Overview

Passpoint – also known as Hotspot 2.0 – is an industry-wide next generation approach to public internet access driven by the Wi-Fi Alliance that brings the following benefits:

- Frictionless onboarding and roaming, thanks to a one-time registration followed by automatic access to interconnected hotspots
- More secure and private Wi-Fi connections, compared with general visitor networks

Passpoint is based on the IEEE 802.11u standard, which is a set of protocols enabling cellular-like roaming. Following the initial enrollment, frequent users such as visitors, guests or employees bypass repeated logins, forms and passwords, as their mobile devices automatically join the Wi-Fi subscriber service when they return to a venue or roam between inter-linked Passpoint enabled hotspots and providers, while being better protected against potential cyber threats.

If a device supports 802.11u and is enrolled to a service, it automatically communicates with the Wi-Fi infrastructure via the access points to discover the network SSID and connects securely to it by presenting its access credentials. Upon successful authentication, the device is provisioned with Passpoint standards-based management objects - known as Per-Provider Subscription Management Objects (PPS-MO).

GlobalReach - an ASSA ABLOY company - has been involved with Passpoint since its inception and even contributed to the creation and initial pilot testing of the standard. As a result, it is one of the few trusted worldwide experts on this topic, with a proven platform backed by real-world operational experiences at scale.

The best user experience is to offer Passpoint through a customer/brand mobile app integration, as it further simplifies the onboarding process, while incentivizing app downloads and customer loyalty, leading to further engagement and monetization opportunities. To this effect, Trusted WiFi offers a Software Development Kit (SDK) for easy app integration.

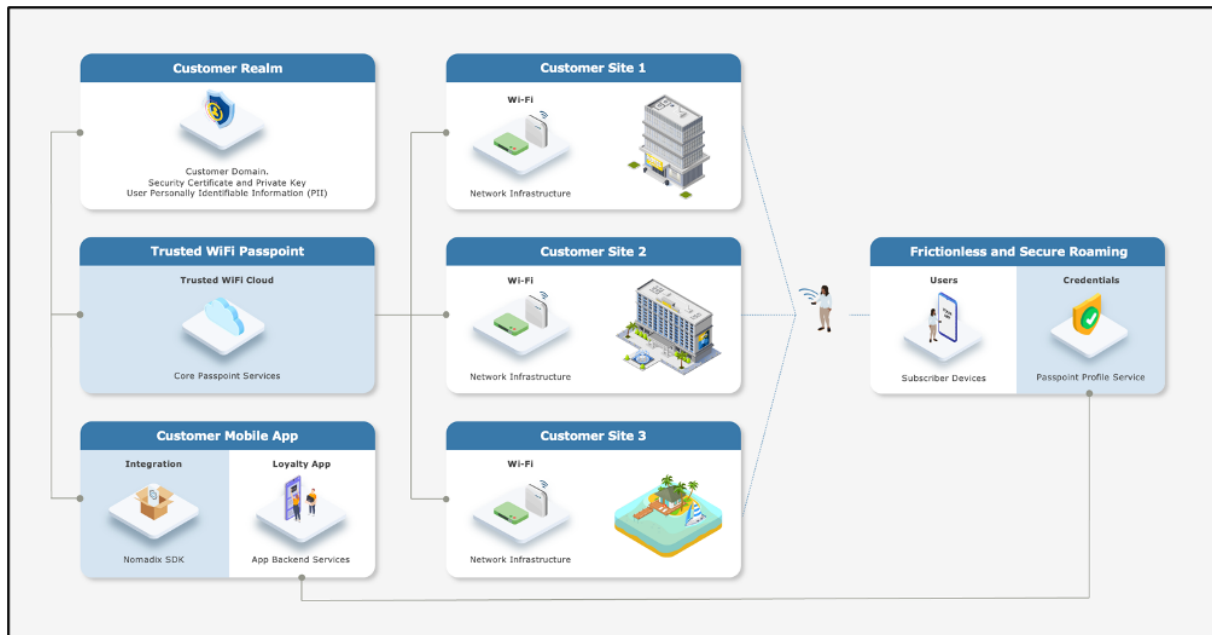
End-to-End Service Components

Implementing an end-to-end Trusted WiFi Passpoint service requires a combination of the following software and hardware components:

- **Trusted WiFi Passpoint:** the core services are offered and managed centrally via the **Trusted WiFi Passpoint Module** (hosted and operated by GlobalReach). Following an initial setup performed by the GlobalReach Operations team, Managed Service Providers (MSPs) can then add sites to customer realms and monitor the solution.
- **Customer Realm:** the service requires connectivity to a realm including domain, security certificate and private key, as well as the database holding the users' Personally Identifiable Information (PII). It is the responsibility of the Identity Provider – typically the customer/brand – to make this part available.
- **Mobile App:** the best user experience is to offer Passpoint through a customer/brand mobile app integration. Trusted WiFi offers a **Software Development Kit (SDK)** for easy implementation. It is the responsibility of the app owner – typically the customer/brand – to perform the integration.
- **Local Networks:** the local networks must be compliant with Passpoint (Hotspot 2.0). – Most recent Wi-Fi access points and controllers from major vendors support Passpoint today, however older models or products from more exotic manufacturers might not do. The configuration of the local infrastructure is typically handled by the Managed Service Providers (MSPs).
- **User Devices:** the subscribers' mobile devices (belonging to visitors, guests, employees, etc.) must be compliant with Passpoint (Hotspot 2.0) – All recent iOS and Android-based smartphones or tablets support Passpoint today, however older models or products running less popular operating systems might not do. Laptops compatibility is also more erratic. It is therefore essential to maintain a traditional onboarding service in parallel with Passpoint to handle non-compatible devices.

High-Level Topology

The diagram below illustrates the high-level topology for the end-to-end Trusted WiFi Passpoint service:



Glossary

The following is a glossary of the most common terms used regarding this solution.

Term	Abbreviation	Description
Deployment	-	Enabling of a Trusted WiFi product module for a property via the Trusted WiFi interface.
Module	-	Product or service purchased from Trusted WiFi that is managed through its own sub-section via the Trusted WiFi interface.
License	-	Legal agreement that grants users the right to use specific software, outlining terms and conditions for its usage, distribution, and potential modifications, while protecting the intellectual property of the software developer. GlobalReach licenses comprise of two different types: one-off licenses - typically to initially enable a software module. recurring licenses - typically including software updates and technical support, or more in the case of OpEx consumption commercial models based on price per month/quarter/year. Trusted WiFi is sold as a combination of one-off licenses to activate the service and recurring licenses based on a price per Wi-Fi access point per month.
Managed Service Provider	MSP	The third-party company that remotely manages and monitors a client's IT infrastructure and end-user systems, offering services like network and infrastructure management, security, and 24/7 technical support.
Organization	Org	A company account in Trusted WiFi.
Operator	-	An organization type account in Trusted WiFi that is used by MSPs to manage a property's Wi-Fi network.
Customer	-	An organization type account in Trusted WiFi typically used for customers/brands that allows grouping to view all properties belonging to the same company even if managed by several different MSPs.
Linked Organization	Linked Org	A link creating a relationship between an operator and a customer account, allowing a customer to view a property while allowing an operator to manage it.
User	-	An individual accessing a product, service or system. In the context of Trusted WiFi, a user is setup to access products and deployments information at Admin/Editor/Viewer levels, according to their relevant role permissions. In the context of a given product or service, a user is another term referring to a subscriber: the person at the end of the chain interacting with that product or service – usually through a device.
Property	-	Trusted WiFi concept representing an Individual site or location where products are deployed.
Linked Property	-	Site or location shared between operator and customer accounts.

Passpoint Software Development Kit	Passpoint SDK	The service that sits within the customer's mobile app and that is connected to the Trusted WiFi RADIUS infrastructure allowing Passpoint profiles to be created for a given Passpoint realm.
Passpoint Realm	-	The customer specific domain that is used to provision Passpoint profiles that are approved for connection to any associated network.
Passpoint Profile	-	The security certified profile that sits on a subscriber's device. If installed correctly it allows seamless authentication to the secure Passpoint SSID.
Secure Passpoint Service Set Identifier	Secure Passpoint SSID	The Wi-Fi network that is associated to the Passpoint realm configured to allow subscriber devices with a valid Passpoint profile to seamlessly connect to the Passpoint network at a property.
Subscriber	-	An individual person using a service.
Subscriber Device	-	The equipment – typically a smartphone, tablet or computer – the subscriber is using to connect to the service.
Collision-Resistant Unique Identifier	CUID	A unique identifier designed to be collision-resistant – meaning engineered to minimize the likelihood of generating duplicate IDs even in distributed systems – and more efficient in terms of space and database indexing performance due to its sequential nature. In the context of Passpoint, a CUID is delivered to a subscriber device when it requests a Passpoint profile.
Customer Loyalty Mobile Application	Mobile App	The iOS and/or Android digital application used by businesses to engage and reward their customers through loyalty programs. In the context of Passpoint, the best user experience is to offer Passpoint through a customer/brand mobile app integration using the Trusted WiFi SDK.

UNIFI NETWORK SERVER CONFIGURATION

Supported Versions

This document is based on firmware 9.3.45.

Prerequisites

It is assumed the following prerequisites are met before configuring UniFi Network Server for a Trusted WiFi Passpoint service:

- A supported UniFi account activated and licensed.
- A Trusted WiFi account with operator permissions.
- A core Trusted WiFi Passpoint service configured and tested.
- A property in Trusted WiFi with deployed Passpoint modules.
- A deployed and configured Wi-Fi network.

Notes

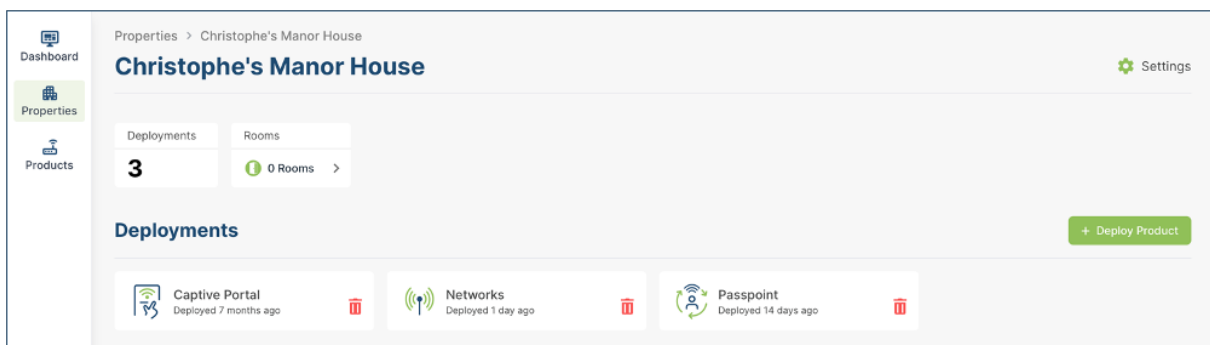
This document focuses on a specific part of the Ubiquiti UniFi Network Server configuration only. Please refer to the Passpoint Administration Guide for instructions on how to configure the end-to-end Passpoint service. Please refer to the original Ubiquiti documentation for complete instructions on Ubiquiti configuration.

Warning

All properties must share the same NAI realm, RADIUS IP / port settings and SSID name for the Passpoint service (CustomerPasspoint). Each property requires a separate RADIUS secret and NAS identifier, both generated when a configuration is activated within the Trusted WiFi management platform.

Core Passpoint Settings

- Log into Trusted WiFi, then click on the **Properties** icon in the left menu to view the properties list.
- Select or search for the property you wish to work on. This will open its deployments page:



- Click on the **Passpoint** tile and then click on the **Configuration** option in the left menu to display the Passpoint settings summary as per the example below:

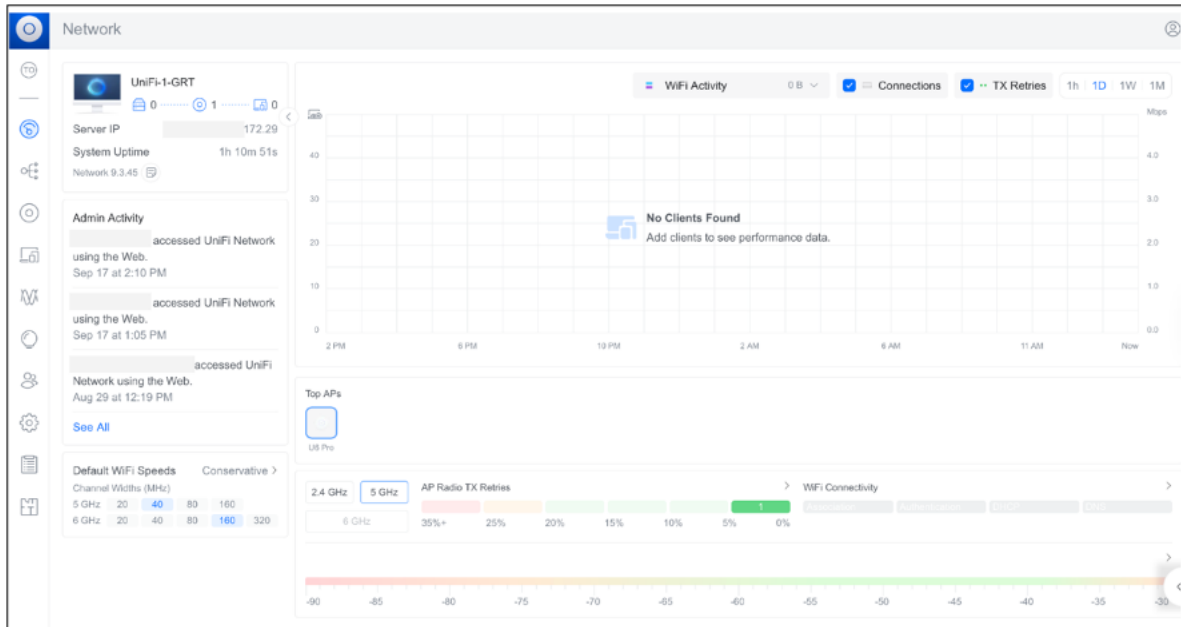
Configuration	
NAI Realm	customer.cloud.global
Friendly Name	customer.cloud.global
Primary RADIUS IP	116.214.120.1
Secondary RADIUS IP	116.214.121.1
RADIUS Authentication Port	1812
RADIUS Accounting Port	1813
NAS Identifier	1ft386a9
RADIUS Secret	5dc7b90b296789h08a6ef764b1e3d

- Take note of the details for your respective installation as they will be required at a later step.

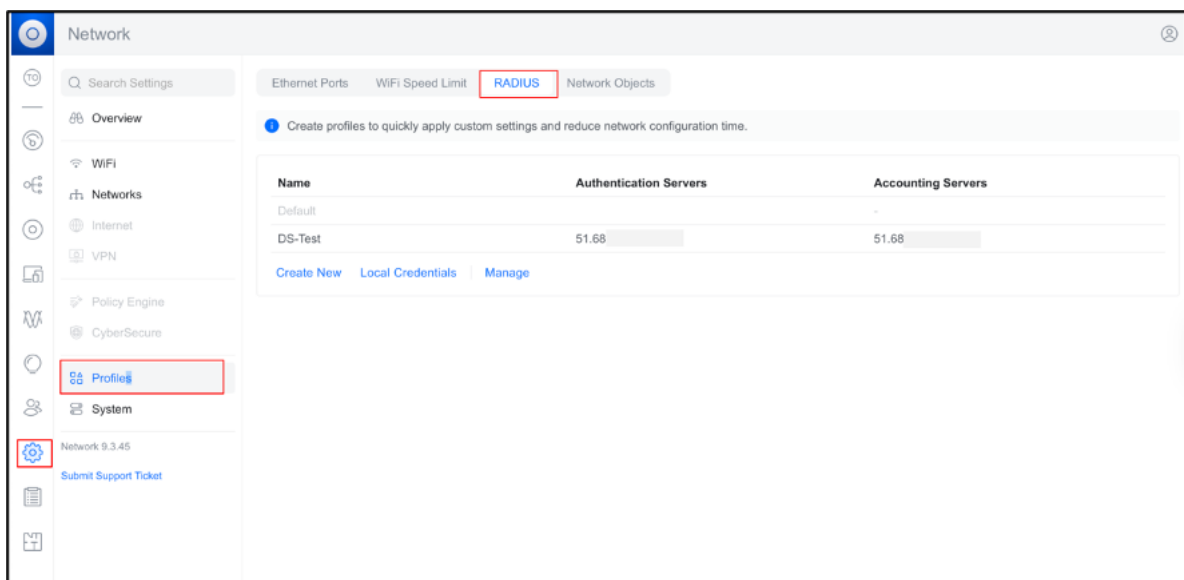
UniFi Network Server Configuration

RADIUS Profile

- Log into the UniFi Dashboard with your credentials to display the page below:



- On the left-hand side menu items, click on **Settings > Profiles > RADIUS**. The following screen will appear:



- Click **Create New** and the following pop up will appear on the right:

- Complete the fields, as per the description:

Field	Description
Name	Provide a RADIUS name
RADIUS Assigned VLAN Support	Select the Wireless Network checkbox
TLS	Leave unchecked
IP Address	Enter the <Primary RADIUS IP> from Trusted WiFi
Port	Enter the <RADIUS Authentication Port> from Trusted WiFi
Shared Secret	Enter the <RADIUS Shared Secret> from Trusted WiFi
Add	Select Add and to add the Secondary RADIUS Sever IP from Trusted WiFi
IP Address	Enter the <Secondary RADIUS IP> from Trusted WiFi
Port	Enter the < RADIUS Authentication Port> from Trusted WiFi
Shared Secret	Enter the < RADIUS Shared Secret> from Trusted WiFi
Accounting Servers	Enable this check box
IP Address	Enter the <Primary RADIUS IP> from Trusted WiFi
Port	Enter the < RADIUS Accounting Port> from Trusted WiFi
Shared Secret	Enter the < RADIUS Shared Secret> from Trusted WiFi
Interim Update Interval	Enable this checkbox and enter a value or leave the default as 3600 seconds

RADIUS

Name
RADIUS

RADIUS Assigned VLAN Support

☐ Wired Networks ⓘ

☒ Wireless Networks ⓘ

☐ TLS ⓘ

IP Address

Port
1812

Shared Secret

Add

IP Address	Port	Shared Secret	Edit
116.214.120.1	1812	*****	ⓘ
116.214.121.1	1812	*****	ⓘ

☒ Accounting Servers

IP Address

Port
1813

Shared Secret

Add

IP Address	Port	Shared Secret	Edit
116.214.120.1	1813	*****	ⓘ
116.214.121.1	1813	*****	ⓘ

☒ Interim Update Interval
3600 Seconds

- Click **Add** to save this RADIUS profile configuration and add it to the account:

Network

Ethernet Ports WiFi Speed Limit **RADIUS** Network Objects

Create profiles to quickly apply custom settings and reduce network configuration time.

Name	Authentication Servers	Accounting Servers
Default	-	-
DS-Test	51.68	51.68
RADIUS	2 Servers ⓘ	2 Servers ⓘ

Create New Local Credentials Manage

Passpoint (Hotspot 2.0)

- Proceed to the **WiFi** page on the left of the screen to display the view below:

Name	Network	Broadcasting APs	WiFi Band	Clients	Security
UBNT-GRT-CP-Test	Native Network	All APs	2.4 GHz 5 GHz	-	WPA2
UBNT-8021X	Native Network	U6 Pro	2.4 GHz 5 GHz	-	WPA2 Enterprise
UBNT-GRT-CP-STAGING-TEST	Native Network	All APs	2.4 GHz 5 GHz	-	WPA2

[Create New](#) [Manage](#)

Channel Plan
Click on a channel to exclude it from use

2.4 GHz 2412-2484 MHz
Channels 1 2 3 4 5 6 7 8 9 10 11 12 13 14
20 MHz [X] [] [] [] [] [] [] [] [] [] [] [] [] []
40 MHz [] [] [] [] [] [] [] [] [] [] [] [] [] []

5 GHz 5180-5885 MHz
Channels 36 40 44 48 52 56 60 64 100 104 108 112 116 120 124 128 132 136 140 144 149 153 157 161 165
20 MHz []
40 MHz []
80 MHz []
160 MHz []

☒ In Use ☐ Enabled ☐ DFS ☒ Not available ⓘ

[Restore to Defaults](#)

- Click **Create New** and the following screen will be displayed:

< Name

Password Must have at least 8 characters.

Broadcasting APs ⓘ ☒ All ☐ Specific ☐ Groups

Advanced Auto Manual

Private Pre-Shared Keys ⓘ ☐

Hotspot ⓘ ☒ Off ☐ Captive Portal ☐ Passpoint

Enhanced IoT Connectivity ⓘ ☐

WiFi Band ⓘ ☒ 2.4 GHz ☒ 5 GHz ☐ 6 GHz

Band Steering ⓘ ☒

Hide WiFi Name ☐

Client Device Isolation ⓘ ☐

Proxy ARP ⓘ ☐

BSS Transition ⓘ ☒

UAPSD ⓘ ☐

Fast Roaming ⓘ ☐

WiFi Speed Limit ⓘ ☐

Multicast Enhancement ⓘ ☐

Multicast and Broadcast Control ⓘ ☐

802.11 DTIM Period ⓘ ☒ Auto 1 3

Network Name Broadcast Control ⓘ ☒ Auto ☐ Manual

[Add WiFi Network](#)

- Complete the fields, as per the description below:

Field	Description
Name	Enter a name for the SSID
Password	Leave blank
Broadcasting APs	Select All, Specific or Group radio button
Advanced	Select Manual
Private Pre-Shared Keys	Leave unchecked
Hotspot	2.0 Select Passpoint radio button
Venue Name	Provide your venue name
Venue Type	Select the appropriate venue type from the drop down. e.g. Assembly Area
Network Type	Select appropriate network type from the drop down
NAI Realm	Click Add and enter the name realm. e.g. customer.cloud.global
EAP Method	EAP-TTLS
Sub Method	MSCHAP v2
Roaming Consortium List	Leave blank
3GPP Cellular Network	Leave blank
Domain List	Add the domain e.g. customer.cloud.global
Operator Friendly Name	Provide the operator's friendly name
Enhanced IoT Connectivity	Leave unchecked
WiFi Band	Select 2.4 GHZ and 5 GHZ
Band Steering	Enable this checkbox
Hide WiFi Name	Leave unchecked
Client Device Isolation	You can toggle the checkbox on or off as needed
Proxy ARP	You can toggle the checkbox on or off as needed
BSS Transition	Enable this checkbox
UAPSD	You can toggle the checkbox on or off as needed
Fast Roaming	You can toggle the checkbox on or off as needed
WiFi Speed Limit	You can toggle the checkbox on or off as needed
Multicast Enhancement	You can toggle the checkbox on or off as needed

neededMulticast and Broadcast ControlYou can toggle the checkbox on or off as needed802.11
 DTIM PeriodEnable the Auto checkboxMinimum Data Rate ControlSelect the Auto radio
 buttonMAC Address FilterLeave uncheckedRADIUS MAC AuthenticationLeave
 uncheckedSecurity ProtocolSelect WPA2 EnterpriseRADIUS ProfileSelect the RADIUS Profile
 you configured from the step aboveNAS IDSelect Custom. Enter the <NAS ID> from Trusted
 WiFiDAS/DAC(CoA)Leave uncheckedPMFSelect DisabledGroup Rekey IntervalLeave
 uncheckedWiFi Blackout SchedulerSelect Off

- Your configuration will appear similar to the below:

The screenshot displays the configuration page for a network named "Passpoint Demo UniFi". The interface is organized into several sections:

- Header:** Name: Passpoint Demo UniFi
- Broadcasting APs:** All (selected), Specific, Groups
- Advanced:** Auto, Manual (selected)
- Private Pre-Shared Keys:** [checkbox]
- Hotspot:** Off, Captive Portal, Passpoint (selected)
- Venue Information:**
 - Venue Name: Passpoint Demo
 - Venue Type: Unspecified
 - Network Type: Free Public Network
 - IP Address Type Availability: IPv4 (Single NATed pri...), IPv6 (Unavailable)
 - NAI Realm: Add
- Roaming Consortium List:**

Name	Organization ID	Action
customer.cloud.global		Add
- 3GPP Cellular Network:**

MCC	MNC	Action
		Add
- Domain List:**

Domain Name	Action
customer.cloud.global	Remove
- Operator Friendly Name:** Operator 1
- WAN Metrics:** [checkbox]
- Connection Capability:**

IP Protocol	Action	Action
ICMP	Allow	Add

Enhanced IoT Connectivity ⓘ	☐
WiFi Band ⓘ	☒ 2.4 GHz ☒ 5 GHz ☐ 6 GHz
Band Steering ⓘ	☒
Hide WiFi Name	☐
Client Device Isolation ⓘ	☐
Proxy ARP ⓘ	☐
BSS Transition ⓘ	☒
UAPSD ⓘ	☐
Fast Roaming ⓘ	☐
WiFi Speed Limit ⓘ	☐
Multicast Enhancement ⓘ	☐
Multicast and Broadcast Control ⓘ	☐
802.11 DTIM Period ⓘ	☒ Auto ☐ Manual 2.4 GHz: 1 5 GHz: 3
Minimum Data Rate Control ⓘ	☒ Auto ☐ Manual
MAC Address Filter ⓘ	☐
Security Protocol ⓘ	WPA2 Enterprise
RADIUS Profile	RADIUS
NAS ID	☐ AP MAC ☐ Site Name ☐ AP Name ☐ BSSID ☒ Custom 2024444b
DAS/DAC (CoA) ⓘ	☐
PMF ⓘ	☐ Required ☐ Optional ☒ Disabled
Group Rekey Interval ⓘ	☐
WiFi Blackout Scheduler ⓘ	Off On
Add WiFi Network Cancel	

- Select **Add WiFi Network**:

Network						
Name	Network	Broadcasting APs	WiFi Band		Clients	Security
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Passpoint Demo UniFi	Native Network	All APs	2.4 GHz	5 GHz	-	WPA2 Enterprise
Create New Manage						

- The configuration is now completed.

PDF Downloads

Step-by-step resources for IT teams and installers to correctly deploy and manage Trusted WiFi Passpoint.

Latest product guide

Trusted WiFi Passpoint Administration Guide (v1.0)

Trusted WiFi Passpoint Hardware Configuration Guide for Aruba Central (v1.0)

Trusted WiFi Passpoint Hardware Configuration Guide for Aruba Mobility Controllers (v1.0)

Trusted WiFi Passpoint Hardware Configuration Guide for Cisco Meraki (v1.0)

Trusted WiFi Passpoint Hardware Configuration Guide for Nomadix WLAN Controllers (v1.0)

Trusted WiFi Passpoint Hardware Configuration Guide for Ruckus SmartZone (v1.0)

Trusted WiFi Passpoint Hardware Configuration Guide for Ruckus ZoneDirector (v1.0)

Trusted WiFi Passpoint - Hardware Configuration Guide for ExtremeCloud IQ (v1.0)

Trusted WiFi Passpoint - Hardware Configuration Guide for Ruckus One (v1.0)

Trusted WiFi Passpoint - Hardware Configuration Guide for Ubiquiti Unifi Network Server (v1.0)

Ruckus ZoneDirector

Download this guide 

Introduction

Scope and Purpose

Thank you for purchasing the Trusted WiFi solution. This document is a hardware configuration guide describing how to setup a Ruckus ZoneDirector WLAN Controller for a Trusted WiFi Passpoint service.

- For more information on how to setup an end-to-end Trusted WiFi Passpoint service, please refer to the Trusted WiFi Passpoint Administration Guide.
- For complete information on how to setup a Ruckus ZoneDirector WLAN Controller, please refer to the vendor's original documentation.

Documentation Conventions

{{snippet.Documentation Conventions}}

Passpoint Overview

Passpoint – also known as Hotspot 2.0 – is an industry-wide next generation approach to public internet access driven by the Wi-Fi Alliance that brings the following benefits:

- Frictionless onboarding and roaming, thanks to a one-time registration followed by automatic access to interconnected hotspots
- More secure and private Wi-Fi connections, compared with general visitor networks

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The best user experience is to offer Passpoint through a customer/brand mobile app integration, as it further simplifies the onboarding process, while incentivizing app downloads and customer loyalty, leading to further engagement and monetization opportunities. To this effect, Trusted WiFi offers a Software Development Kit (SDK) for easy app integration.

Seamless Onboarding and Roaming

Best user experience via brand mobile loyalty app integration:

- One-time setup in 3 "taps" via a mobile app
- Followed by automatic connection for returning visitors and roaming within any of the participating venues
- No manual SSID selection, no login/password, no reset after a given period
- No impact from MAC address randomization
- Incentive to download the app, leading to visitor engagement opportunities
- Non-compatible devices can still use the legacy onboarding process

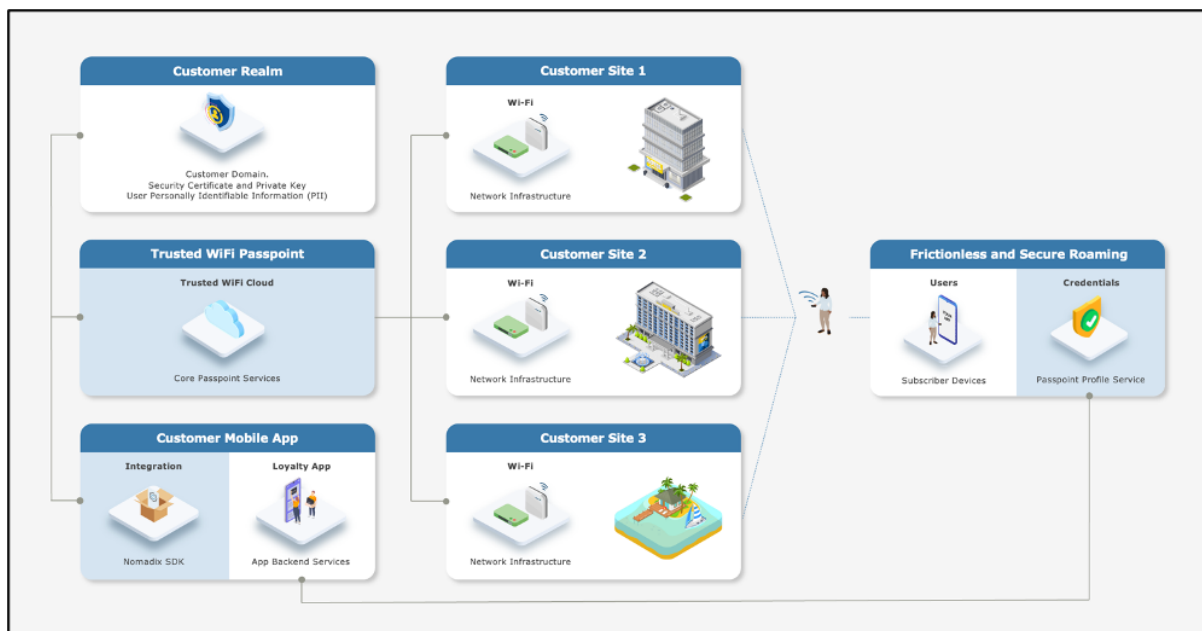
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Linked Organization	Linked Org	A link creating a relationship between an operator and a customer account, allowing a customer to view a property while allowing an operator to manage it.
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Property	-	Trusted WiFi concept representing an Individual site or location where products are deployed.
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Passpoint Realm	-	The customer specific domain that is used to provision Passpoint profiles that are approved for connection to any associated network.
Passpoint Profile	-	The security certified profile that sits on a subscriber's device. If installed correctly it allows seamless authentication to the secure Passpoint SSID.
Secure Passpoint Service Set Identifier	Secure Passpoint SSID	The Wi-Fi network that is associated to the Passpoint realm configured to allow subscriber devices with a valid Passpoint profile to seamlessly connect to the Passpoint network at a property.
Subscriber	-	An individual person using a service.
Subscriber Device	-	The equipment – typically a smartphone, tablet or computer – the subscriber is using to connect to the service.
Collision-Resistant Unique Identifier	CUID	A unique identifier designed to be collision-resistant – meaning engineered to minimize the likelihood of generating duplicate IDs even in distributed systems – and more efficient in terms of space and database indexing performance due to its sequential nature. In the context of Passpoint, a CUID is delivered to a subscriber device when it requests a Passpoint profile.
Customer Loyalty Mobile Application	Mobile App	The iOS and/or Android digital application used by businesses to engage and reward their customers through loyalty programs. In the context of Passpoint, the best user experience is to offer Passpoint through a customer/brand mobile app integration using the Trusted WiFi SDK.

RUCKUS ZONEDIRECTOR CONFIGURATION

Supported Versions

Ruckus ZoneDirector version 10.0 or later is recommended to run Passpoint. This document is using screens from v10.5. While the user interface might vary a little between versions, the requirements are the same.

Warning

The ZoneDirector 1200 is now end of life and should be replaced with a supported alternative.

Prerequisites

It is assumed the following prerequisites are met before configuring a Ruckus ZoneDirector WLAN Controller for a Trusted WiFi Passpoint service:

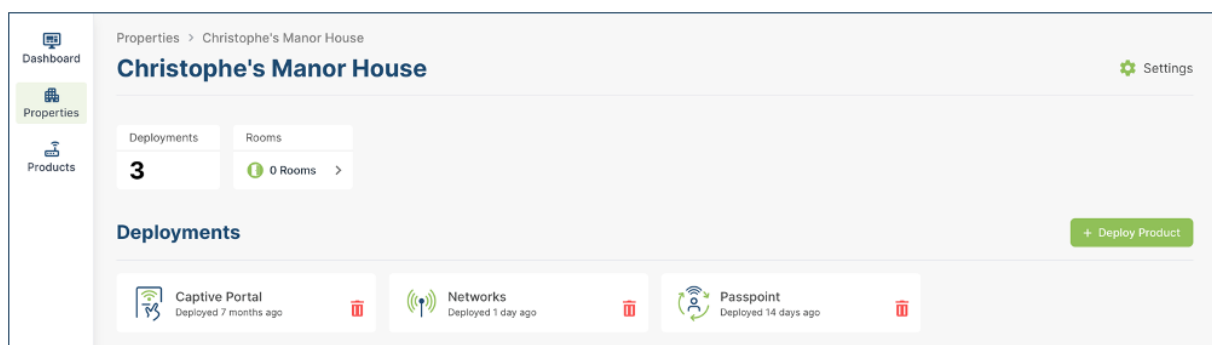
- A supported Ruckus ZoneDirector running version 10.0 or greater, activated and licensed.
- A Trusted WiFi account with operator permissions.
- A core Trusted WiFi Passpoint service configured and tested.
- A property in Trusted WiFi with deployed Gateway and Passpoint modules.
- A deployed and configured Wi-Fi network.

Warning

All properties must share the same NAI realm, RADIUS IP / port settings and SSID name for the Passpoint service (CustomerPasspoint). Each property requires a separate RADIUS secret and NAS identifier, both generated when a configuration is activated within the Trusted WiFi management platform.

Core Passpoint Settings

- Log into Trusted WiFi, then click on the **Properties** icon in the left menu to view the properties list.
- Select or search for the property you wish to work on. This will open its deployments page:



- Click on the **Passpoint** tile and then click on the **Configuration** option in the left menu to display the Passpoint settings summary as per the example below:

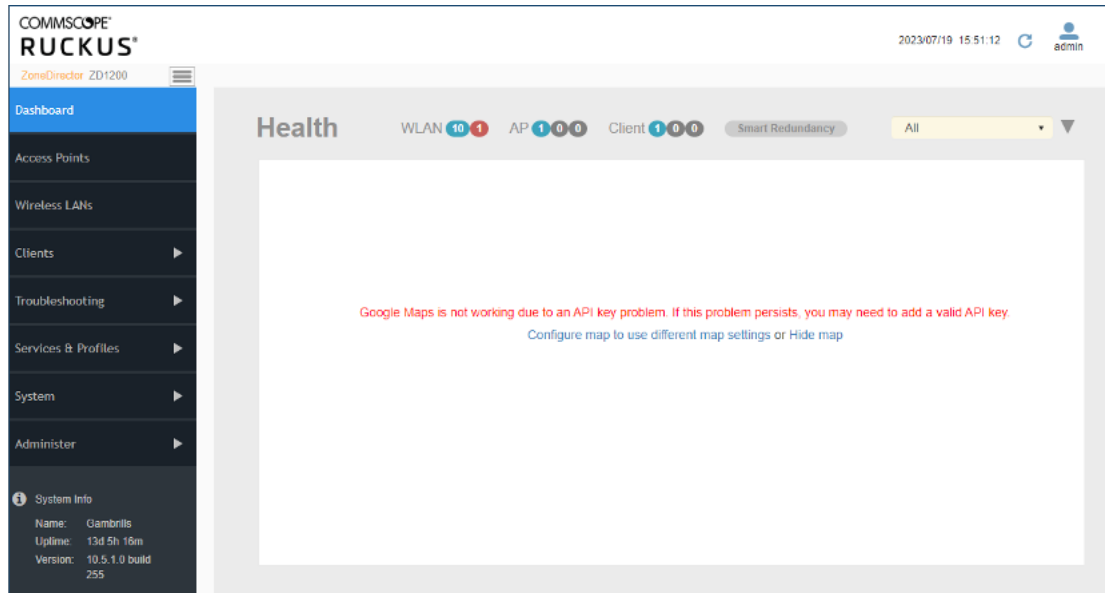
Configuration	
NAI Realm	customer.cloud.global
Friendly Name	customer.cloud.global
Primary RADIUS IP	116.214.120.1
Secondary RADIUS IP	116.214.121.1
RADIUS Authentication Port	1812
RADIUS Accounting Port	1813
NAS Identifier	1f386a9
RADIUS Secret	5dc7b90b296789h08a6ef764b1e3d

- Take note of the details for your respective installation as they will be required at a later step.

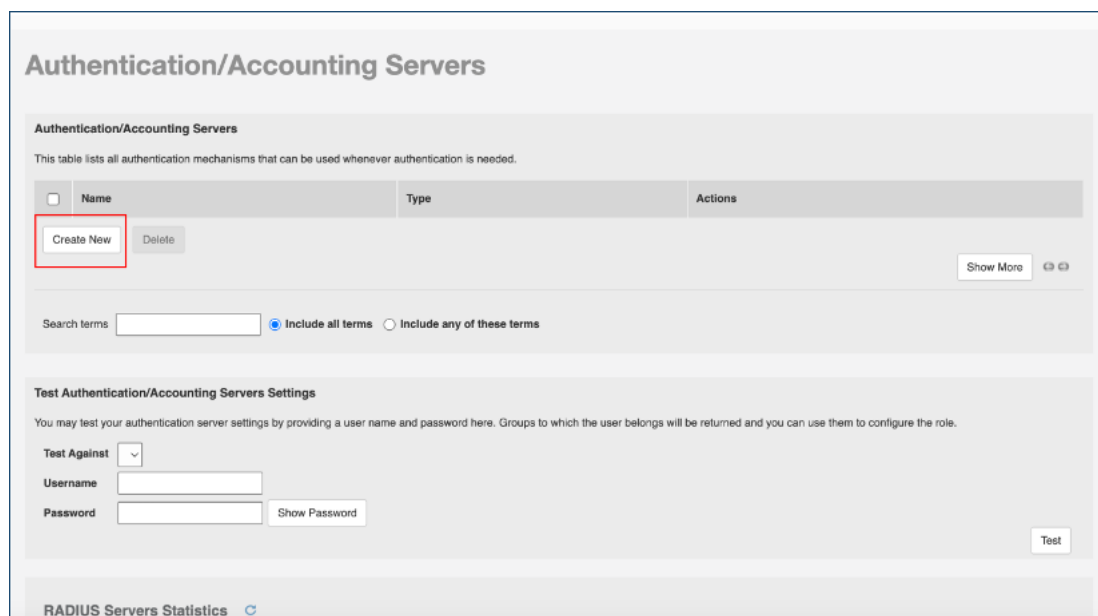
Ruckus ZoneDirector Configuration

RADIUS

- Log into the Ruckus ZoneDirector with your credentials to display the page below:



- On the left-hand navigation menu, click on the **Services & Profiles > AAA Servers** to display the page below:



- Click on **Create New** in the upper left part of the screen to display the pages below:

Authentication/Accounting Servers

Authentication/Accounting Servers

This table lists all authentication mechanisms that can be used whenever authentication is needed.

Name	Type	Actions
Create New Name: New Name Type: ☒ Active Directory ☐ LDAP ☐ RADIUS ☐ RADIUS Accounting ☐ TACACS+ Global Catalog: ☐ Enable Global Catalog support Encryption: ☐ Enable TLS encryption IP Address*: Port*: 389 Windows Domain Name: example: domain.ruckuswireless.com OK Cancel		

Create New

Name	RADIUS NC Demo
Type	☐ Active Directory ☐ LDAP ☒ RADIUS ☐ RADIUS Accounting ☐ TACACS+
Encryption	☐ TLS
Auth Method	☒ PAP ☐ CHAP
Backup RADIUS	☒ Enable Backup RADIUS support
First Server	
IP Address*	116.214.120.1
Port*	1812
Shared Secret*	
Confirm Secret*	
Second Server	
IP Address*	116.214.121.1
Port*	1812
Shared Secret*	
Confirm Secret*	
Retry Policy	
Request Timeout*	3 seconds
Max Number of Retries*	2 times
Max Number of Consecutive Drop Packets*	1
Reconnect Primary*	5 minutes
OK Cancel	

- Complete the fields, as per the description below:
 FieldDescriptionNameEnter a name.TypeSelect RADIUS.EncryptionLeave unchecked.Auth MethodSelect PAP.Backup RADIUSSelect this checkbox.Primary SeverIP AddressesEnter the <Primary RADIUS IP> from Trusted WiFi.PortEnter the < RADIUS Authentication Port> from Trusted WiFi.Shared SecretEnter the < RADIUS Shared Secret> from Trusted WiFi.Confirm Shared SecretRe-enter the < RADIUS Shared Secret> from Trusted WiFi.Secondary ServerIP AddressesEnter the <Secondary RADIUS IP> from Trusted WiFi.PortEnter the < RADIUS Authentication Port> from Trusted WiFi.Shared SecretEnter the < RADIUS Shared Secret> from Trusted WiFi.Confirm Shared SecretRe-enter the < RADIUS Shared Secret> from Trusted WiFi.

- Click on **OK** at the bottom of the screen to save the RADIUS configuration.
- Once back on the AAA screen, click **Create New** again to configure the RADIUS accounting settings and open the page below:

Authentication/Accounting Servers

Authentication/Accounting Servers
This table lists all authentication mechanisms that can be used whenever authentication is needed.

Name	Type	Actions
Create New Name: New Name Type: ☒ Active Directory ☐ LDAP ☐ RADIUS ☐ RADIUS Accounting ☐ TACACS+ Global Catalog: ☐ Enable Global Catalog support Encryption: ☐ Enable TLS encryption IP Address*: Port*: 389 Windows Domain Name: (example: domain.nutanixwireless.com) OK Cancel		

Create New

Name	RADIUS Accounting NC Demc
Type	☐ Active Directory ☐ LDAP ☐ RADIUS ☒ RADIUS Accounting ☐ TACACS+
Encryption	☐ TLS
Backup RADIUS	☒ Enable Backup RADIUS Accounting support
First Server	
IP Address*	116.214.120.1
Port*	1813
Shared Secret*	
Confirm Secret*	
Second Server	
IP Address*	116.214.121.1
Port*	1813
Shared Secret*	
Confirm Secret*	
Retry Policy	
Request Timeout*	3 seconds
Max Number of Retries*	2 times
Max Number of Consecutive Drop Packets*	1
Reconnect Primary*	5 minutes

OK **Cancel**

- Complete the fields, as per the description below:
 FieldDescriptionNameEnter a name.
 TypeSelect RADIUS Accounting.
 EncryptionLeave unchecked.
 Auth MethodSelect PAP.
 Backup RADIUSSelect this checkbox.
 Primary SeverIP AddressesEnter the <Primary

RADIUS IP> from Trusted WiFi.PortEnter the < RADIUS Authentication Port> from Trusted WiFi.Shared SecretEnter the < RADIUS Shared Secret> from Trusted WiFi.Confirm Shared SecretRe-enter the < RADIUS Shared Secret> from Trusted WiFi.Secondary ServerIP AddressesEnter the <Secondary RADIUS IP> from Trusted WiFi.PortEnter the < RADIUS Authentication Port> from Trusted WiFi.Shared SecretEnter the < RADIUS Shared Secret> from Trusted WiFi.Confirm Shared SecretRe-enter the < RADIUS Shared Secret> from Trusted WiFi.

- Click on **OK** at the bottom of the screen to save the RADIUS Accounting configuration.

Warning

If the ZoneDirector controller is behind a Nomadix Gateway, the address, port, and shared secret settings will be associated with the gateway.

Passpoint (Hotspot 2.0)

- Click on the **Services & Profiles > Hotspot 2.0 Services** on the left of the screen to display the page below:

Hotspot 2.0 Services

Operator Profiles

Name	Description	Actions
Nomadix		Edit Clone

[Create New](#) [Delete](#)

Search terms ☒ Include all terms ☐ Include any of these terms

Service Provider Profiles

Name	Description	Actions
Nomadix2		Edit Clone

[Create New](#) [Delete](#)

Search terms ☒ Include all terms ☐ Include any of these terms

- Under **Service Provider Profiles**, click on **Create New** to display the page below:

Create New

Name*

Description

NAI Realm List

Name	Encoding	EAP Method	Action
Create New	Delete		

Domain Name List

Domain Name	Action
Create New	Delete

[Advanced Options](#)

OK **Cancel**

- Enter a name and optional description for this Service Provider profile.
- Under **NAI Realm List**, click **Create New** to display the page below:

Create New

Name*	Service Provider Name																																																
Description																																																	
NAI Realm List	<table border="1"> <thead> <tr> <th>☐</th> <th>Name</th> <th>Encoding</th> <th>EAP Method</th> <th>Action</th> </tr> </thead> <tbody> <tr> <td colspan="5">Create New</td> </tr> <tr> <td></td> <td>customer.cloud.gl</td> <td>RFC-4282</td> <td>#1</td> <td></td> </tr> <tr> <td></td> <td colspan="3">EAP Method : EAP-TLS</td> <td></td> </tr> <tr> <td>☐</td> <td>Auth Info</td> <td>Auth Type</td> <td colspan="2">Action</td> </tr> <tr> <td colspan="5">Create New Delete</td> </tr> <tr> <td colspan="5"></td> </tr> <tr> <td colspan="5">Done Cancel</td> </tr> <tr> <td colspan="5">Create New Delete</td> </tr> </tbody> </table>				☐	Name	Encoding	EAP Method	Action	Create New						customer.cloud.gl	RFC-4282	#1			EAP Method : EAP-TLS				☐	Auth Info	Auth Type	Action		Create New Delete										Done Cancel					Create New Delete				
☐	Name	Encoding	EAP Method	Action																																													
Create New																																																	
	customer.cloud.gl	RFC-4282	#1																																														
	EAP Method : EAP-TLS																																																
☐	Auth Info	Auth Type	Action																																														
Create New Delete																																																	
Done Cancel																																																	
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Domain Name List	<table border="1"> <thead> <tr> <th>☐</th> <th>Domain Name</th> <th>Action</th> </tr> </thead> <tbody> <tr> <td colspan="3">Create New Delete</td> </tr> </tbody> </table>				☐	Domain Name	Action	Create New Delete																																									
☐	Domain Name	Action																																															
Create New Delete																																																	

[Advanced Options](#)

OK Cancel

- Complete the fields, as per the description below:
 FieldDescriptionNameEnter the NAI realm name.
 EncodingSelect the appropriate encoding type from the drop-down list.
 EAP MethodSelect EAP-TTLS from the drop down list.
- Click **Done** to display the page below:

Create New

Name*	Service Provider Name																		
Description																			
NAI Realm List	<table border="1"> <thead> <tr> <th>☐</th> <th>Name</th> <th>Encoding</th> <th>EAP Method</th> <th>Action</th> </tr> </thead> <tbody> <tr> <td>☐</td> <td>customer.cloud.global</td> <td>RFC-4282</td> <td>#1:N/A, #2:N/A, #3:N/A, #4:N/A</td> <td>Edit Clone</td> </tr> <tr> <td colspan="5">Create New Delete</td> </tr> </tbody> </table>				☐	Name	Encoding	EAP Method	Action	☐	customer.cloud.global	RFC-4282	#1:N/A, #2:N/A, #3:N/A, #4:N/A	Edit Clone	Create New Delete				
☐	Name	Encoding	EAP Method	Action															
☐	customer.cloud.global	RFC-4282	#1:N/A, #2:N/A, #3:N/A, #4:N/A	Edit Clone															
Create New Delete																			
Domain Name List	<table border="1"> <thead> <tr> <th>☐</th> <th>Domain Name</th> <th>Action</th> </tr> </thead> <tbody> <tr> <td colspan="3">Create New Delete</td> </tr> </tbody> </table>				☐	Domain Name	Action	Create New Delete											
☐	Domain Name	Action																	
Create New Delete																			

[Advanced Options](#)

OK Cancel

- Under the **Domain Name List**, click **Create New** to display the page below and enter the **Domain Name**:

Create New

Name*	Service Provider Name				
Description					
NAI Realm List	☐	Name	Encoding	EAP Method	Action
	☐	customer.cloud.global	RFC-4282	#1:N/A, #2:N/A, #3:N/A, #4:N/A	Edit Clone
	Create New Delete				
Domain Name List	☐	Domain Name	Action		
	☐	customer.cloud.gl	Save Cancel		
	Create New Delete				

[Advanced Options](#)

- Click **Save** to save the Domain Name.
- Repeat this step for any additional **Domain Names** that need to be added:

Create New

Name*	Service Provider Name				
Description					
NAI Realm List	☐	Name	Encoding	EAP Method	Action
	☐	customer.cloud.global	RFC-4282	#1:N/A, #2:N/A, #3:N/A, #4:N/A	Edit Clone
	Create New Delete				
Domain Name List	☐	Domain Name	Action		
	☐	customer.cloud.global	Edit Clone		
	Create New Delete				

[Advanced Options](#)

- Click on **OK** to save the Service Provider profile.
- Under the **Operator Profiles**, click **Create New** to display the page below:

Create New

Name*	New Name						
Description							
Venue Information	Group Unspecified	Type Unspecified					
ASRA Option	☐ Additional step required for access						
Internet Option	☐ Specified with connectivity to internet						
Access Network Type	Private						
IP Address Type	IPv4 Address: Not available IPv6 Address: Not available						
Operator Friendly Name	☐ Language Name Action Create New Delete						
Service Provider Profiles							
	<table> <thead> <tr> <th>Name</th> <th>Description</th> </tr> </thead> <tbody> <tr> <td>☐ Service Provider Name</td> <td></td> </tr> </tbody> </table> 1-1 (1) Search terms Ser ☒ Include all terms ☐ Include any of these terms			Name	Description	☐ Service Provider Name	
Name	Description						
☐ Service Provider Name							
Advanced Options							

- Complete the fields, as per the description below:
FieldDescriptionNameEnter the name of the Operator Profile. (provided by the installer).
ASRA OptionsLeave unchecked.
Internet OptionEnable this checkbox.
Access Network TypeSelect Free Public from the drop down list.
IP Address TypeSelect the appropriate IPv4 & IPv6 Address from the drop down list.
- Under the **Operator Friendly Name**, click on **Create New**, select **English** and enter a **Name** followed by **Save**.
- Select the **Service Provider** by clicking the checkbox corresponding to the desired name:

Create New

Name*	Operator Profile Name		
Description			
Venue Information	Group	Unspecified	Type
ASRA Option	☐ Additional step required for access		
Internet Option	☐ Specified with connectivity to internet		
Access Network Type	Free Public		
IP Address Type	IPv4 Address:	Single NATed private address	
	IPv6 Address:	Not available	
Operator Friendly Name	☒ Language	Name	Action
	☐ English	customer.cloud.global	Edit Clone
	Create New	Delete	

Service Provider Profiles

Name	Description
☒ Service Provider Name	

1-1 (1)

Search terms ☒ Include all terms ☐ Include any of these terms

[Advanced Options](#)

[OK](#) [Cancel](#)

- Click **OK** to save the configuration.

Wi-Fi Network

- To create the Hotspot 2.0 WLAN SSID, click on **Wireless LANs** on the left of the screen to display the page below:

Wireless LANs

View Mode: [List](#) [Group](#)

[+](#) [-](#) [↺](#) [↻](#) [+](#) Create [✎](#) Edit [📄](#) Clone [🗑️](#) Delete

Search

Name	ESSID	Description	Type	Auth
No data available.				

0-0 of 0 shown [1](#)

- Click on **Create** to display the page below:

Create WLAN

General

Name:

ESSID:

Description:

WLAN Usages

Type: ☐ Standard Usage (For most regular wireless network usages.)
☐ Guest Access (Guest access policies and access control will be applied.)
☐ Hotspot Service (WISPr)
☒ Hotspot 2.0
☐ Autonomous

Hotspot 2.0 Operator: +

OK Cancel

- Complete the fields, as per the description below:
 FieldDescriptionGeneral OptionsNameEnter the customer SSID name.ESSIDEnter the customer ESSID name.DescriptionEnter an optional description.WLAN UsagesTypeSelect the Hotspot 2.0 radio button.Hotspot 2.0 OperatorSelect the operator that you have previously configured from the drop down list.Authentication ServerSelect the Authentication server that you have previously configured from the drop down.
- Scroll down to the bottom of the screen to display the page below:

Create WLAN

Authentication

Method: ☒ 802.1x EAP

Fast BSS Transition: ☒ Enable 802.11r FT Roaming (Recommended to enable 802.11k Neighbor-list Report for assistant.)

Authentication Server: +

Zero-IT Activation™: ☐ Enable Zero-IT Activation
 (WLAN users are provided with wireless configuration installer after they log in.)

Encryption

Method: ☒ WPA2 ☐ WPA3

Algorithm: ☒ AES

802.11w MFP: ☒ Disabled ☐ Optional ☐ Required

Advanced Options

OK Cancel

- Complete the fields, as per the description below:
 FieldDescriptionAuthentication OptionsMethodSelect the 802.1x EAP radio button.Fast BSS TransitionsEnable the 802.1rFT Roaming check box.Authentication ServerSelect the Authentication Server you have already configured from the drop-down list.Encryption OptionsLeave the default enabled.
- Scroll down to the bottom of the screen to display the **Advanced Options**:

Create WLAN

Advanced Options

Wireless Client Isolation: ☒ Isolate wireless client traffic from other clients on the same AP.
☐ Isolate wireless client traffic from all hosts on the same VLAN/subnet.

No WhiteList

(Requires whitelist for gateway and other allowed hosts.)

WLAN Priority: ☒ High ☐ Low

Accounting Server: NMX_Acct Send Interim-Update every 10 minutes

Access Control: L2/MAC No ACLs

L3/4/IP address No ACLs

Device Policy None Precedence Policy Default

☐ Enable Role based Access Control Policy
 Requires enable Role based Access Control Policy in role under "Services & Profiles" -> "Roles"

Application Recognition & Control: ☐ Enable Application Recognition & Control

OK Cancel

- Complete the fields, as per the description below:FieldDescriptionWireless Client IsolationEnable the Isolate wireless client traffic from other clients on the same AP. checkbox.WhitelistAdd the desired whitelist by clicking the Create New button.Accounting ServerSelect the Accounting Server you have configured from the drop down list.
- Scroll down to the bottom of the screen to display the following options:

Create WLAN

Per STA rate limiting will not work if SSID rate limiting is enabled.

Multicast Filter: ☐ Drop multicast packets from associated clients

Access VLAN: VLAN ID 1006 ☐ Enable Dynamic VLAN

Hide SSID: ☐ Hide SSID in Beacon Broadcasting (Closed System)

Tunnel Mode: ☐ Tunnel WLAN traffic to ZoneDirector
 (Recommended for VoIP clients and PDA devices.)

Proxy ARP: ☐ Enable Proxy ARP

Background Scanning: ☐ Do not perform background scanning for this WLAN service.
 (Any radio that supports this WLAN will not perform background scanning)

Load Balancing: ☒ Do not perform client load balancing for this WLAN service.
 (Applies to this WLAN only. Load balancing may be active on other WLANs)

Band Balancing: ☐ Do not perform Band Balancing on this WLAN service.
 (Applies to this WLAN only. Band Balancing might be enabled on other WLANs)

Max Clients: 100 clients per AP radio to associate with this WLAN

802.11d: ☐ Support for 802.11d (only applies to radios configured to operate in 2.4 GHz band)

OK Cancel

- Set the **Access VLAN** to **1006** and click **OK** to save the configuration.

NAS ID

- To configure the NAS ID, log into the ZoneDirector controller via the CLI and enter the following commands:

Commands:Please login: admin

]Password:

Welcome to the Ruckus Wireless ZoneDirector 1200 Command Line Interface

ruckus> en

ruckus# conf

```
You have all rights in this mode.  
ruckus(config)# wlan [customer SSID]  
The WLAN service '[customer SSID]' has been loaded. To save the WLAN service, type 'end' or  
'exit'.  
ruckus(config-wlan)# nasid-type user-define <WORD> (Where WORD is the NAD-ID)  
ruckus(config-wlan)# called-station-id-type wlan-bssid  
ruckus(config-wlan)# exit
```

Hardware Configuration Guide for ExtremeCloud IQ

Download this guide 

INTRODUCTION

Scope and Purpose

Thank you for purchasing Trusted WiFi solution. This document is a hardware configuration guide describing how to setup an ExtremeCloud IQ management platform for a Trusted WiFi Passpoint service.

- For more information on how to setup an end-to-end Trusted WiFi Passpoint service, please refer to the Trusted WiFi Passpoint Administration Guide.
- For complete information on how to setup an ExtremeCloud IQ management platform, please refer to the vendor's original documentation.

Documentation Conventions

{{snippet.Documentation Conventions}}

Passpoint Overview

Passpoint – also known as Hotspot 2.0 – is an industry-wide next generation approach to public internet access driven by the Wi-Fi Alliance that brings the following benefits:

- Frictionless onboarding and roaming, thanks to a one-time registration followed by automatic access to interconnected hotspots
- More secure and private Wi-Fi connections, compared with general visitor networks

Passpoint is based on the IEEE 802.11u standard, which is a set of protocols enabling cellular-like roaming. Following the initial enrollment, frequent users such as visitors, guests or employees bypass repeated logins, forms and passwords, as their mobile devices automatically join the Wi-Fi subscriber service when they return to a venue or roam between inter-linked Passpoint enabled hotspots and providers, while being better protected against potential cyber threats.

If a device supports 802.11u and is enrolled to a service, it automatically communicates with the Wi-Fi infrastructure via the access points to discover the network SSID and connects securely to it by presenting its access credentials. Upon successful authentication, the device is provisioned with Passpoint standards-based management objects - known as Per-Provider Subscription Management Objects (PPS-MO).

GlobalReach - an ASSA ABLOY company - has been involved with Passpoint since its inception and even contributed to the creation and initial pilot testing of the standard. As a result, it is one of the few trusted worldwide experts on this topic, with a proven platform backed by real-world operational experiences at scale.

The best user experience is to offer Passpoint through a customer/brand mobile app integration, as it further simplifies the onboarding process, while incentivizing app downloads and customer loyalty, leading to further engagement and monetization opportunities. To this effect, Trusted WiFi offers a Software Development Kit (SDK) for easy app integration.

Seamless Onboarding and Roaming

Best user experience via brand mobile loyalty app integration:

- One-time setup in 3 "taps" via a mobile app
- Followed by automatic connection for returning visitors and roaming within any of the participating venues
- No manual SSID selection, no login/password, no reset after a given period
- No impact from MAC address randomization
- Incentive to download the app, leading to visitor engagement opportunities
- Non-compatible devices can still use the legacy onboarding process

End-to-End Service Components

Implementing an end-to-end Trusted WiFi Passpoint service requires a combination of the following software and hardware components:

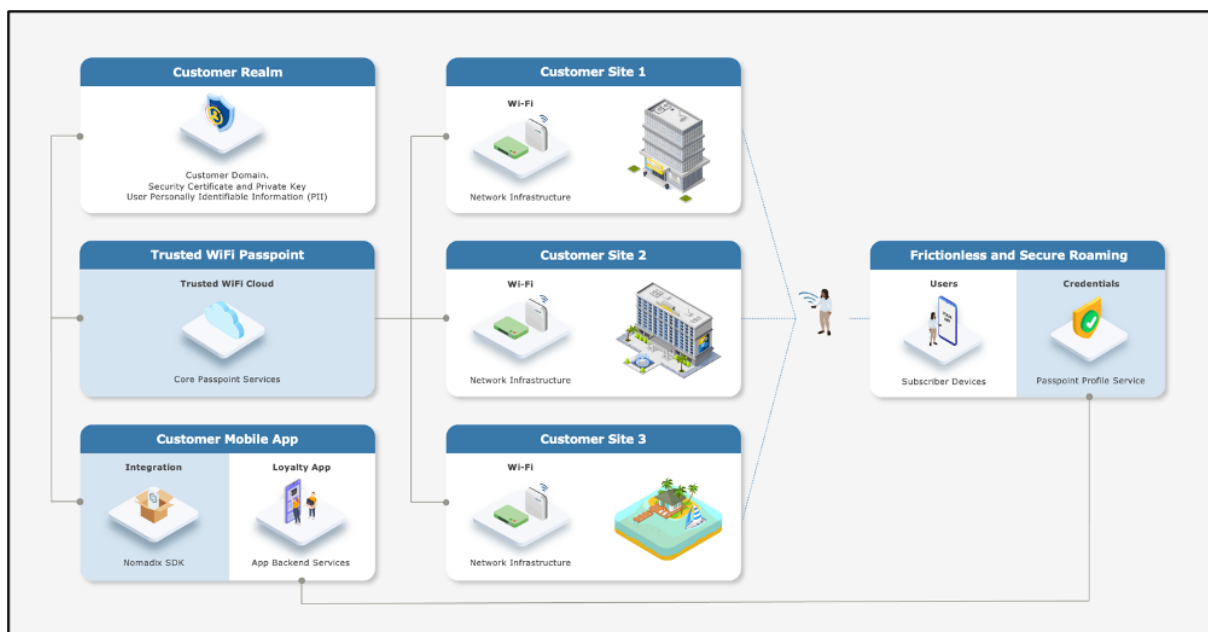
- **Trusted WiFi Passpoint:**
 - The core services are offered and managed centrally via the Trusted WiFi(hosted and operated by GlobalReach),
 - Each site requires acting as a local an edge connector. At the time of writing, the supported models include the EG 1000, EG 3000/L and EG 6000.Following an initial setup performed by the GlobalReach Operations team, Managed Service Providers (MSPs) can then add sites to customer realms and monitor the solution.
- **Customer Realm:** the service requires connectivity to a realm including domain, security certificate and private key, as well as the database holding the users' Personally Identifiable

Information (PII). It is the responsibility of the Identity Provider – typically the customer/brand – to make this part available.

- **Mobile App:** the best user experience is to offer Passpoint through a customer/brand mobile app integration. Trusted WiFi offers a **Software Development Kit (SDK)** for easy implementation. It is the responsibility of the app owner – typically the customer/brand – to perform the integration.
- **Local Networks:** the local networks must be compliant with Passpoint (Hotspot 2.0). – Most recent Wi-Fi access points and controllers from major vendors support Passpoint today, however older models or products from more exotic manufacturers might not do. The configuration of the local infrastructure is typically handled by the Managed Service Providers (MSPs).
- **User Devices:** the subscribers' mobile devices (belonging to visitors, guests, employees, etc.) must be compliant with Passpoint (Hotspot 2.0) – All recent iOS and Android-based smartphones or tablets support Passpoint today, however older models or products running less popular operating systems might not do. Laptops compatibility is also more erratic. It is therefore essential to maintain a traditional onboarding service in parallel with Passpoint to handle non-compatible devices.

High-Level Topology

The diagram below illustrates the high-level topology for the end-to-end Trusted WiFi Passpoint service:



Glossary

The following is a glossary of the most common terms used regarding this solution.

Term	Abbreviation	Description
Deployment	-	Enabling of a Trusted WiFi product module for a property via the Trusted WiFi interface.
Module	-	Product or service purchased from Trusted WiFi that is managed through its own sub-section via the Trusted WiFi interface.
License	-	Legal agreement that grants users the right to use specific software, outlining terms and conditions for its usage, distribution, and potential modifications, while protecting the intellectual property of the software developer. GlobalReach licenses comprise of two different types: one-off licenses - typically to initially enable a software module. recurring licenses - typically including software updates and technical support, or more in the case of OpEx consumption commercial models based on price per month/quarter/year. Trusted WiFi is sold as a combination of one-off licenses to activate the service and recurring licenses based on a price per Wi-Fi access point per month.
Managed Service Provider	MSP	The third-party company that remotely manages and monitors a client's IT infrastructure and end-user systems, offering services like network and infrastructure management, security, and 24/7 technical support.
Organization	Org	A company account in Trusted WiFi.
Operator	-	An organization type account in Trusted WiFi that is used by MSPs to manage a property's Wi-Fi network.
Customer	-	An organization type account in Trusted WiFi typically used for customers/brands that allows grouping to view all properties belonging to the same company even if managed by several different MSPs.
Linked Organization	Linked Org	A link creating a relationship between an operator and a customer account, allowing a customer to view a property while allowing an operator to manage it.
User	-	An individual accessing a product, service or system. In the context of Trusted WiFi, a user is setup to access products and deployments information at Admin/Editor/Viewer levels, according to their relevant role permissions. In the context of a given product or service, a user is another term referring to a subscriber: the person at the end of the chain interacting with that product or service – usually through a device.
Property	-	Trusted WiFi concept representing an Individual site or location where products are deployed.
Linked Property	-	Site or location shared between operator and customer accounts.

Passpoint Software Development Kit	Passpoint SDK	The service that sits within the customer's mobile app and that is connected to the Trusted WiFi RADIUS infrastructure allowing Passpoint profiles to be created for a given Passpoint realm.
Passpoint Realm	-	The customer specific domain that is used to provision Passpoint profiles that are approved for connection to any associated network.
Passpoint Profile	-	The security certified profile that sits on a subscriber's device. If installed correctly it allows seamless authentication to the secure Passpoint SSID.
Secure Passpoint Service Set Identifier	Secure Passpoint SSID	The Wi-Fi network that is associated to the Passpoint realm configured to allow subscriber devices with a valid Passpoint profile to seamlessly connect to the Passpoint network at a property.
Subscriber	-	An individual person using a service.
Subscriber Device	-	The equipment – typically a smartphone, tablet or computer – the subscriber is using to connect to the service.
Collision-Resistant Unique Identifier	CUID	A unique identifier designed to be collision-resistant – meaning engineered to minimize the likelihood of generating duplicate IDs even in distributed systems – and more efficient in terms of space and database indexing performance due to its sequential nature. In the context of Passpoint, a CUID is delivered to a subscriber device when it requests a Passpoint profile.
Customer Loyalty Mobile Application	Mobile App	The iOS and/or Android digital application used by businesses to engage and reward their customers through loyalty programs. In the context of Passpoint, the best user experience is to offer Passpoint through a customer/brand mobile app integration using the Trusted WiFi SDK.

EXTREMECLOUD IQ CONFIGURATION

Supported Versions

This document is based on ExtremeCloud IQ firmware 25.4.3-40.

Prerequisites

It is assumed the following prerequisites are met before configuring ExtremeCloud IQ for a Trusted WiFi Passpoint service:

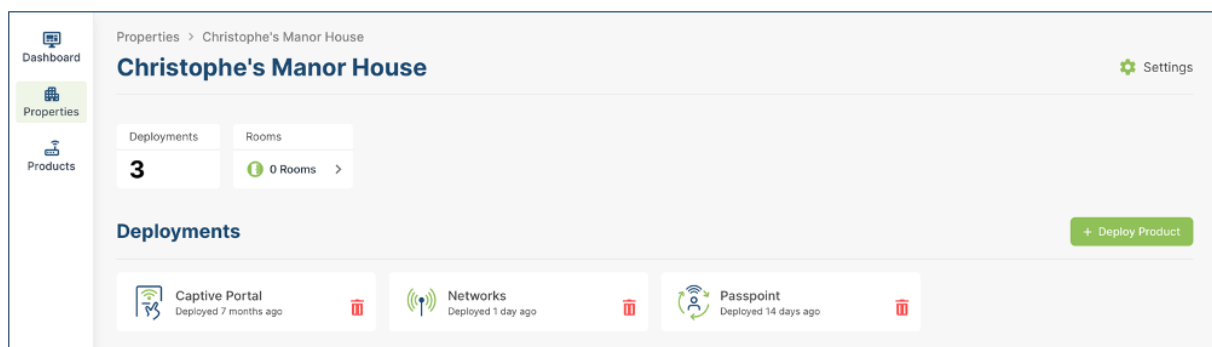
- A supported ExtremeCloud IQ platform, activated and licensed.
- A Trusted WiFi account with operator permissions.
- A core Trusted WiFi Passpoint service configured and tested.
- A property in Trusted WiFi with deployed Passpoint modules.
- A deployed and configured Wi-Fi network.

Warning

All properties must share the same NAI realm, RADIUS IP / port settings and SSID name for the Passpoint service (CustomerPasspoint). Each property requires a separate RADIUS secret and NAS identifier, both generated when a configuration is activated within the Trusted WiFi management platform.

Core Passpoint Settings

- Log into Trusted WiFi, then click on the **Properties** icon in the left menu to view the properties list.
- Select or search for the property you wish to work on. This will open its deployments page:



- Click on the **Passpoint** tile and then click on the **Configuration** option in the left menu to display the Passpoint settings summary as per the example below:

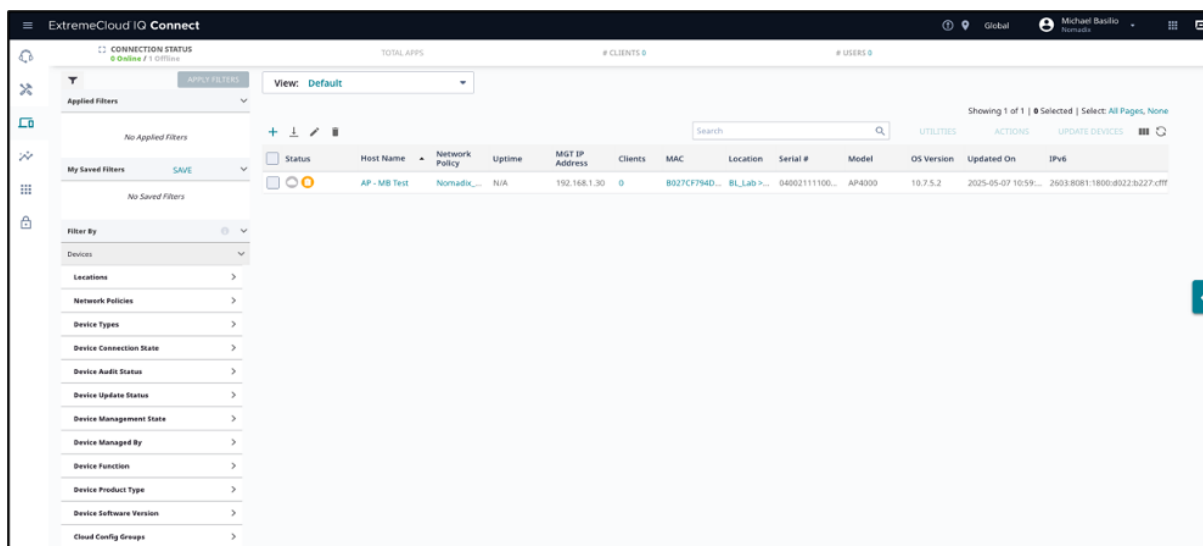
Configuration	
NAI Realm	customer.cloud.global
Friendly Name	customer.cloud.global
Primary RADIUS IP	116.214.120.1
Secondary RADIUS IP	116.214.121.1
RADIUS Authentication Port	1812
RADIUS Accounting Port	1813
NAS Identifier	1ft386a9
RADIUS Secret	5dc7b90b296789h08a6ef764b1e3d

- Take note of the details for your respective installation as they will be required at a later step.

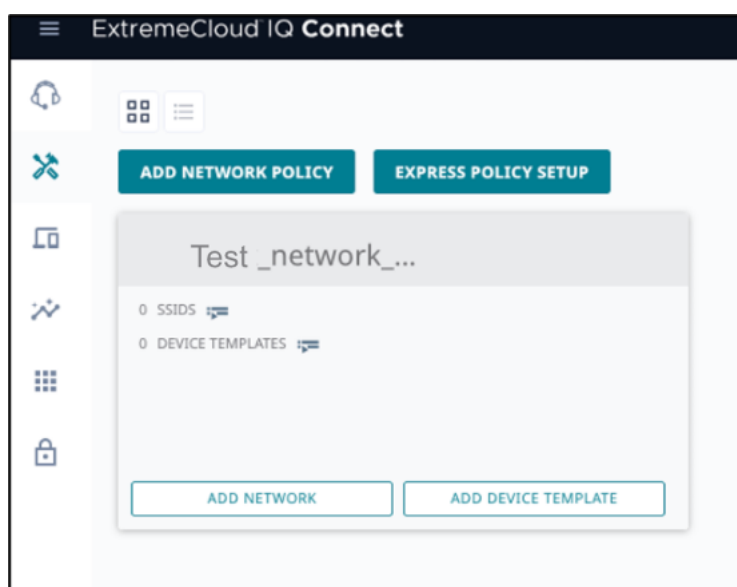
ExtremeCloud IQ Configuration

RADIUS Profile

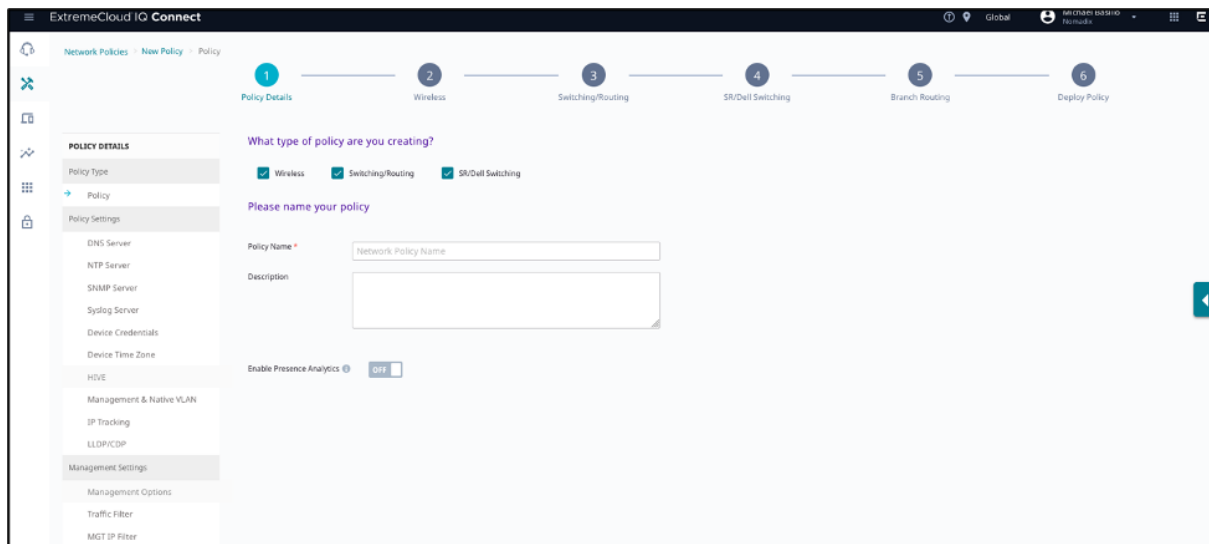
- Log into the ExtremeCloud IQ dashboard with your credentials to display the page below:



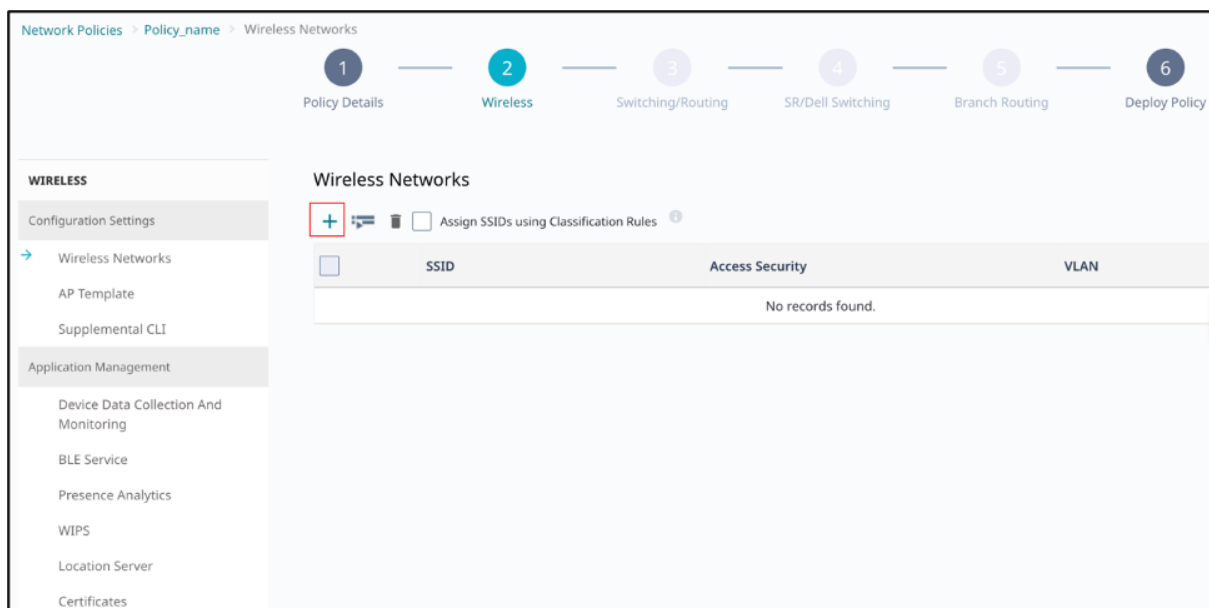
- On the left-hand side menu items, click on **Configure > Network Policies**. The following screen will appear:



- Click the **Add Network Policy** button and the following page will appear:



- On the **Policy Details** tab, fill-in the screen as follows:
- Under **Policy Type**, check **Wireless** and uncheck the others.
- Under **Policy Settings**, provide a name and optional description.
- Click **Next**, then click **Deploy**.
- The **Wireless** tab will appear. Under **Wireless Networks**, click **+** as shown below:



- The following page will appear:

1 Policy Details 2 Wireless 3 Switching/Routing 4 SR/Dell Switching 5 Branch Routing 6 Deploy Policy

Wireless Network

Name (SSID) *

Broadcast Name *

Broadcast SSID Using

☒ 2.4 GHz

☒ 5 GHz

☐ 6 GHz ⓘ

SSID Usage

SSID AUTHENTICATION MAC AUTHENTICATION

Enterprise WPA / WPA2 / WPA3 Personal WPA / WPA2 / WPA3 Private Pre-Shared Key Open Unsecured Enhanced Open Secured

Key Management

Encryption Method

Custom (non-zero) setting in the user group overrides this setting.

☐ Set the maximum number of clients per private PSK Range: 0-15, 0=no limit

This setting is only supported with local PPSK.

☐ Set the MAC binding numbers per private PSK Range: 1-5

- Complete the fields, as per the description below:
 FieldDescriptionWireless NetworkName (SSID)Enter a internal SSID nameBroadcast NameEnter the SSID you want your users to see.SSID UsageSSID AuthenticationSelect Enterprise.Key ManagementSelect WPA2-802.1XEncryption MethodSelect CCMP (AES)Enable Captive Web PortalLeave disabled.

Wireless Network

Name (SSID) *

Broadcast Name *

Broadcast SSID Using

☒ 2.4 GHz

☒ 5 GHz

☐ 6 GHz ⓘ

Protected Management Frames

☐ OFF

802.11w Prevents forgery and retransmission of management frames

SSID Usage

SSID AUTHENTICATION MAC AUTHENTICATION

Enterprise WPA / WPA2 / WPA3 Personal WPA / WPA2 / WPA3 Private Pre-Shared Key Open Unsecured Enhanced Open Secured

Hotspot

Key Management

Encryption Method

Enable Captive Web Portal ☐ OFF

- Once the above is completed, scroll down to the **Authentication Settings** section.
- Under the **Authenticate via RADIUS Server**, click **+** to add a RADIUS server group as shown below:

Authentication Settings

Authentication with ExtremeCloud IQ Authentication Service ☐ OFF

Authenticate via RADIUS Server

Default RADIUS Server Group  

Name	Type	IP/Host Name	Order
No records found.			

Configure RADIUS Servers ✕

RADIUS Server Group Name * RADIUS Server Group Description 

EXTERNAL RADIUS SERVER (0) EXTREME NETWORKS A3 (0) EXTREME NETWORKS RADIUS SERVER (0) EXTREME NETWORKS RADIUS PROXY (0) EXTREME NETWORKS RADSEC PROXY (0)

Name	IP/Host Name
☐ secondary_radius	116.214.121.1
☐ Primary_radius	116.214.120.1

CANCEL SAVE RADIUS

- Enter a **RADIUS Server Group Name** and an optional **RADIUS Server Group Description**.
- Select the **Settings** icon next to the description field box as shown below.

Configure RADIUS Servers

RADIUS Server Group Name * RADIUS Server Group Description 

PrimaryRADIUS

- The **Select RADIUS Settings** dialog box appears:

Configure RADIUS Servers

SelectRadiusSettings

Note: These settings only apply for IQ Engine devices. These settings are ignored for non-IQ Engine devices.

Retry Interval

600

Range: 60 - 100000000 (seconds)

Accounting Interim Update Interval

600

Range: 10 - 100000000 (seconds)

☐ Permit Dynamic Change Of Authorization Messages (RFC 3576)

☐ Inject Operator-Name attribute

☐ Message Authenticator attribute

Not Supported for Extreme Networks RADIUS Proxy

CANCEL

SAVE RADIUS SETTINGS

- Change the **Accounting interim update interval** to 300 (seconds).
- Click **Save RADIUS Settings** on the bottom right.
- Return to the **Configure RADIUS Servers** dialog box.
- Click **+** under External RADIUS Server to **add a RADIUS Server** to the server group.
- The following screen will appear:

Configure RADIUS Servers

RADIUS Server Group Name *

PrimaryRADIUS

RADIUS Server Group Description

EXTERNAL RADIUS SERVER (0)

EXTREME NETWORKS A3 (0)

EXTREME NETWORKS RADIUS SERVER (0)

EXTREME NETWORKS RADIUS PROXY (0)

EXTREME NETWORKS RADSEC PROXY (0)

New External RADIUS Server

Name *

Description

☐ Peer Discovery (RFC 7585) Enabled

Type *

Standard

IP/Host Name *

Server Type *

☒ Authentication

Port: * 1812

☒ Accounting

Port: * 1813

CANCEL

SAVE EXTERNAL RADIUS

- Enter a name, e.g. Primaryradius.

- Click + next to IP/Host Name.

IP/Host Name *

Server Type *

Authentication ☒ Port: 1812

Accounting ☒ Port: 1813

Shared Secret

Show Password ☐

- Select **IP Address**. The **New IP Address or Host Name** dialog box appears:

New IP Address or Host Name

New IP Address or Host Name

Name * Primary

IP Address

IP Address	Type	Classification Rules	Assignment Description
116.214.120.1	Global		

CANCEL SAVE IP OBJECT

- Enter the **Name**, such as **“Primary”**.
- Enter the **<Primary RADIUS IP>** from Trusted WiFi in the **IP Address** field.
- Click **SAVE IP OBJECT** on the bottom right.
- Return to the **New External RADIUS Server** section. You see the name of the object you created in the **IP/Host Name** field as shown below:

Configure RADIUS Servers

New External RADIUS Server

Name *

Description

☐ Peer Discovery (RFC 7585) Enabled

Type *

IP/Host Name *   ⓘ This field is required.

Server Type *

☒ Authentication	Port: * 1812
☒ Accounting	Port: * 1813

Shared Secret

☐ Show Password

- Enter the <RADIUS Authentication Port> from Trusted WiFi.
- Enter the <RADIUS Authentication Port> from Trusted WiFi.
- Enter the <RADIUS Authentication Port> from Trusted WiFi.

Configure RADIUS Servers

New External RADIUS Server

Name *

Primaryradius

Description

☐ Peer Discovery (RFC 7585) Enabled

Type *

Standard

IP/Host Name *

Primary1

+ 

 This field is required

Server Type *

☒ Authentication

Port: * 1812

☒ Accounting

Port: * 1813

Shared Secret

.....

☐ Show Password

- Click **Save External RADIUS** on the bottom right.
- Return to the **Configure RADIUS Servers** page where you see the server you added.
- Click + under External RADIUS Server to **add a RADIUS Server** to the server group.
- Repeat the above process for the **Secondary RADIUS Server IP**.
- Check the box next to the server you added. This indicates you want to add it to the server group.

Configure RADIUS Servers

RADIUS Server Group Name *
RADIUS Server Group Description

PrimaryRADIUS

EXTERNAL RADIUS SERVER (2) EXTREME NETWORKS A3 (0) EXTREME NETWORKS RADIUS SERVER (0) EXTREME NETWORKS RADIUS PROXY (0) EXTREME NETWORKS RADSEC PROXY (0)

Name	IP/Host Name
☒ secondary_radius	116.214.121.1
☒ Primary_radius	116.214.120.1
☐ Secondaryradius	116.214.121.1
☐ Primaryradius	116.214.120.1

CANCEL
SAVE RADIUS

- Return to the **Authenticate via RADIUS Server** section of the **Wireless Networks** page. You see the RADIUS server group and server you created.

Authentication Settings

Authentication with ExtremeCloud IQ Authentication Service OFF

Authenticate via RADIUS Server

Default RADIUS Server Group PrimaryRADIUS

Name	Type	IP/Host Name	Order
secondary_radius	External RADIUS Server	116.214.121.1	↑ ↓
Primary_radius	External RADIUS Server	116.214.120.1	↑ ↓

☐ Apply RADIUS server groups to devices via classification

- Click **Save** on the bottom right to save your network policy configuration.
- Return to the **Wireless Networks** page where you see the SSID you created:

Wireless Networks

☐ Assign SSIDs using Classification Rules

SSID	Access Security	VLAN
PasspointDemo	WPA / WPA2 / WPA3 802.1X	1

- The Network policy configuration is completed.

Passpoint (Hotspot 2.0)

Use the supplemental CLI option to configure Hotspot 2.0. When you enable supplemental CLI, you enter the commands into the GUI. For that reason, we recommend composing the commands in a text file beforehand, so you have them ready when enabling the supplemental CLI.

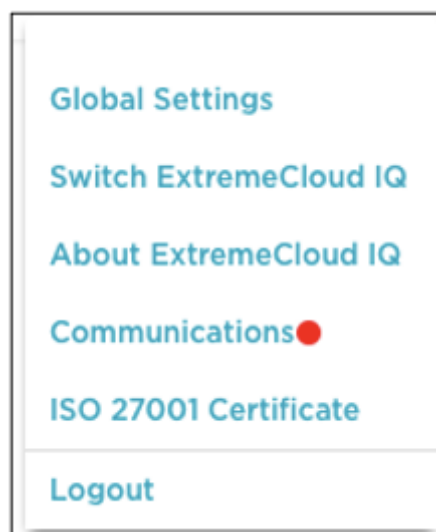
Compose Your CLI

Create a text file with the commands that link your network policy to Hotspot 2.0.

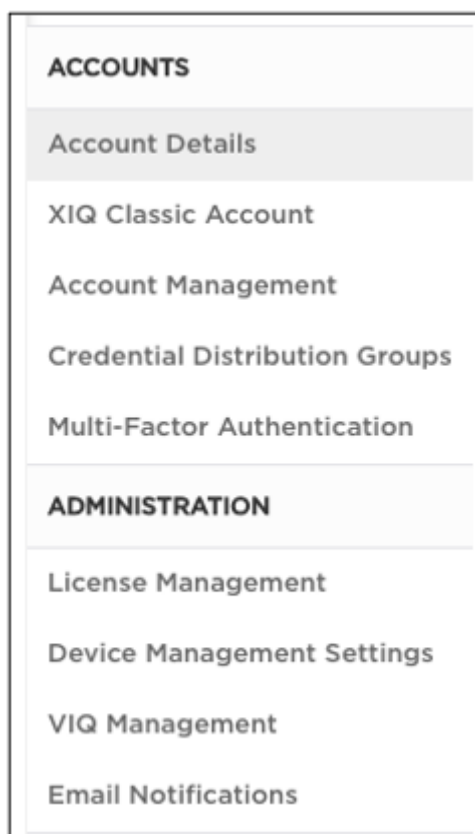
- Create a hotspot profile with a name “**DemoPasspoint-profile**”, anqp domain ID, and network type. anqp-domain-id default is 0, which means that the ANQP information is unique to this access point. A network type of 1 indicates a private network.hotspot profile DemoPasspoint-profile
hotspot profile DemoPasspoint -profile anqp-domain-id 0
hotspot profile DemoPasspoint -profile network-type 1 access-internet
- Configure the operator name “**OperatorPasspoint**” and the language (English).hotspot profile Cloud4Wi-profile operator-name OperatorPasspoint language-code eng
- Configure the hotspot to support IPv4 with a single NAT private IPv4 address by configuring **ip-type ipv4 3 ipv6 0** indicating that IPv6 is not available.hotspot profile DemoPasspoint-profile ip-type ipv4 3 ipv6 0
- Configure the domain name as set in your DemoPasspoint configuration page, for example: “**<customer.cloud.global>**”.hotspot profile DemoPasspoint-profile domain-name customer.cloud.global
- Create the NAI-realm “**customer.cloud.global**” by specifying these parameters:
 - Encoding type—“0” (the default)
 - EAP method—“21” for EAP-TTLS
 - Inner authentication—“4” for MS-chapv2
- Configure **AAA attribute NAS-Identifier** from your Trusted WiFi settings.aaa attribute NAS-Identifier 2045454b
- Configure **wan-metrics**.hotspot profile DemoPasspoint-profile wan-metrics link-up symmetric link-speed 10000
hotspot profile DemoPasspoint-profile wan-metrics link-down symmetric link-speed 10000
- **Apply** the DemoPasspoint hotspot-profile to the DemoPasspoint SSID.ssid DemoPasspoint hotspot-profile DemoPasspoint-profile
- **Save** the configuration.save configuration

Enable the supplemental CLI

- Select Global **Settings** on the top right of the Dashboard under your user icon.



- On the left side of the Dashboard, click **VIQ Management** under **Administration**.



- The **VIQ Management** page appears.

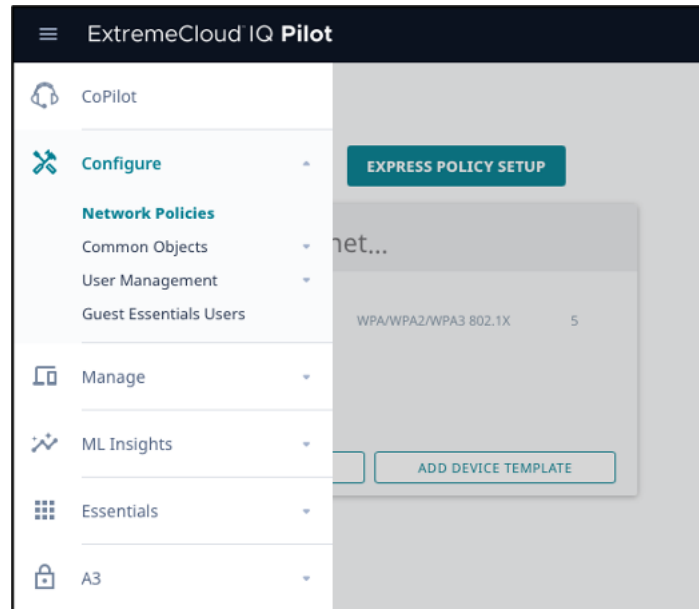
The screenshot shows the 'VIQ Management' page. It has a title 'VIQ Management' at the top. Below the title, there are several rows of settings and actions:

VIQ Name:	VHM-XDZQSZKS
Last backed up on	BACK UP NOW
Current Status	ACTIVE
Delete Data	RESET VIQ
Export VIQ	EXPORT VIQ
Import VIQ	IMPORT VIQ
Supplemental CLI	☒ ON
AP Out-of-the-box Wireless Onboarding	☒ ON

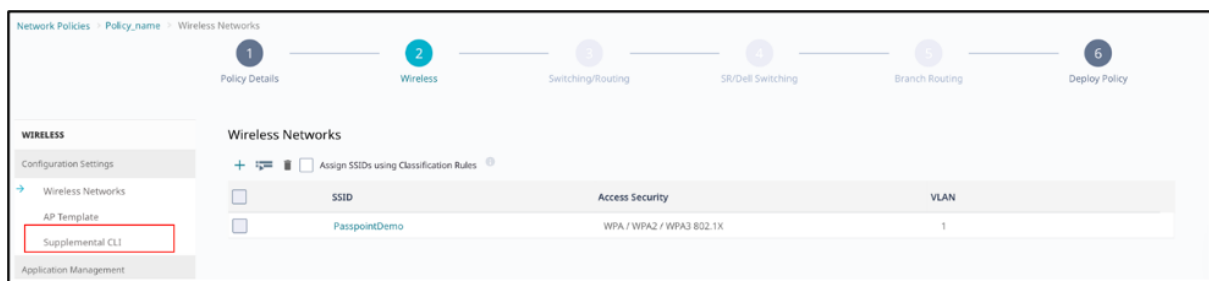
- Verify that Supplemental CLI is **ON**. If not, **enable it**.

Add the Hotspot 2.0 configuration to the network policy

- On the left-hand side menu items, click on **Configure > Network Policies**.
- The **Network Policies** page will appear:



- Click the name of the SSID you created in the above steps.
- Click **Next** to the **Wireless Network** tab.
- Under **Policy Settings** in the menu bar on the left, click **Supplemental CLI**.



- The **Supplemental CLI** page appears:
- Verify that **Supplemental CLI** is **ON**. If not, enable it
- Enter a **Name**, such as “**Hotspot**”.
- Paste the CLI commands in your text file into the **CLI Commands** box.

Supplemental CLI

NOTE: Policy-level Supplemental CLI currently only applies to IQ Engine devices. For Dell and Extreme Networks devices that do not run IQ Engine, please use per-device Supplemental CLI.

Supplemental CLI ☒ ON

Re-use Supplemental CLI Settings (Pick existing settings) ☒ UI overrides Supplemental CLI ☐ Supplemental CLI overrides UI

Name *

Description

CLI Commands *

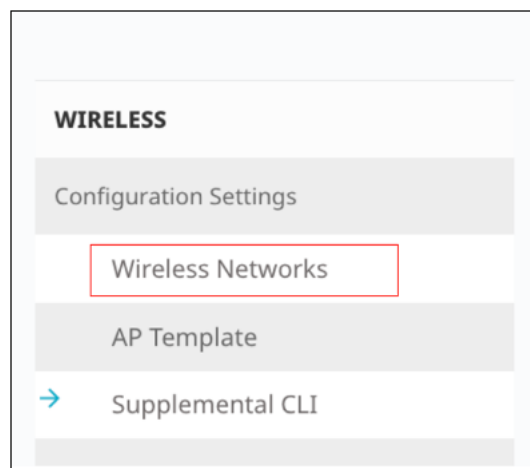
```

hotspot profile DemoPasspoint-profile
hotspot profile DemoPasspoint-profile anqp-domain-id 0
hotspot profile DemoPasspoint-profile network-type 1 access-internet
hotspot profile Cloud4Wi-profile operator-name OperatorPasspoint
language-code eng
hotspot profile DemoPasspoint-profile ip-type ipv4 3 ipv6 0
hotspot profile DemoPasspoint-profile domain-name customer.cloud.global
Hotspot profile DemoPasspoint-profile nai-realm customer.cloud.global
encoding-type 0
hotspot profile DemoPasspoint-profile nai-realm customer.cloud.global
eap-method 21 inner-auth 4
aaa attribute NAS-Identifier 2045454b
  
```

Note: 1. You can enter multiple CLI commands, one command per line, not exceeding a maximum total of 8192 characters. CLI commands should be limited to configuration commands. "Show" command or other commands which are used to display information should be excluded.
2. You can use CLI Commands that contain IP and VLAN objects: \$(ip:ip_object_name) and \$(vlan:vlan_object_name).
3. You must perform a complete configuration update each time commands are entered to device configurations.

CANCEL **SAVE**

- Click **Save** on the bottom right. A message appears on the top left indicating that the supplemental CLI was saved.
- On the left-hand side, click **Wireless Networks** to go back to the tab:



- Click **Next** on the bottom right hand side. The following tab will appear:

Network Policies > Policy_name > Deploy Policy

1 Policy Details 2 **Wireless** 3 Switching/Routing 4 SR/Dell Switching 5 Branch Routing 6 **Deploy Policy**

Apply the network policy to selected devices

Only show devices that are:
☒ Assigned ☐ Eligible ☐ Filtered

Status	Device Name	Device Model	IP Address	MAC Address	Serial # / Service Tag	Last Updated On
No records found.						

- Click **Eligible** to display your access points:

POLICY DETAILS WIRELESS NETWORKS DEVICE TEMPLATES ROUTER SETTINGS ADDITIONAL SETTINGS **DEPLOY POLICY**

Apply the network policy to selected devices

Only show devices that are:

☐	STATUS	DEVICE NAME	DEVICE MODEL	IP ADDRESS	MAC ADDRESS	SERIAL # / SERVICE TAG	LAST UPDATED ON
☐		AccessPoint	AP230	10.32.128.171	C8675E8591C0	02301806050427	2020-10-05 17:52:45

- Select your access points by checking the box next to them in the Status column.
- Click **Upload** on the bottom right. The **Device Update** dialog box appears:

Device Update [X]

1 device will be updated

☒ **Update Network Policy and Configuration**

- ☒ **Delta Configuration Update**
Update device with changed configuration.
- ☐ **Complete Configuration Update**
Update device with all configurations. Used to reset device to ExtremeCloud IQ configuration settings. (Supported only on devices running HOS or IQE Firmware)

☐ **Upgrade IQ Engine and Extreme Network Switch Images**

Activation Time for Extreme Networks Devices Running Images

☐ **Activate at next reboot (requires rebooting manually)**

[SAVE AS DEFAULTS] [CLOSE] **[PERFORM UPDATE]**

- Under **Update Network Policy and Configuration**, select **Complete Configuration Update**. (Delta Configuration Update is the default; you want a complete update.)
- Click **Perform Update** on the bottom right of the dialog box to save your configuration.
- The access points are rebooted (this can take a few minutes). You see a message on the upper left indicating that the devices are successfully deployed.
- The configuration is now completed, and you can now proceed with the testing.

Cisco Meraki

Download this guide 

Introduction

Scope and Purpose

Thank you for purchasing the Trusted WiFi solution. This document is a hardware configuration guide describing how to setup Cisco Meraki for a Trusted WiFi Passpoint service.

- For more information on how to setup an end-to-end Trusted WiFi Passpoint service, please refer to the Trusted WiFi Passpoint Administration Guide.
- For complete information on how to setup Cisco Meraki, please refer to the vendor's original documentation.

Documentation Conventions

{{snippet.Documentation Conventions}}

Passpoint Overview

Passpoint – also known as Hotspot 2.0 – is an industry-wide next generation approach to public internet access driven by the Wi-Fi Alliance that brings the following benefits:

- Frictionless onboarding and roaming, thanks to a one-time registration followed by automatic access to interconnected hotspots
- More secure and private Wi-Fi connections, compared with general visitor networks

Passpoint is based on the IEEE 802.11u standard, which is a set of protocols enabling cellular-like roaming. Following the initial enrollment, frequent users such as visitors, guests or employees bypass repeated logins, forms and passwords, as their mobile devices automatically join the Wi-Fi subscriber service when they return to a venue or roam between inter-linked Passpoint enabled hotspots and providers, while being better protected against potential cyber threats.

If a device supports 802.11u and is enrolled to a service, it automatically communicates with the Wi-Fi infrastructure via the access points to discover the network SSID and connects securely to it by presenting its access credentials. Upon successful authentication, the device is provisioned with Passpoint standards-based management objects - known as Per-Provider Subscription Management Objects (PPS-MO).

GlobalReach - an ASSA ABLOY company - has been involved with Passpoint since its inception and even contributed to the creation and initial pilot testing of the standard. As a result, it is one of the few trusted worldwide experts on this topic, with a proven platform backed by real-world operational experiences at scale.

The best user experience is to offer Passpoint through a customer/brand mobile app integration, as it further simplifies the onboarding process, while incentivizing app downloads and customer loyalty, leading to further engagement and monetization opportunities. To this effect, Trusted WiFi offers a Software Development Kit (SDK) for easy app integration.

Seamless Onboarding and Roaming

Best user experience via brand mobile loyalty app integration:

- One-time setup in 3 "taps" via a mobile app
- Followed by automatic connection for returning visitors and roaming within any of the participating venues
- No manual SSID selection, no login/password, no reset after a given period
- No impact from MAC address randomization
- Incentive to download the app, leading to visitor engagement opportunities
- Non-compatible devices can still use the legacy onboarding process

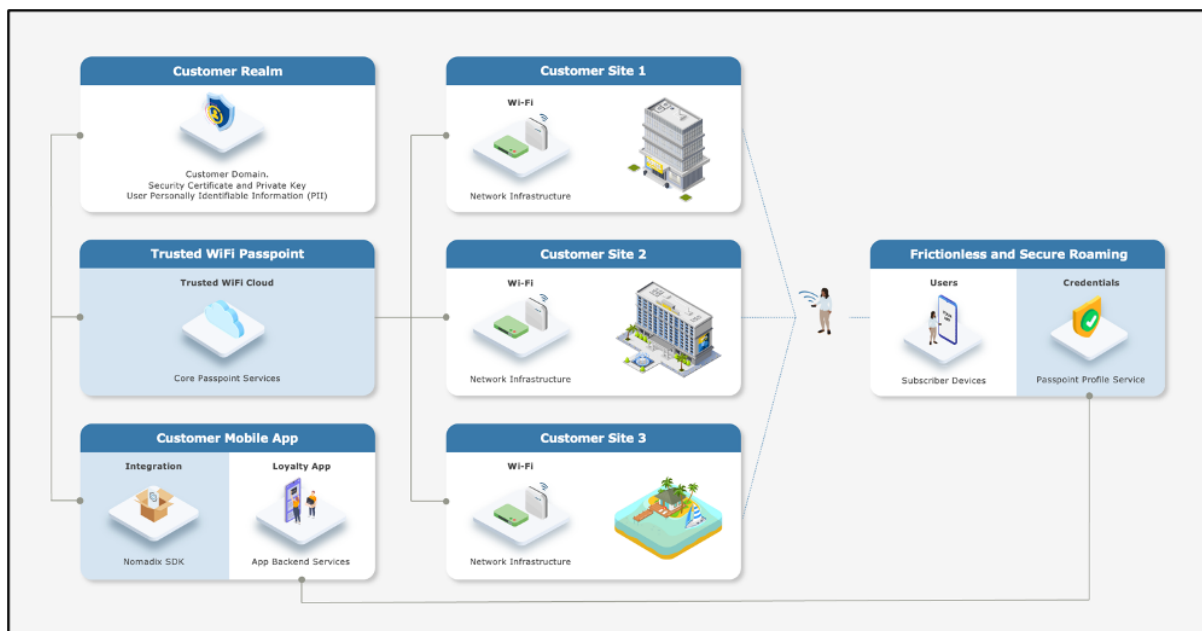
End-to-End Service Components

Implementing an end-to-end Trusted WiFi Passpoint service requires a combination of the following software and hardware components:

- **Trusted WiFi Passpoint:** the core services are offered and managed centrally via the Trusted WiFi Passpoint Module (hosted and operated by GlobalReach). Following an initial setup performed by the GlobalReach Operations team, Managed Service Providers (MSPs) can then add sites to customer realms and monitor the solution.
- **Customer Realm:** the service requires connectivity to a realm including domain, security certificate and private key, as well as the database holding the users' Personally Identifiable Information (PII). It is the responsibility of the Identity Provider – typically the customer/brand – to make this part available.
- **Mobile App:** the best user experience is to offer Passpoint through a customer/brand mobile app integration. Trusted WiFi offers a **Software Development Kit (SDK)** for easy implementation. It is the responsibility of the app owner – typically the customer/brand – to perform the integration.
- **Local Networks:** the local networks must be compliant with Passpoint (Hotspot 2.0). – Most recent Wi-Fi access points and controllers from major vendors support Passpoint today, however older models or products from more exotic manufacturers might not do. The configuration of the local infrastructure is typically handled by the Managed Service Providers (MSPs).
- **User Devices:** the subscribers' mobile devices (belonging to visitors, guests, employees, etc.) must be compliant with Passpoint (Hotspot 2.0) – All recent iOS and Android-based smartphones or tablets support Passpoint today, however older models or products running less popular operating systems might not do. Laptops compatibility is also more erratic. It is therefore essential to maintain a traditional onboarding service in parallel with Passpoint to handle non-compatible devices.

High-Level Topology

The diagram below illustrates the high-level topology for the end-to-end Trusted WiFi Passpoint service:



Glossary

The following is a glossary of the most common terms used regarding this solution.

Term	Abbreviation	Description
Deployment	-	Enabling of a Trusted WiFi product module for a property via the Trusted WiFi interface.
Module	-	Product or service purchased from Trusted WiFi that is managed through its own sub-section via the Trusted WiFi interface.
License	-	Legal agreement that grants users the right to use specific software, outlining terms and conditions for its usage, distribution, and potential modifications, while protecting the intellectual property of the software developer. GlobalReach licenses comprise of two different types: one-off licenses - typically to initially enable a software module. recurring licenses - typically including software updates and technical support, or more in the case of OpEx consumption commercial models based on price per month/quarter/year. Trusted WiFi is sold as a combination of one-off licenses to activate the service and recurring licenses based on a price per Wi-Fi access point per month.
Managed Service Provider	MSP	The third-party company that remotely manages and monitors a client's IT infrastructure and end-user systems, offering services like network and infrastructure management, security, and 24/7 technical support.
Organization	Org	A company account in Trusted WiFi.
Operator	-	An organization type account in Trusted WiFi that is used by MSPs to manage a property's Wi-Fi network.
Customer	-	An organization type account in Trusted WiFi typically used for customers/brands that allows grouping to view all properties belonging to the same company even if managed by several different MSPs.
Linked Organization	Linked Org	A link creating a relationship between an operator and a customer account, allowing a customer to view a property while allowing an operator to manage it.
User	-	An individual accessing a product, service or system. In the context of Trusted WiFi, a user is setup to access products and deployments information at Admin/Editor/Viewer levels, according to their relevant role permissions. In the context of a given product or service, a user is another term referring to a subscriber: the person at the end of the chain interacting with that product or service – usually through a device.
Property	-	Trusted WiFi concept representing an Individual site or location where products are deployed.
Linked Property	-	Site or location shared between operator and customer accounts.

Passpoint Software Development Kit	Passpoint SDK	The service that sits within the customer's mobile app and that is connected to the Trusted WiFi RADIUS infrastructure allowing Passpoint profiles to be created for a given Passpoint realm.
Passpoint Realm	-	The customer specific domain that is used to provision Passpoint profiles that are approved for connection to any associated network.
Passpoint Profile	-	The security certified profile that sits on a subscriber's device. If installed correctly it allows seamless authentication to the secure Passpoint SSID.
Secure Passpoint Service Set Identifier	Secure Passpoint SSID	The Wi-Fi network that is associated to the Passpoint realm configured to allow subscriber devices with a valid Passpoint profile to seamlessly connect to the Passpoint network at a property.
Subscriber	-	An individual person using a service.
Subscriber Device	-	The equipment – typically a smartphone, tablet or computer – the subscriber is using to connect to the service.
Collision-Resistant Unique Identifier	CUID	A unique identifier designed to be collision-resistant – meaning engineered to minimize the likelihood of generating duplicate IDs even in distributed systems – and more efficient in terms of space and database indexing performance due to its sequential nature. In the context of Passpoint, a CUID is delivered to a subscriber device when it requests a Passpoint profile.
Customer Loyalty Mobile Application	Mobile App	The iOS and/or Android digital application used by businesses to engage and reward their customers through loyalty programs. In the context of Passpoint, the best user experience is to offer Passpoint through a customer/brand mobile app integration using the Trusted WiFi SDK.

CISCO MERAKI CONFIGURATION

Supported Versions

This document is based on Cisco Meraki MR 30.7.

Prerequisites

It is assumed the following prerequisites are met before configuring Cisco Meraki for a Trusted WiFi Passpoint service:

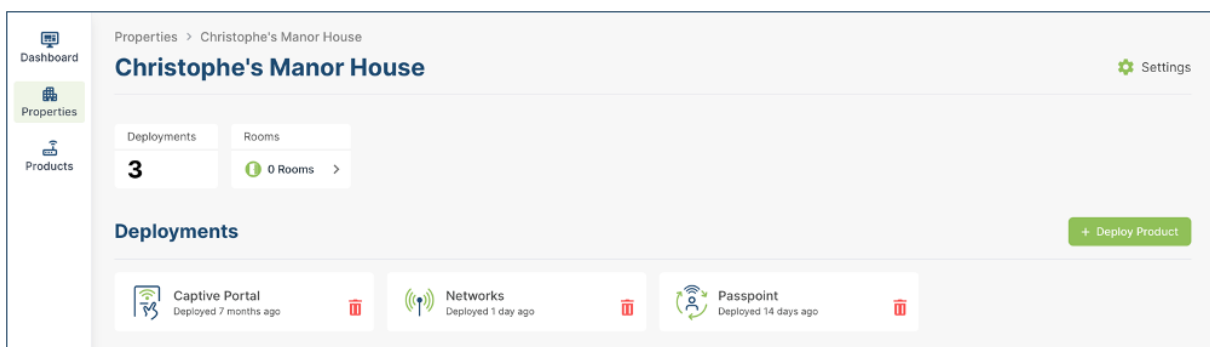
- A supported Cisco Meraki account activated and licensed.
- A Trusted WiFi account with operator permissions.
- A core Trusted WiFi Passpoint service configured and tested.
- A property in Trusted WiFi with deployed Passpoint modules.
- A deployed and configured Wi-Fi network.

Warning

All properties must share the same NAI realm, RADIUS IP / port settings and SSID name for the Passpoint service (CustomerPasspoint). Each property requires a separate RADIUS secret and NAS identifier, both generated when a configuration is activated within the Trusted WiFi management platform.

Core Passpoint Settings

- Log into Trusted WiFi, then click on the **Properties** icon in the left menu to view the properties list.
- Select or search for the property you wish to work on. This will open its deployments page:



- Click on the **Passpoint** tile and then click on the **Configuration** option in the left menu to display the Passpoint settings summary as per the example below:

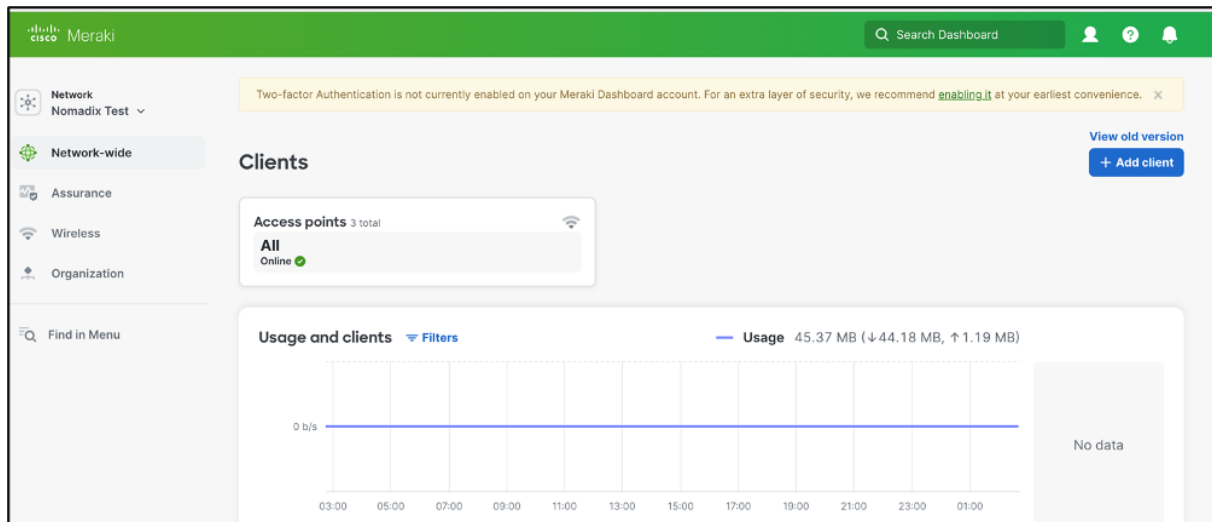
Configuration	
NAI Realm	customer.cloud.global
Friendly Name	customer.cloud.global
Primary RADIUS IP	116.214.120.1
Secondary RADIUS IP	116.214.121.1
RADIUS Authentication Port	1812
RADIUS Accounting Port	1813
NAS Identifier	1f1386a9
RADIUS Secret	5dc7b90b296789h08a6ef764b1e3d

- Take note of the details for your respective installation as they will be required at a later step.

Cisco Meraki Configuration

SSID & Access Control

- Log into the Cisco Meraki Dashboard with your credentials to display the page below:



- On the left-hand side menu items, click on **Wireless** -> **SSID**. Under an unused SSID click **edit settings**. The following screen will appear:

- Complete the fields, as per the description below:
 FieldDescriptionBasic InformationSSID (name)Provide an SSID name.SSID StatusSelect the Enabled option for SSID status.SecuritySelect Enterprise with radio box and select my RADIUS server.Wi-Fi Personal Network (WPN)Select Disabled.WPA encryptionSelect WPA2 only.802.11wSelect the Disabled (never use) radio button.Mandatory DHCPSelect Disabled.Splash pageSelect None (direct access).

Basic info

SSID (name)

SSID status Enabled Disabled

☐ Hide SSID

Security WPA2 Enterprise with 0 RADIUS servers

☐ Open (no encryption)
Any user can associate

☐ Opportunistic Wireless Encryption (OWE)
Any user can associate with data encryption

☐ Password
Users must enter a passphrase to associate

☐ MAC-based access control (no encryption)
RADIUS server is queried at association time

☒ Enterprise with

User credentials are validated with RADIUS at association time

☐ Identity PSK with RADIUS

RADIUS server is queried at association time to obtain a passphrase for a device based on its MAC address

☐ Identity PSK without RADIUS
Devices are assigned a group policy based on its passphrase

Wi-Fi Personal Network (WPN) Enabled Disabled

WPA encryption

802.11w ☐ Enabled (allow unsupported clients)
☐ Required (reject unsupported clients)
☒ Disabled (never use)

Mandatory DHCP Enabled Disabled

Splash page None

Not all splash authentication methods are compatible with WPA2-Enterprise authentication

☒ None (direct access)
Users can access the network as soon as they associate

- Scroll down to **RADIUS**.
- Under RADIUS servers, click **Add Server** to configure your Authentication RADIUS settings. You will need to add two servers, for Primary and Secondary settings.

RADIUS 0 RADIUS servers

RADIUS servers

#	Host IP or FQDN	Auth port	Secret	RadSec	Test	Actions
You have no servers defined						

[Add server](#) 3 max.

- Complete the fields, as per the description below:
FieldDescriptionPrimary ServerIP
AddressesEnter the <Primary RADIUS IP> from Trusted WiFi.PortEnter the < RADIUS Authentication Port> from Trusted WiFi.Shared SecretEnter the < RADIUS Shared Secret> from Trusted WiFi.Secondary ServerIP AddressesEnter the <Secondary RADIUS IP> from Trusted WiFi.PortEnter the < RADIUS Authentication Port> from Trusted WiFi.Shared SecretEnter the < RADIUS Shared Secret> from Trusted WiFi.
- Click on **Done**.
- Under RADIUS accounting servers, click **Add Server** to configure your RADIUS Accounting settings. You will need to add two servers for both Primary and Secondary settings.

RADIUS accounting servers

#	Host IP or FQDN	Acct port	Secret	RadSec ⓘ	Actions
You have no servers defined					

[Add server](#) 3 max.

- Complete the fields, as per the description below:

Field	Description
Primary SeverIP	Addresses
Port	Enter the < RADIUS Authentication Port> from Trusted WiFi.
Shared Secret	Enter the < RADIUS Shared Secret> from Trusted WiFi.
Secondary ServerIP	Addresses
Port	Enter the < RADIUS Authentication Port> from Trusted WiFi.
Shared Secret	Enter the < RADIUS Shared Secret> from Trusted WiFi.

RADIUS 2 RADIUS servers, 2 accounting servers

RADIUS servers

#	Host IP or FQDN	Auth port	Secret	RadSec ⓘ	Test	Actions
1	116.214.120.1	1812	*****	☐	Test	...
2	116.214.121.1	1812	*****	☐	Test	...

[Add server](#) 3 max.

RADIUS accounting servers

#	Host IP or FQDN	Acct port	Secret	RadSec ⓘ	Actions
1	116.214.120.1	1813	*****	☐	...
2	116.214.121.1	1812	*****	☐	...

[Add server](#) 3 max.

Accounting interim minutes

- Scroll Down to **Advanced RADIUS Settings**.
- Under **NAS ID** Category 1, select Custom from the drop down.
- Enter the **NAS ID** from the Trusted WiFi configuration settings.

Advanced RADIUS settings

(NAS ID, Called-station-ID, RADIUS timeout, retry count, fallback, EAP timers)

Called-station-ID

#	Category
1	AP MAC address
2	SSID name

[Add identifier](#) 4 max.

NAS ID

#	Category	Custom
1	Custom	2024664b
2	SSID number	

[Add identifier](#) 4 max.

Server timeout second(s)

Retry count

RADIUS fallback Active Off

- Click **Save** on the bottom of the screen.

Passpoint (Hotspot 2.0)

- Click on the **Wireless > Hotspot 2.0** on the left of the screen to display the page below:

Hotspot 2.0

SSID:

Hotspot 2.0

Operator name

Venue name

Venue type

Network type

Domain list
One domain per line

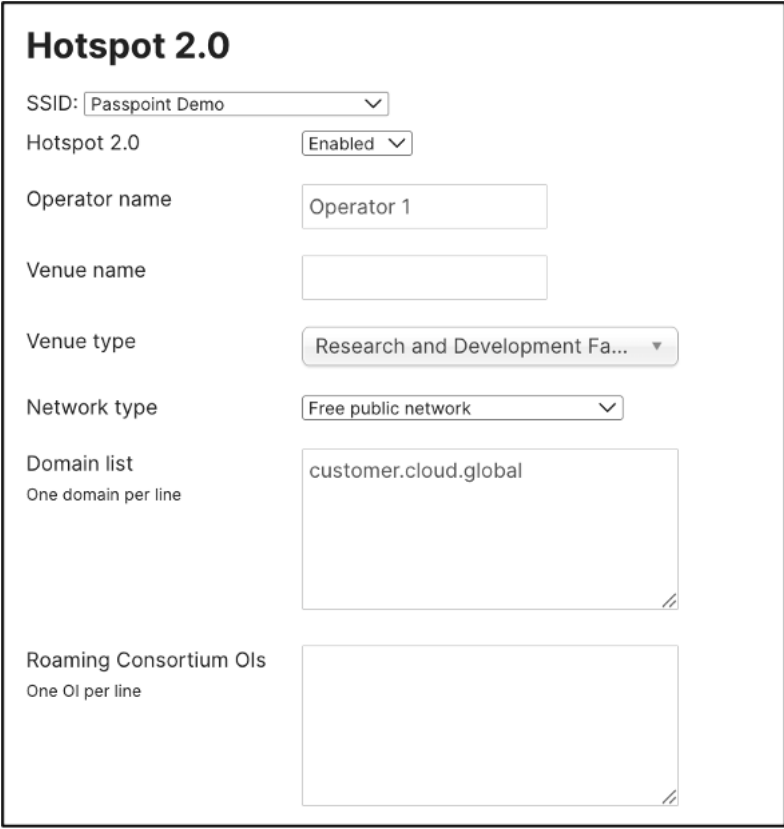
Roaming Consortium OIs
One OI per line

NAI Realms

MCC/MNCs
One MCC/MNC pair per line,
like:
123 456

- Complete the fields, as per the description below:

Field	Description
Hotspot 2.0	Select Enabled from the dropdown.
Operator name	Enter an operator name.
Venue name	Enter a venue name.
Venue type	Select a venue type from the drop-down.
Network type	Select a network type from the drop-down.
Domain list	Enter the domain as shown in your configuration in Trusted WiFi e.g. customer.cloud.gobal
Roaming Consortium OIs	Leave blank.



Hotspot 2.0

SSID:

Hotspot 2.0

Operator name

Venue name

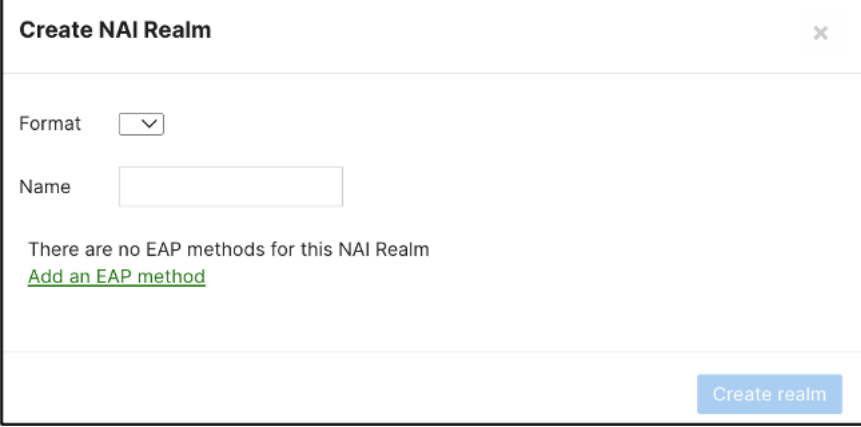
Venue type

Network type

Domain list
One domain per line

Roaming Consortium Ols
One Ol per line

- Click on **Create realm**. The following screen will display:



Create NAI Realm ✕

Format

Name

There are no EAP methods for this NAI Realm

[Add an EAP method](#)

- Complete the fields, as per the description below:
FieldDescriptionFormatSelect 0.NameEnter the name realm. e.g. customer.cloud.globalMethod IDSelect 21.EAP-TTLS from the drop-down.Authentication MethodsSelect PAP, MSCHAP and MSCHAPV2.

Create NAI Realm

Format 0

Name

Method ID	Authentication Methods	Actions
21 EAP-TTLS	PAP CHAP MSCHAPV2	X

[Add an EAP method](#)

Create realm

- Click on **Create realm**.

Hotspot 2.0

SSID: Passpoint Demo

Hotspot 2.0 Enabled

Operator name

Venue name

Venue type Research and Development Fa...

Network type Free public network

Domain list
One domain per line

Roaming Consortium OIs
One OI per line

NAI Realms
Create realm Delete

Format	Name	Methods
☐ 0	customer.cloud.global	21: PAP, CHAP, MSCHAPV2

1 total

MCC/MNCs
One MCC/MNC pair per line,
like:
123 456

- Click **Save changes**.

Aruba Mobility Controllers

Download this guide 

Introduction

Scope and Purpose

Thank you for purchasing the Trusted WiFi solution. This document is a hardware configuration guide describing how to setup Aruba Mobility Controllers for a Trusted WiFi Passpoint service.

- For more information on how to setup an end-to-end Trusted WiFi Passpoint service, please refer to the Trusted WiFi Passpoint Administration Guide.
- For complete information on how to setup Aruba Mobility Controllers, please refer to the vendor's original documentation.

Documentation Conventions

{{snippet.Documentation Conventions}}

Passpoint Overview

Passpoint – also known as Hotspot 2.0 – is an industry-wide next generation approach to public internet access driven by the Wi-Fi Alliance that brings the following benefits:

- Frictionless onboarding and roaming, thanks to a one-time registration followed by automatic access to interconnected hotspots
- More secure and private Wi-Fi connections, compared with general visitor networks

Passpoint is based on the IEEE 802.11u standard, which is a set of protocols enabling cellular-like roaming. Following the initial enrollment, frequent users such as visitors, guests or employees bypass repeated logins, forms and passwords, as their mobile devices automatically join the Wi-Fi subscriber service when they return to a venue or roam between inter-linked Passpoint enabled hotspots and providers, while being better protected against potential cyber threats.

If a device supports 802.11u and is enrolled to a service, it automatically communicates with the Wi-Fi infrastructure via the access points to discover the network SSID and connects securely to it by presenting its access credentials. Upon successful authentication, the device is provisioned with Passpoint standards-based management objects - known as Per-Provider Subscription Management Objects (PPS-MO).

GlobalReach - an ASSA ABLOY company - has been involved with Passpoint since its inception and even contributed to the creation and initial pilot testing of the standard. As a result, it is one of the few trusted worldwide experts on this topic, with a proven platform backed by real-world operational experiences at scale.

The best user experience is to offer Passpoint through a customer/brand mobile app integration, as it further simplifies the onboarding process, while incentivizing app downloads and customer loyalty, leading to further engagement and monetization opportunities. To this effect, Trusted WiFi offers a Software Development Kit (SDK) for easy app integration.

Seamless Onboarding and Roaming

Best user experience via brand mobile loyalty app integration:

- One-time setup in 3 "taps" via a mobile app
- Followed by automatic connection for returning visitors and roaming within any of the participating venues
- No manual SSID selection, no login/password, no reset after a given period
- No impact from MAC address randomization
- Incentive to download the app, leading to visitor engagement opportunities
- Non-compatible devices can still use the legacy onboarding process

End-to-End Service Components

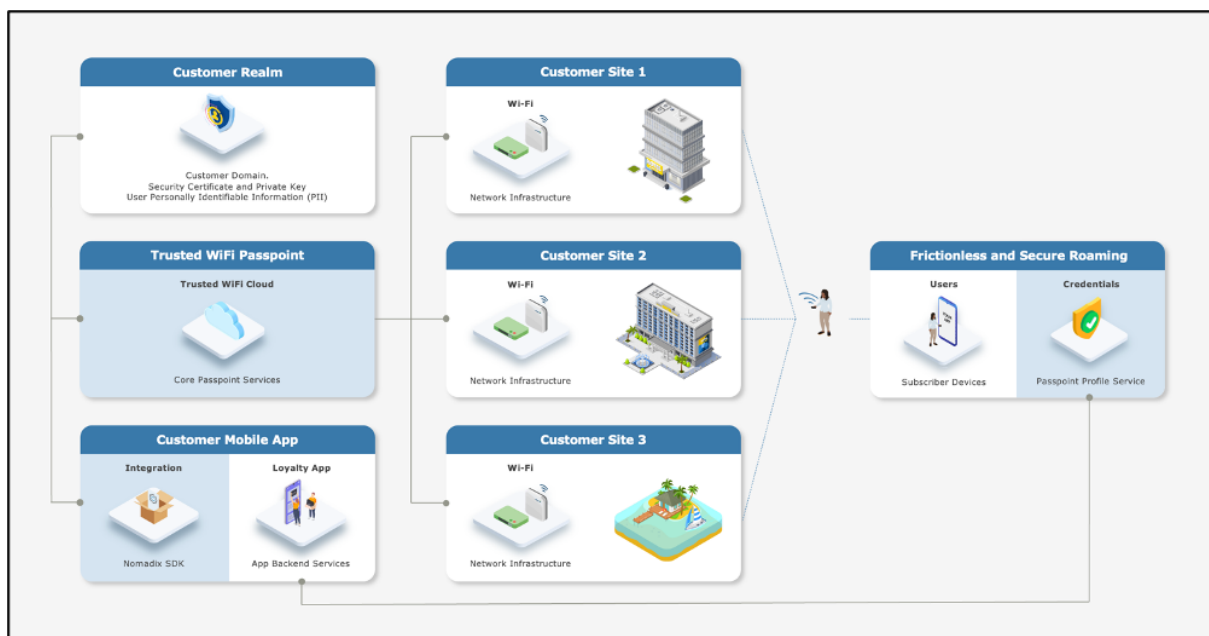
Implementing an end-to-end Trusted WiFi Passpoint service requires a combination of the following software and hardware components:

- **Trusted WiFi Passpoint:** the core services are offered and managed centrally via the Trusted WiFi Passpoint Module (hosted and operated by GlobalReach). Following an initial setup performed by the GlobalReach Operations team, Managed Service Providers (MSPs) can then add sites to customer realms and monitor the solution.
- **Customer Realm:** the service requires connectivity to a realm including domain, security certificate and private key, as well as the database holding the users' Personally Identifiable Information (PII). It is the responsibility of the Identity Provider – typically the customer/brand – to make this part available.
- **Mobile App:** the best user experience is to offer Passpoint through a customer/brand mobile app integration. Trusted WiFi offers a **Software Development Kit (SDK)** for easy implementation. It is the responsibility of the app owner – typically the customer/brand – to perform the integration.
- **Local Networks:** the local networks must be compliant with Passpoint (Hotspot 2.0). – Most recent Wi-Fi access points and controllers from major vendors support Passpoint today, however older models or products from more exotic manufacturers might not do. The configuration of the local infrastructure is typically handled by the Managed Service Providers (MSPs).

- **User Devices:** the subscribers' mobile devices (belonging to visitors, guests, employees, etc.) must be compliant with Passpoint (Hotspot 2.0) – All recent iOS and Android-based smartphones or tablets support Passpoint today, however older models or products running less popular operating systems might not do. Laptops compatibility is also more erratic. It is therefore essential to maintain a traditional onboarding service in parallel with Passpoint to handle non-compatible devices.

High-Level Topology

The diagram below illustrates the high-level topology for the end-to-end Trusted WiFi Passpoint service:



Glossary

The following is a glossary of the most common terms used regarding this solution.

Term	Abbreviation	Description
Deployment	-	Enabling of a Trusted WiFi product module for a property via the Trusted WiFi interface.
Module	-	Product or service purchased from Trusted WiFi that is managed through its own sub-section via the Trusted WiFi interface.
License	-	Legal agreement that grants users the right to use specific software, outlining terms and conditions for its usage, distribution, and potential modifications, while protecting the intellectual property of the software developer. GlobalReach licenses comprise of two different types: one-off licenses - typically to initially enable a software module. recurring licenses - typically including software updates and technical support, or more in the case of OpEx consumption commercial models based on price per month/quarter/year. Trusted WiFi is sold as a combination of one-off licenses to activate the service and recurring licenses based on a price per Wi-Fi access point per month.
Managed Service Provider	MSP	The third-party company that remotely manages and monitors a client's IT infrastructure and end-user systems, offering services like network and infrastructure management, security, and 24/7 technical support.
Organization	Org	A company account in Trusted WiFi.
Operator	-	An organization type account in Trusted WiFi that is used by MSPs to manage a property's Wi-Fi network.
Customer	-	An organization type account in Trusted WiFi typically used for customers/brands that allows grouping to view all properties belonging to the same company even if managed by several different MSPs.
Linked Organization	Linked Org	A link creating a relationship between an operator and a customer account, allowing a customer to view a property while allowing an operator to manage it.
User	-	An individual accessing a product, service or system. In the context of Trusted WiFi, a user is setup to access products and deployments information at Admin/Editor/Viewer levels, according to their relevant role permissions. In the context of a given product or service, a user is another term referring to a subscriber: the person at the end of the chain interacting with that product or service – usually through a device.
Property	-	Trusted WiFi concept representing an Individual site or location where products are deployed.
Linked Property	-	Site or location shared between operator and customer accounts.

Passpoint Software Development Kit	Passpoint SDK	The service that sits within the customer's mobile app and that is connected to the Trusted WiFi RADIUS infrastructure allowing Passpoint profiles to be created for a given Passpoint realm.
Passpoint Realm	-	The customer specific domain that is used to provision Passpoint profiles that are approved for connection to any associated network.
Passpoint Profile	-	The security certified profile that sits on a subscriber's device. If installed correctly it allows seamless authentication to the secure Passpoint SSID.
Secure Passpoint Service Set Identifier	Secure Passpoint SSID	The Wi-Fi network that is associated to the Passpoint realm configured to allow subscriber devices with a valid Passpoint profile to seamlessly connect to the Passpoint network at a property.
Subscriber	-	An individual person using a service.
Subscriber Device	-	The equipment – typically a smartphone, tablet or computer – the subscriber is using to connect to the service.
Collision-Resistant Unique Identifier	CUID	A unique identifier designed to be collision-resistant – meaning engineered to minimize the likelihood of generating duplicate IDs even in distributed systems – and more efficient in terms of space and database indexing performance due to its sequential nature. In the context of Passpoint, a CUID is delivered to a subscriber device when it requests a Passpoint profile.
Customer Loyalty Mobile Application	Mobile App	The iOS and/or Android digital application used by businesses to engage and reward their customers through loyalty programs. In the context of Passpoint, the best user experience is to offer Passpoint through a customer/brand mobile app integration using the Trusted WiFi SDK.

ARUBA MOBILITY CONTROLLERS CONFIGURATION

Supported Versions

Aruba Mobility Controllers support Passpoint from AOS v8.6. Aruba recommends upgrading to the latest version of AOS.

Warning

The configuration of the RADIUS is documented using the WLAN configuration option. The alternative method, under Configuration > Authentication, is not covered in this document.

Prerequisites

It is assumed the following prerequisites are met before configuring Aruba Mobility Controllers for a Trusted WiFi Passpoint service:

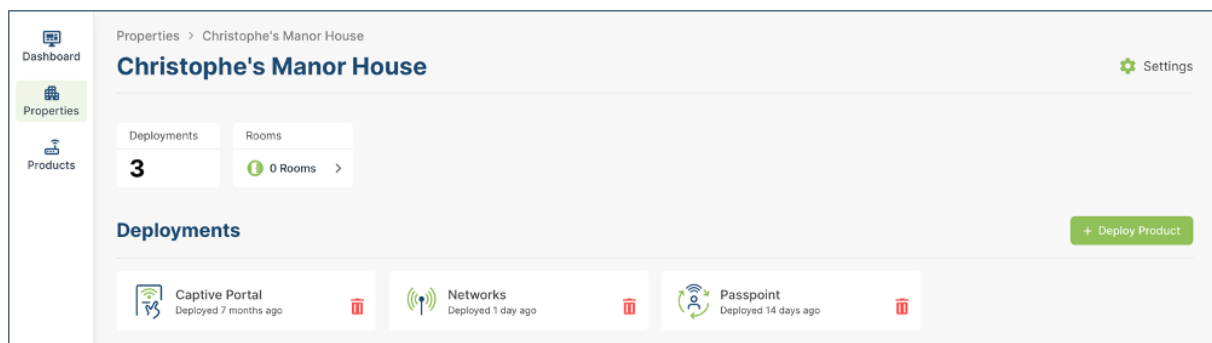
- A supported Aruba Mobility Controller running version 8.6 or greater, activated and licensed.
- A Trusted WiFi account with operator permissions.
- A core Trusted WiFi Passpoint service configured and tested.
- A property in Trusted WiFi with deployed Gateway and Passpoint modules.
- A deployed and configured Wi-Fi network.

Warning

The Aruba configuration is focusing on Passpoint v1.0. All properties must share the same NAI realm, RADIUS IP / port settings and SSID name for the Passpoint service (CustomerPasspoint). Each property requires a separate RADIUS secret and NAS identifier, both generated when a configuration is activated within the Trusted WiFi management platform.

Core Passpoint Settings

- Log into Trusted WiFi, then click on the **Properties** icon in the left menu to view the properties list.
- Select or search for the property you wish to work on. This will open its deployments page:



- Click on the **Passpoint** tile and then click on the **Configuration** option in the left menu to display the Passpoint settings summary as per the example below:

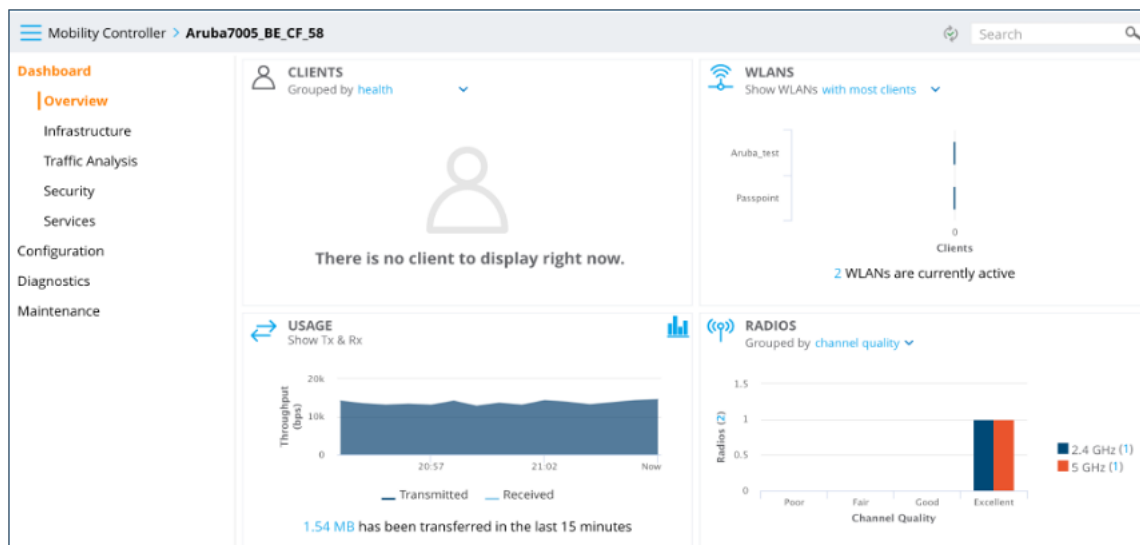
Configuration	
NAI Realm	customer.cloud.global
Friendly Name	customer.cloud.global
Primary RADIUS IP	116.214.120.1
Secondary RADIUS IP	116.214.121.1
RADIUS Authentication Port	1812
RADIUS Accounting Port	1813
NAS Identifier	1f386a9
RADIUS Secret	5dc7b90b296789h08a6ef764b1e3d

- Take note of the details for your respective installation as they will be required at a later step.

Aruba Mobility Controllers Configuration

Wi-Fi Network

- Log into the Aruba Mobility Controller with your credentials to display the page below:



- In the left-hand navigation menu, proceed to **Configuration > WLANs** to display the page below:

The screenshot shows the 'WLANs' configuration page. The left navigation menu is expanded to 'Configuration' > 'WLANs'. The main content area displays a table with 2 WLANs:

NAME (SSID)	AP GROUP	KEY MANAGEMENT	INFORMATION
Passpoint	default	WPA2-Enterprise	--
Aruba_test	default	WPA2-Personal	--

A blue plus sign (+) is visible at the bottom left of the table area to add a new WLAN.

- Click on the **+** on the lower left side of the screen and enter the **Name** of the SSID as shown below:

New WLAN

General VLANs Security Access

Name (SSID):

Primary usage: ☒ Employee ☐ Guest

Broadcast on:

Forwarding mode:

- Click on **Next** and enter the **VLAN** to be assigned to this SSID, as shown below:

New WLAN

General VLANs Security Access

VLAN:

[Hide VLAN details](#)

Named VLANs	
NAME	ID(S)
Vlan1000	1000
--	1

[+](#)

- Click on **Next** to display the page below:

- Set the **Key management** to **WPA2 Enterprise** from the drop-down list.
- Click on the **+** symbol in the Auth server's window to display the page below:

- Complete the fields as per the description below:
 FieldDescriptionNameEnter the name of the Server.
 IP addressEnter the <Primary RADIUS IP> from Trusted WiFi.
 Auth PortEnter the <RADIUS Authentication Port> from Trusted WiFi.
 Accounting portEnter the <RADIUS Accounting Port> from Trusted WiFi.
 Shared SecretEnter the <RADIUS Shared Secret> from Trusted WiFi.
 Retype KeyRe-enter the <RADIUS Shared Secret> from Trusted WiFi.
 TimeoutEnter a timeout value.
- Click **Submit** to save, repeat this operation for the **Secondary Server** and click **Submit** to save.

- Click **Next** to display the page below:

New WLAN

General VLANs Security Access

Default role:

Server-derived roles: ☐

Show [roles](#)

- Click on **Finish** to save the newly configured SSID. The new WLAN displays in the WLANs table.

NAS ID

- In the left-hand menu items, proceed to **Configuration > Authentication**, then proceed to the **Auth Servers** tab.
- Under the **All Servers** heading, select the **Auth RADIUS Server** that was created earlier to display the page below:

Mobility Controller > Aruba7005_BE_CF_58

Dashboard

Configuration

WLANs

Roles & Policies

Access Points

AP Groups

Authentication

Services

Interfaces

System

Tasks

Redundancy

Diagnostics

Maintenance

Auth Servers AAA Profiles L2 Authentication L3 Authentication User Rules

Timeout:

Retransmits:

NAS ID:

NAS IP:

Enable IPv6: ☐

NAS IPv6:

Use MD5: ☐

Mode: ☒

Lowercase MAC addresses: ☐

Use IP address for calling station ID: ☐

MAC address delimiter:

Service-type of FRAMED-USER: ☐

CPPM credentials: ☐

CALLED STATION ID

Station ID type:

Station ID delimiter:

Include SSID: ☐

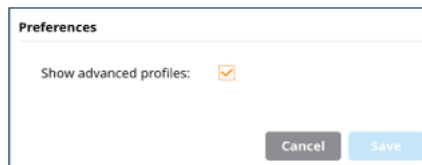
Aruba7005-US, 8.6.0.2

- Complete the fields as per the description below:

Field	Description
NAS ID	Enter the NAS ID from Trusted WiFi.
Called Station ID	Station ID type: MAC Address.
Station ID Delimiter	Colon.
- Verify all information is correct then click **Submit**.

Passpoint (Hotspot 2.0)

- Click on the **Login Name** at the top right of the screen. Select **Preferences** and check the **Show advanced profiles** box.



Preferences

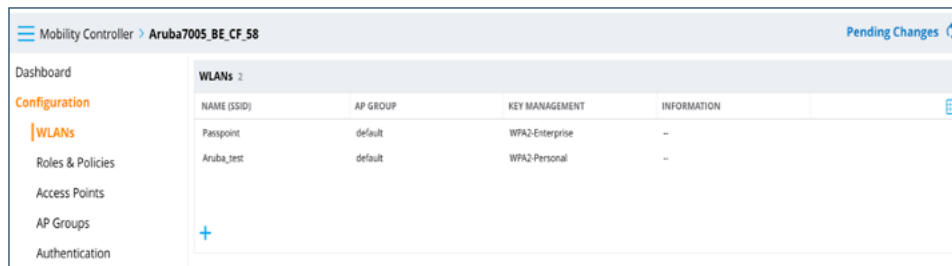
Show advanced profiles: ☒

Cancel Save

Warning

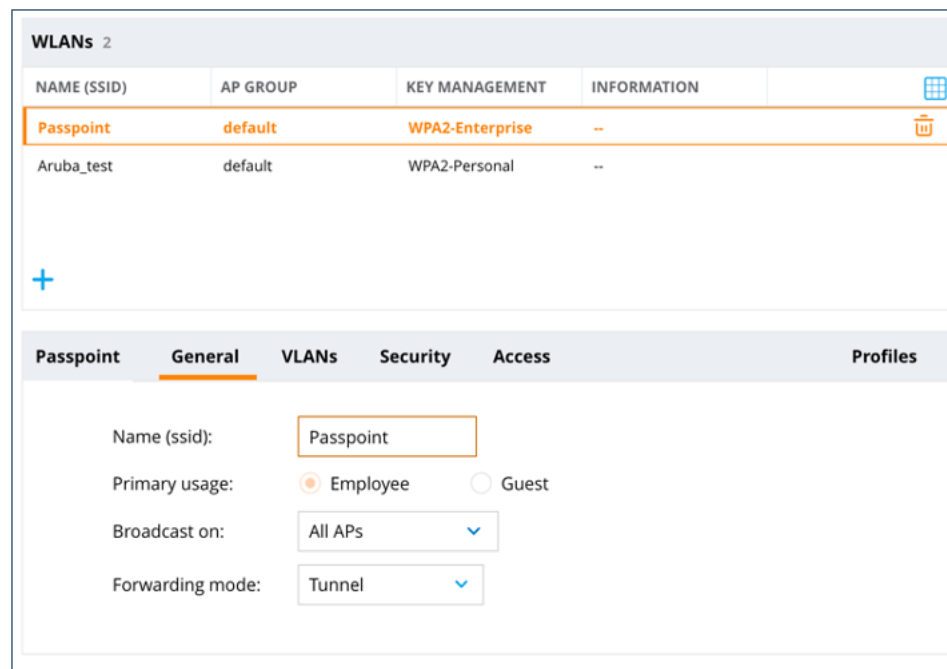
You must be in Advanced mode to configure Passpoint (Hotspot 2.0).

- Click on **Save**.
- In the left-hand menu items, proceed to **Configuration > WLANs** to display the page below:



NAME (SSID)	AP GROUP	KEY MANAGEMENT	INFORMATION
Passpoint	default	WPA2-Enterprise	--
Aruba_test	default	WPA2-Personal	--

- In the **WLANs** table menu, click on the newly created **Passpoint SSID**:



NAME (SSID)	AP GROUP	KEY MANAGEMENT	INFORMATION
Passpoint	default	WPA2-Enterprise	--
Aruba_test	default	WPA2-Personal	--

Passpoint | General | VLANs | Security | Access | Profiles

Name (ssid):

Primary usage: ☒ Employee ☐ Guest

Broadcast on:

Forwarding mode:

- Click on **Profiles** in the SSID section to display the page below and proceed to **Wireless LAN > Virtual AP > [SSID name]**:

Passpoint General VLANs Security Access **Profiles**

Profiles for WLAN Passpoint

- Wireless LAN
- Virtual AP
- Passpoint**
- 802.11k
- AAA
- Anyspot
- Hotspot 2.0
- SSID
- WMM Traffic management

Virtual AP profile: Passpoint

General

Virtual AP enable: ☒

VLAN:

Forward mode:

Openflow Enable: ☒

> RF

> Advanced

> Broadcast/Multicast

Cancel Submit Submit As

- In the side menu tree, select the **Hotspot 2.0** option:

Passpoint General VLANs Security Access **Profiles**

Profiles for WLAN Passpoint

- Wireless LAN
- Virtual AP
- Passpoint
- 802.11k
- AAA
- Anyspot
- Hotspot 2.0**
- SSID
- WMM Traffic management

Hotspot 2.0 Profile: default

Hotspot 2.0 Profile: +

Advertise Hotspot 2.0 Capability: ☐

Enable Hotspot 2.0 OSEN: ☐

ANQP Domain ID:

OSU NAI:

Use GAS Comeback Request/Response: ☐

Additional Steps required for Access Enabled: ☐

Network Internet Access: ☐

GAS Query Response Length Limit: octets

Access network Type:

Roaming Consortium OI value 1:

Cancel Submit Submit As

- To configure a new Hotspot profile, click on the + symbol to display the options below:

Passpoint General VLANs Security Access

Profiles for WLAN Passpoint

- Wireless LAN
- Virtual AP
- Passpoint
- 802.11k
- AAA
- Anyspot
- Hotspot 2.0**
- SSID
- WMM Traffic management

Hotspot 2.0 Profile: New Profile

Profile name:

Advertise Hotspot 2.0 Capability: ☒

Enable Hotspot 2.0 OSEN: ☐

ANQP Domain ID:

OSU NAI:

Use GAS Comeback Request/Response: ☐

Additional Steps required for Access Enabled: ☐

Network Internet Access: ☐

GAS Query Response Length Limit: octets

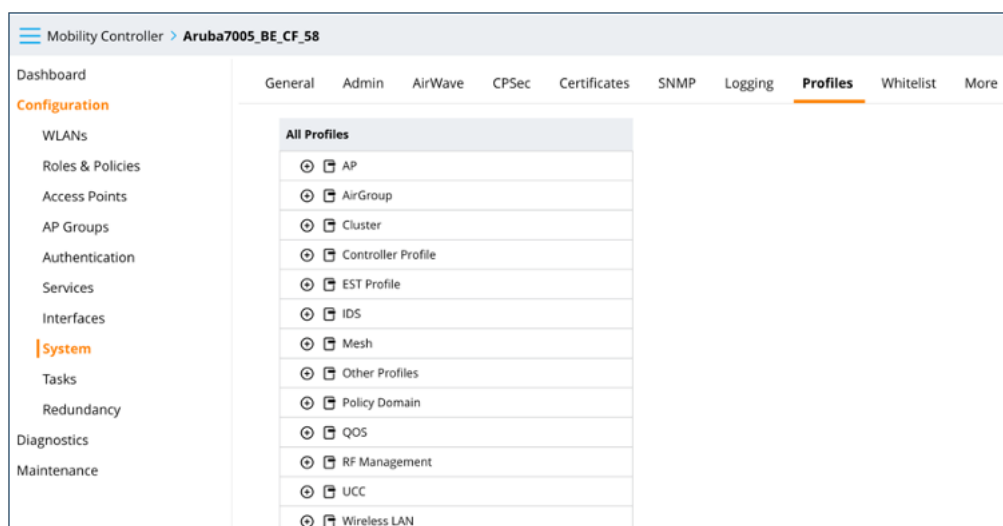
Access network Type:

Roaming Consortium OI value 1:

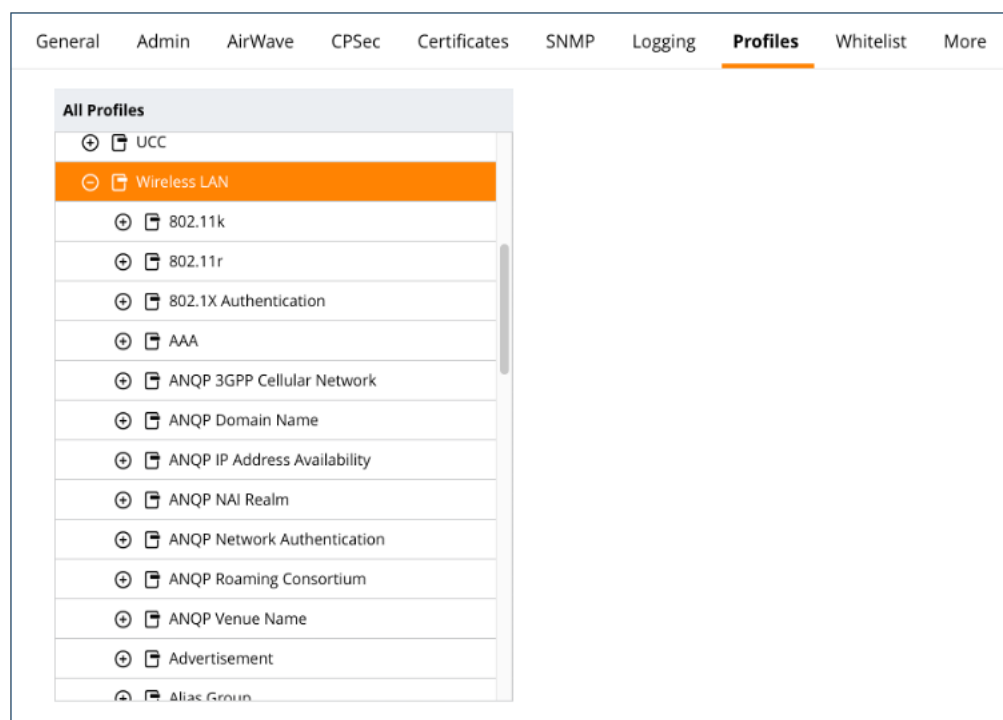
- Complete the fields as per the description below: Field Description Advertise Hotspot 2.0 Capability Check the box. ANQP Domain ID Enter a sequential: 1. OSU NAI Enter the <customer realm> from Trusted WiFi. Access network Type Select public-free from the drop-down list. Roaming Consortium OI value 1 Keep blank. Time zone Enter the time zone (scroll to the bottom of the page).
- Click on **Submit**.

ANQP NAI Realm

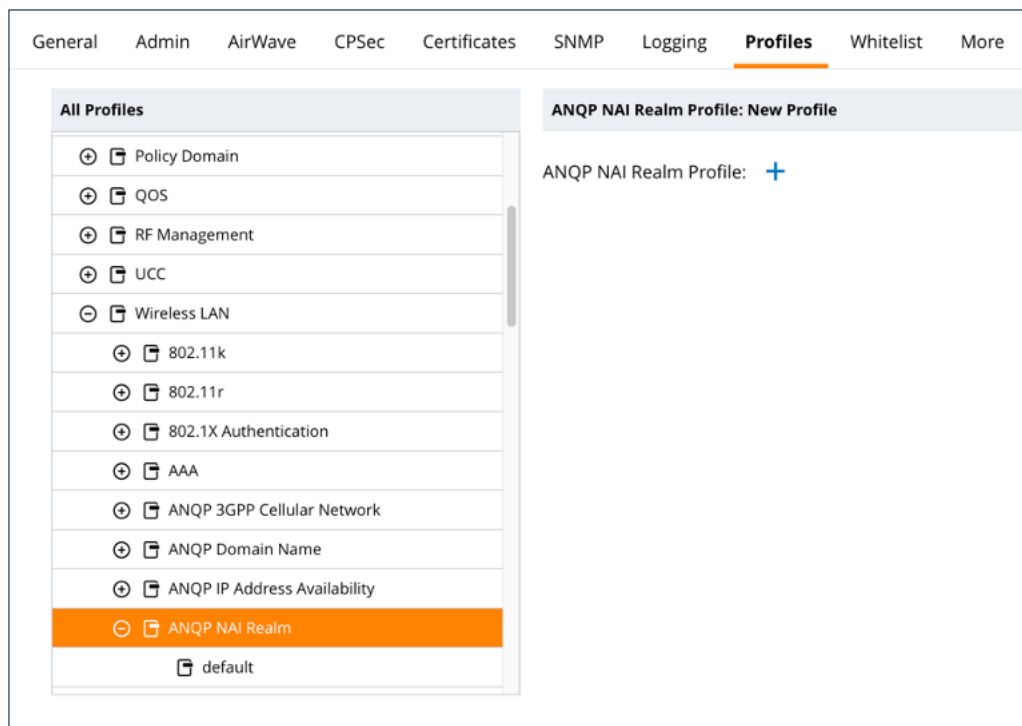
- In the left-hand menu items, proceed to **Configuration > System**, then select the **Profiles** tab:



- Click on the **Wireless LAN** tab to display the following options:



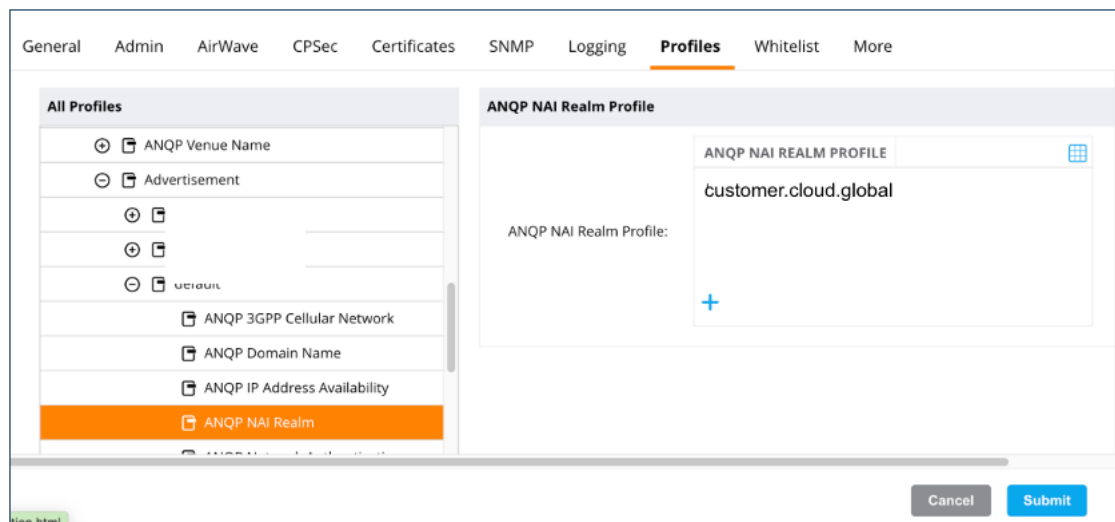
- Click on **ANQP NAI Realm** to display the following options:



- Click on the + to the right of the ANQP NAI Realm Profile to create a new NAI realm:

- Complete the fields as per the description below:

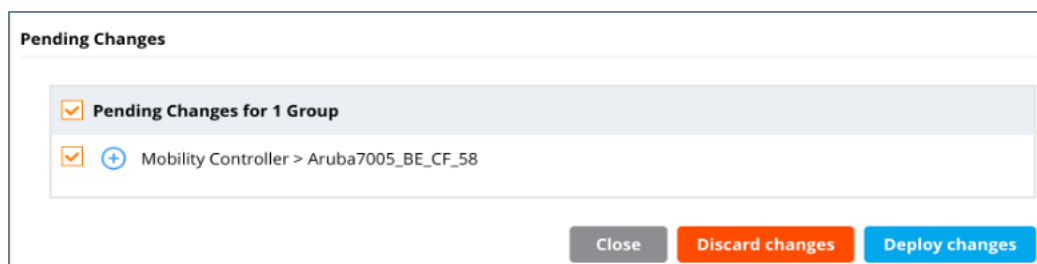
Field	Description				
NAI Realm Name	Enter the <customer realm> from Trusted WiFi.				
NAI Realm EAP Method 1	Select eap-ttls.				
ANQP NAI Realm Authentication Param 1	Click the + icon to create a new one and enter the values as below: <table border="1"><thead><tr><th>ID</th><th>VALUE</th></tr></thead><tbody><tr><td>expanded-eap</td><td>eap-peap-mschapv2</td></tr></tbody></table>	ID	VALUE	expanded-eap	eap-peap-mschapv2
ID	VALUE				
expanded-eap	eap-peap-mschapv2				
NAI Realm EAP Method 2	None.				
- Click on **Submit** to save the realm configuration.
- Go to **Profile > Advertisement > ANPQ NAI Realm** to display the options below:



- Select the **ANQP NAI realm** that was previously created and click **Submit** to save.

Deploy Changes

- To **save** the above Passpoint configuration to the Aruba Mobility Controller, click on the **Pending Changes** button at the top right of the screen and click **Deploy Changes**.



Nomadix WLAN Controllers

Download this guide ☐

Introduction

Scope and Purpose

Thank you for purchasing the Trusted WiFi solution. This document is a hardware configuration guide describing how to setup a Nomadix WLAN Controller (either the AC-1LA physical controller or the cloud controller) for a Trusted WiFi Passpoint service.

- For more information on how to setup an end-to-end Trusted WiFi Passpoint service, please refer to the Trusted WiFi Passpoint Administration Guide.
- For complete information on how to setup a Nomadix WLAN Controller, please refer to the Nomadix Wireless Controller Web-Based Configuration Guide.

Documentation Conventions

{{snippet.Documentation Conventions}}

Passpoint Overview

Passpoint – also known as Hotspot 2.0 – is an industry-wide next generation approach to public internet access driven by the Wi-Fi Alliance that brings the following benefits:

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- More secure and private Wi-Fi connections, compared with general visitor networks

Passpoint is based on the IEEE 802.11u standard, which is a set of protocols enabling cellular-like roaming. Following the initial enrollment, frequent users such as visitors, guests or employees bypass repeated logins, forms and passwords, as their mobile devices automatically join the Wi-Fi subscriber service when they return to a venue or roam between inter-linked Passpoint enabled hotspots and providers, while being better protected against potential cyber threats.

If a device supports 802.11u and is enrolled to a service, it automatically communicates with the Wi-Fi infrastructure via the access points to discover the network SSID and connects securely to it by presenting its access credentials. Upon successful authentication, the device is provisioned with Passpoint standards-based management objects - known as Per-Provider Subscription Management Objects (PPS-MO).

GlobalReach - an ASSA ABLOY company - has been involved with Passpoint since its inception and even contributed to the creation and initial pilot testing of the standard. As a result, it is one of the few trusted worldwide experts on this topic, with a proven platform backed by real-world operational experiences at scale.

The best user experience is to offer Passpoint through a customer/brand mobile app integration, as it further simplifies the onboarding process, while incentivizing app downloads and customer loyalty, leading to further engagement and monetization opportunities. To this effect, Trusted WiFi offers a Software Development Kit (SDK) for easy app integration.

Seamless Onboarding and Roaming

Best user experience via brand mobile loyalty app integration:

- One-time setup in 3 "taps" via a mobile app
- Followed by automatic connection for returning visitors and roaming within any of the participating venues
- No manual SSID selection, no login/password, no reset after a given period
- No impact from MAC address randomization
- Incentive to download the app, leading to visitor engagement opportunities
- Non-compatible devices can still use the legacy onboarding process

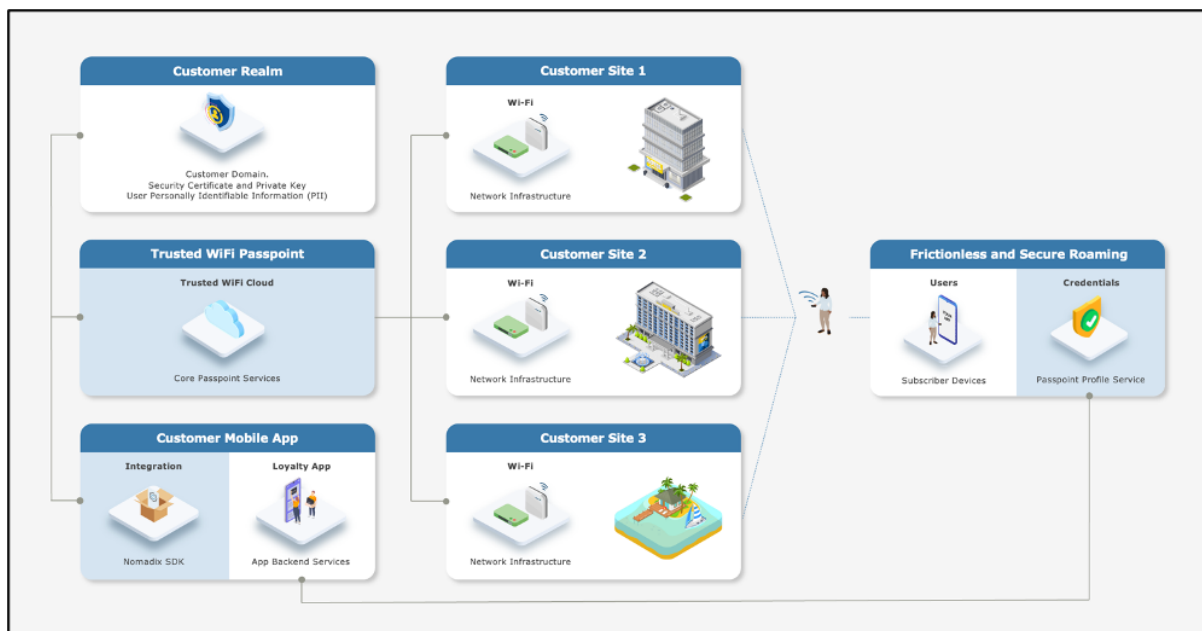
End-to-End Service Components

Implementing an end-to-end Trusted WiFi Passpoint service requires a combination of the following software and hardware components:

- **Trusted WiFi Passpoint:** the core services are offered and managed centrally via the **Trusted WiFi Passpoint Module** (hosted and operated by GlobalReach). Following an initial setup performed by the GlobalReach Operations team, Managed Service Providers (MSPs) can then add sites to customer realms and monitor the solution.
- **Customer Realm:** the service requires connectivity to a realm including domain, security certificate and private key, as well as the database holding the users' Personally Identifiable Information (PII). It is the responsibility of the Identity Provider – typically the customer/brand – to make this part available.
- **Mobile App:** the best user experience is to offer Passpoint through a customer/brand mobile app integration. Trusted WiFi offers a **Software Development Kit (SDK)** for easy implementation. It is the responsibility of the app owner – typically the customer/brand – to perform the integration.
- **Local Networks:** the local networks must be compliant with Passpoint (Hotspot 2.0). – Most recent Wi-Fi access points and controllers from major vendors support Passpoint today, however older models or products from more exotic manufacturers might not do. The configuration of the local infrastructure is typically handled by the Managed Service Providers (MSPs).
- **User Devices:** the subscribers' mobile devices (belonging to visitors, guests, employees, etc.) must be compliant with Passpoint (Hotspot 2.0) – All recent iOS and Android-based smartphones or tablets support Passpoint today, however older models or products running less popular operating systems might not do. Laptops compatibility is also more erratic. It is therefore essential to maintain a traditional onboarding service in parallel with Passpoint to handle non-compatible devices.

High-Level Topology

The diagram below illustrates the high-level topology for the end-to-end Trusted WiFi Passpoint service:



Glossary

The following is a glossary of the most common terms used regarding this solution.

Term	Abbreviation	Description
Deployment	-	Enabling of a Trusted WiFi product module for a property via the Trusted WiFi interface.
Module	-	Product or service purchased from Trusted WiFi that is managed through its own sub-section via the Trusted WiFi interface.
License	-	Legal agreement that grants users the right to use specific software, outlining terms and conditions for its usage, distribution, and potential modifications, while protecting the intellectual property of the software developer. GlobalReach licenses comprise of two different types: one-off licenses - typically to initially enable a software module. recurring licenses - typically including software updates and technical support, or more in the case of OpEx consumption commercial models based on price per month/quarter/year. Trusted WiFi is sold as a combination of one-off licenses to activate the service and recurring licenses based on a price per Wi-Fi access point per month.
Managed Service Provider	MSP	The third-party company that remotely manages and monitors a client's IT infrastructure and end-user systems, offering services like network and infrastructure management, security, and 24/7 technical support.
Organization	Org	A company account in Trusted WiFi.
Operator	-	An organization type account in Trusted WiFi that is used by MSPs to manage a property's Wi-Fi network.
Customer	-	An organization type account in Trusted WiFi typically used for customers/brands that allows grouping to view all properties belonging to the same company even if managed by several different MSPs.
Linked Organization	Linked Org	A link creating a relationship between an operator and a customer account, allowing a customer to view a property while allowing an operator to manage it.
User	-	An individual accessing a product, service or system. In the context of Trusted WiFi, a user is setup to access products and deployments information at Admin/Editor/Viewer levels, according to their relevant role permissions. In the context of a given product or service, a user is another term referring to a subscriber: the person at the end of the chain interacting with that product or service – usually through a device.
Property	-	Trusted WiFi concept representing an Individual site or location where products are deployed.
Linked Property	-	Site or location shared between operator and customer accounts.

Passpoint Software Development Kit	Passpoint SDK	The service that sits within the customer's mobile app and that is connected to the Trusted WiFi RADIUS infrastructure allowing Passpoint profiles to be created for a given Passpoint realm.
Passpoint Realm	-	The customer specific domain that is used to provision Passpoint profiles that are approved for connection to any associated network.
Passpoint Profile	-	The security certified profile that sits on a subscriber's device. If installed correctly it allows seamless authentication to the secure Passpoint SSID.
Secure Passpoint Service Set Identifier	Secure Passpoint SSID	The Wi-Fi network that is associated to the Passpoint realm configured to allow subscriber devices with a valid Passpoint profile to seamlessly connect to the Passpoint network at a property.
Subscriber	-	An individual person using a service.
Subscriber Device	-	The equipment – typically a smartphone, tablet or computer – the subscriber is using to connect to the service.
Collision-Resistant Unique Identifier	CUID	A unique identifier designed to be collision-resistant – meaning engineered to minimize the likelihood of generating duplicate IDs even in distributed systems – and more efficient in terms of space and database indexing performance due to its sequential nature. In the context of Passpoint, a CUID is delivered to a subscriber device when it requests a Passpoint profile.
Customer Loyalty Mobile Application	Mobile App	The iOS and/or Android digital application used by businesses to engage and reward their customers through loyalty programs. In the context of Passpoint, the best user experience is to offer Passpoint through a customer/brand mobile app integration using the Trusted WiFi SDK.

NOMADIX WLAN CONTROLLERS CONFIGURATION

Prerequisites

It is assumed the following prerequisites are met before configuring a Nomadix WLAN Controller (either the AC-1LA physical controller or the cloud controller) for a Trusted WiFi Passpoint service:

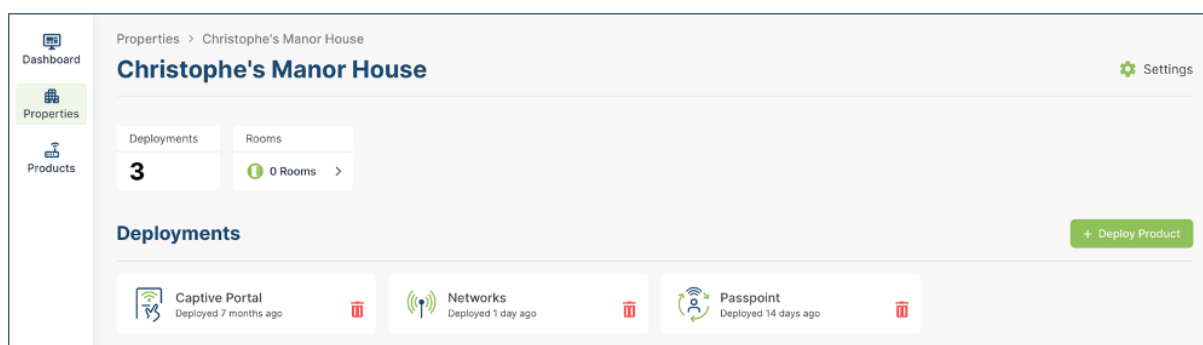
- A supported Nomadix WLAN controller, activated and licensed.
- A Trusted WiFi account with operator permissions.
- A core Trusted WiFi Passpoint service configured and tested.
- A property in Trusted WiFi with deployed Passpoint modules.
- A deployed and configured Wi-Fi network.

Warning

All properties must share the same NAI realm, RADIUS IP/port settings and SSID name for the Passpoint service (CustomerPasspoint). Each property requires a separate RADIUS secret and NAS identifier, both generated when a configuration is activated within the Trusted WiFi management platform.

Core Passpoint Settings

- Log into Trusted WiFi, then click on the **Properties** icon in the left menu to view the properties list.
- Select or search for the property you wish to work on. This will open its deployments page:



- Click on the **Passpoint** tile and then click on the **Configuration** option in the left menu to display the Passpoint settings summary as per the example below:

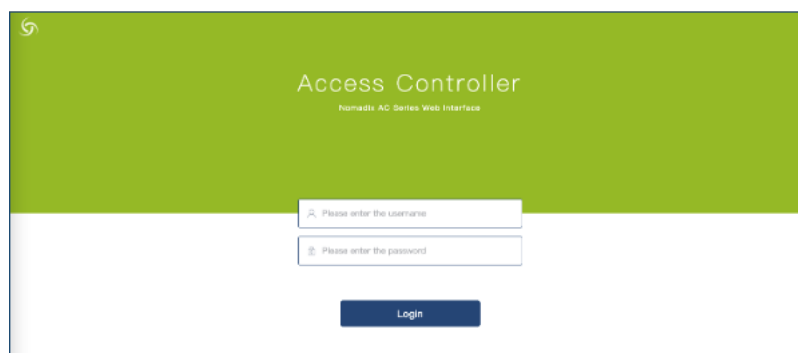
Configuration	
NAI Realm	customer.cloud.global
Friendly Name	customer.cloud.global
Primary RADIUS IP	116.214.120.1
Secondary RADIUS IP	116.214.121.1
RADIUS Authentication Port	1812
RADIUS Accounting Port	1813
NAS Identifier	1ft386a9
RADIUS Secret	5dc7b90b296789h08a6ef764b1e3d

- Take note of the details for your respective installation as they will be required at a later step.

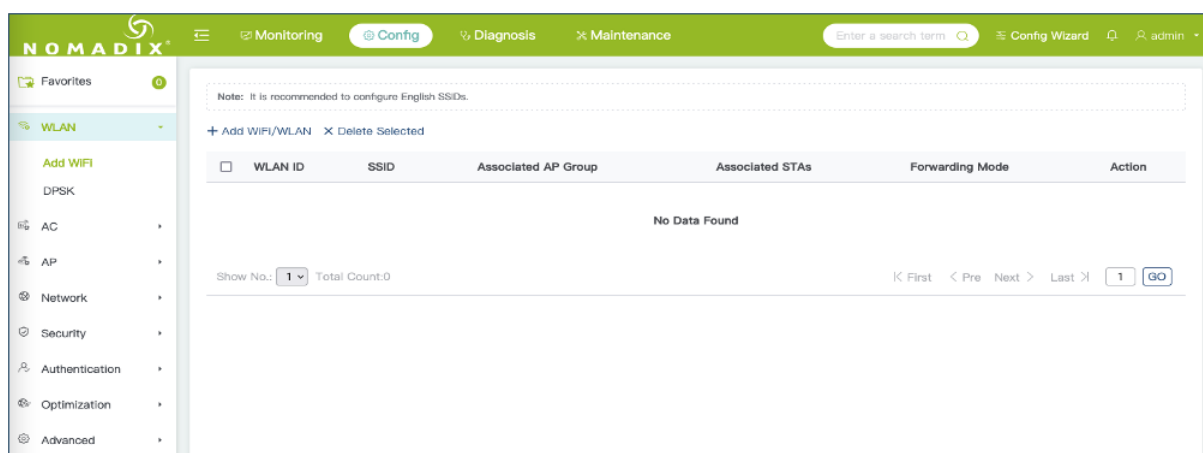
Nomadix WLAN Controllers Configuration

Wi-Fi Network

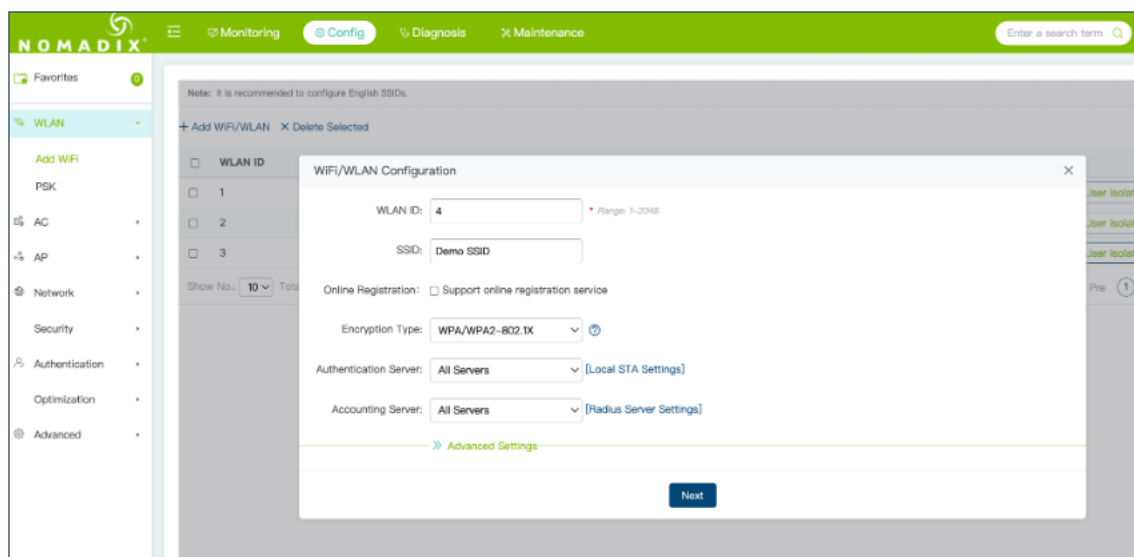
- Log into the Nomadix WLAN controller using your credentials:



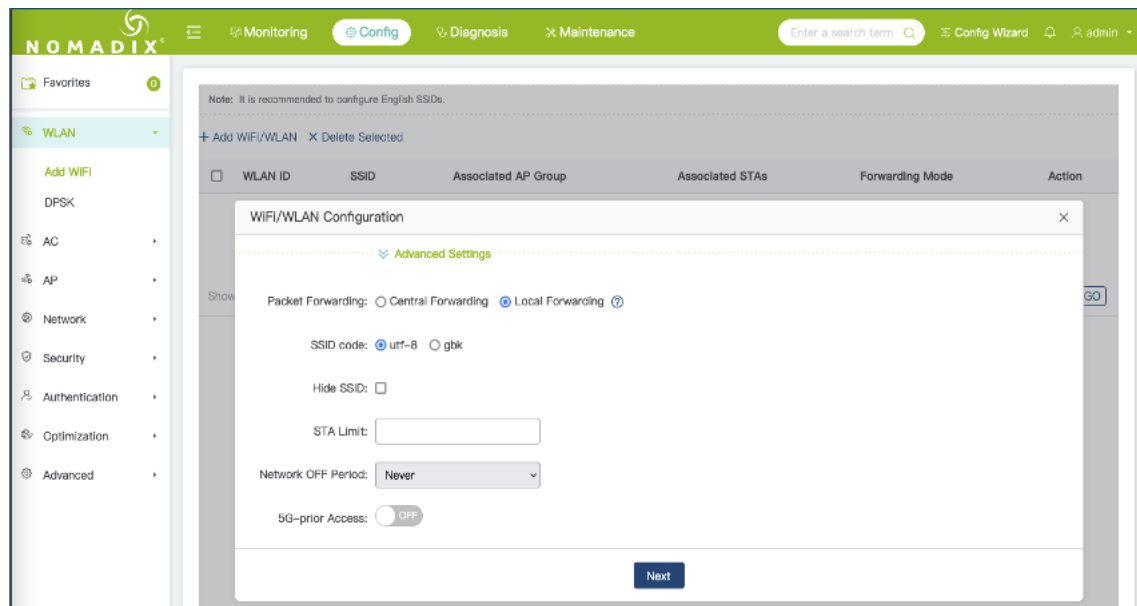
- Click on the **Config** tab at the top of the screen, expand the **WLAN** section and select **Add WiFi** from the left-hand navigation menu.



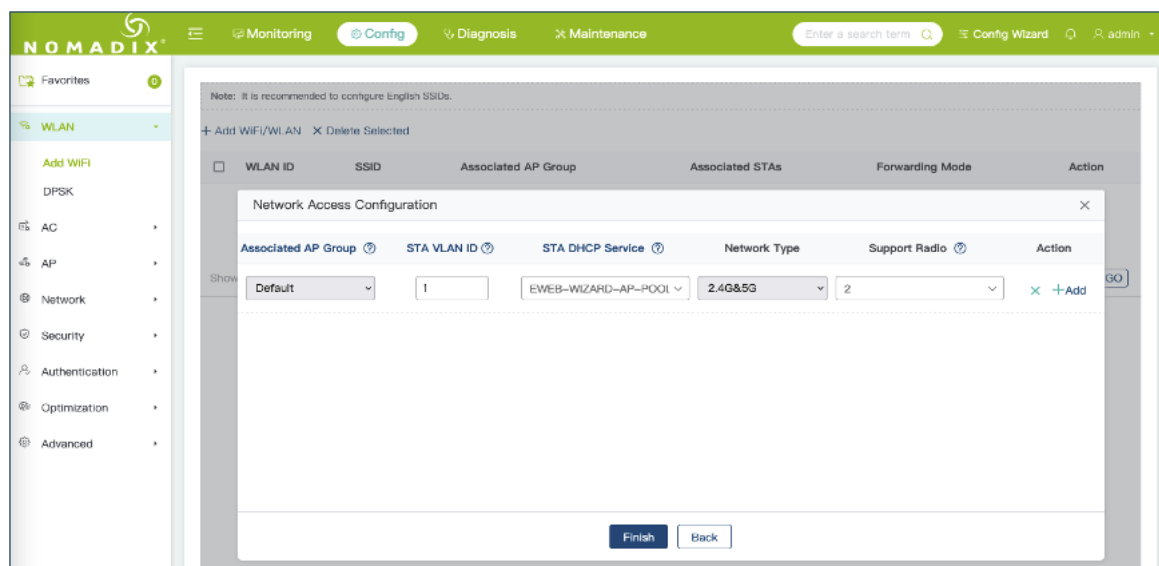
- Click on **Add WiFi/WLAN** in the content area:



- Enter the required information, being sure to select **WPA2-802.1X** for encryption type. Configure any advanced settings that may be required.

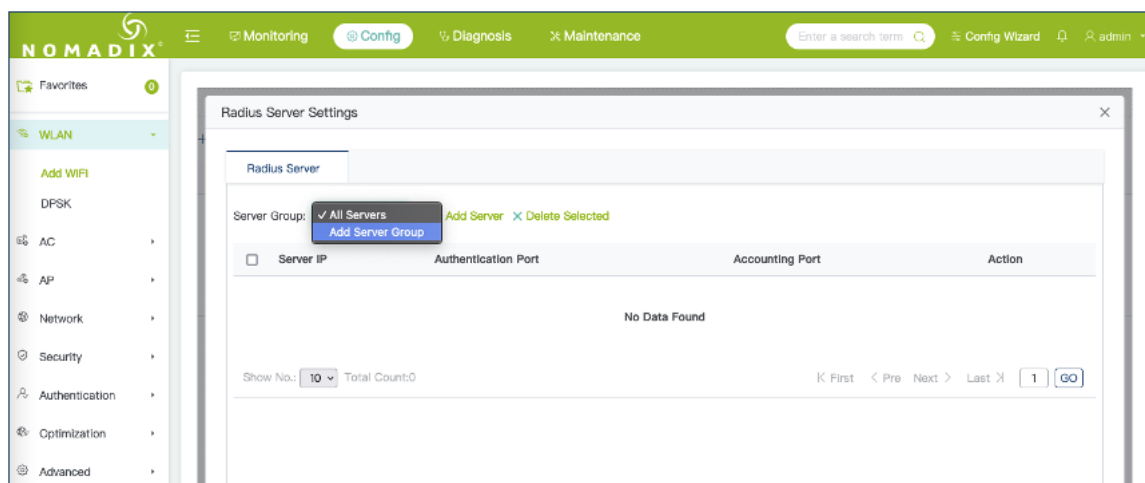


- Click **Next** and specify the AP Group and other information to apply the new WLAN.
- Click **Finish**.

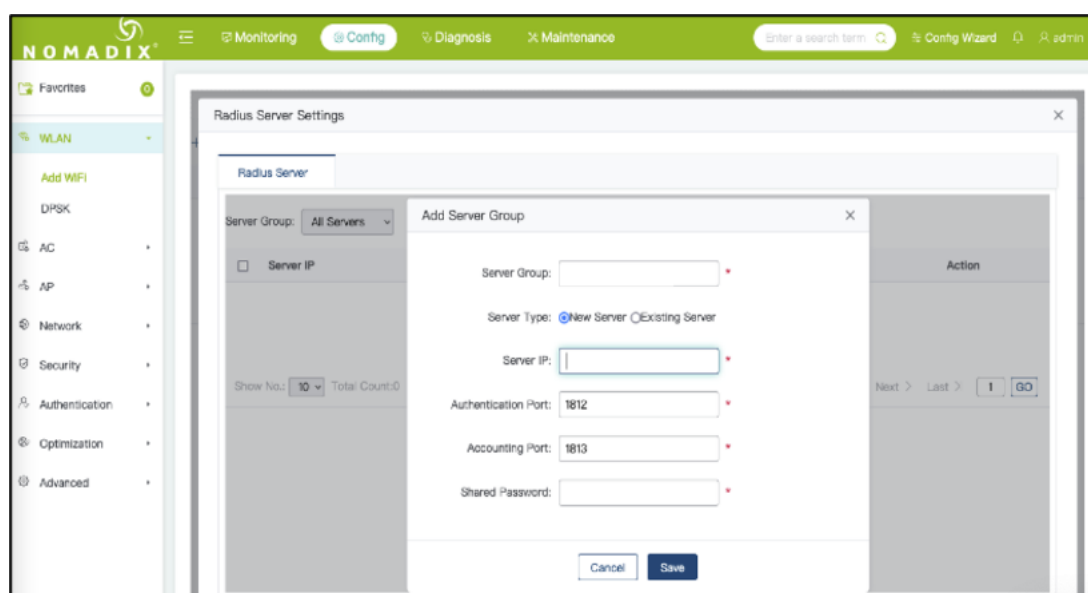


RADIUS

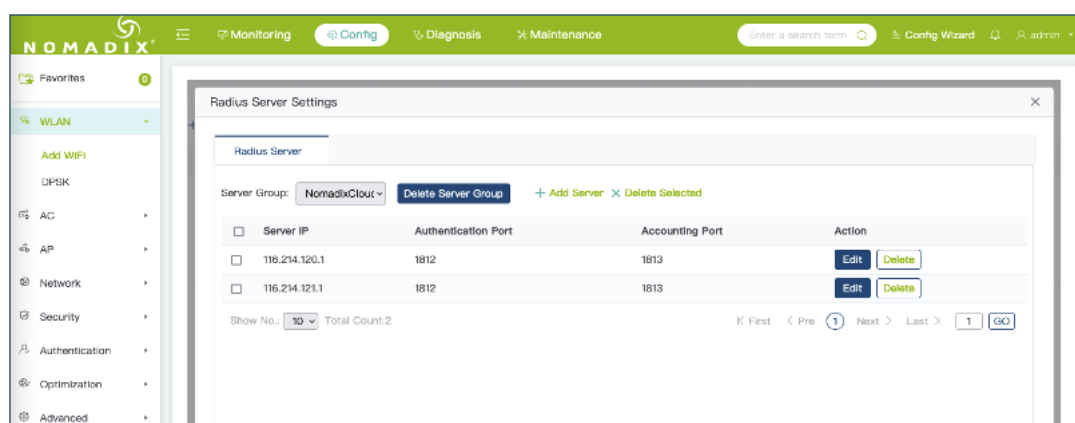
- Click on **Radius Server Settings** and select **Add Server Group** from the drop-down list:



- Create a group called *TrustedWiFi* and enter the information for the primary RADIUS server.



- Click **Save**, then click **Add Server** and enter details for the secondary RADIUS server and click **Save** again.
- Primary and Secondary RADIUS servers should have been configured as shown below:



- Click the **X** in the top right-hand corner to close the RADIUS server configuration panel. The WiFi/WLAN configuration reappears.
- Select *TrustedWiFi* for the **Authentication Server** and **Accounting Server** fields:

Note: It is recommended to configure English SSIDs.

+ Add WIFI/WLAN X Delete Selected

WLAN ID	SSID	Associated AP Group	Associated STAs	Forwarding Mode	Action
WiFi/WLAN Configuration					
WLAN ID: 1	SSID: Demo SSID				
Encryption Type: WPA2-802.1X					
Authentication Server: TrustedWiFi [Radius Server Settings]					
Accounting Server: TrustedWiFi					
Advanced Settings					
Next					

- Click **Next**. Specify the AP group and other information to apply the new WLAN if not already completed and click **Finish**.

Passpoint (Hotspot 2.0)

- Expand the **AP** section in the left-hand navigation menu and select **Hotspot2.0**:

Note: Hotspot 2.0 is a technical standard developed by Wi-Fi Alliance. This standard allows wireless STAs to complete automatic identity identification and seamless switching in a WLAN via 802.11u without using an additional identity, thereby providing wireless users with the access and roaming experience similar to that of cellular network users.

Template Name:

Apply to SSID: [WIFI/WLAN Settings]

[Advanced Settings](#)

Save

- Enter a template name such as Passpoint and select the **SSID** that was created in the previous steps from the drop-down list:

The screenshot shows the Nomadix configuration interface. The top navigation bar includes 'Monitoring', 'Config', 'Diagnosis', and 'Maintenance'. The left sidebar shows the 'AP' (Access Point) section selected. The main content area displays the 'Template-1' configuration for 'Passpoint'. The 'Apply to SSID' is set to 'Demo SSID'. The 'Advanced Settings' section is expanded, showing a 'Save' button.

- Expand the **Advanced Settings** and scroll down to the **Cellular** section. Complete the settings as per the description below:

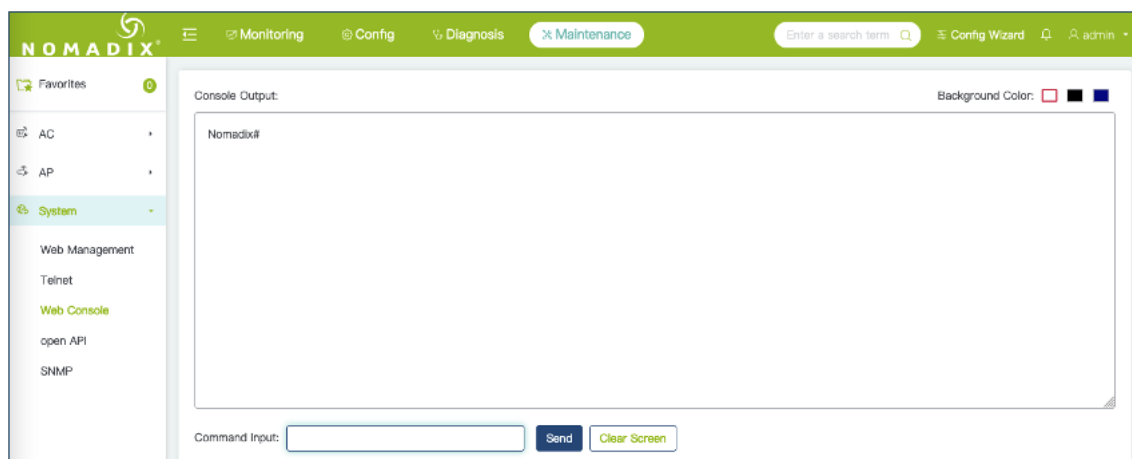
FieldDescriptionEnable this DHCP poolSelect EnableNAI DomainEnter the [customer domain]Auth TypeSelect EAP-TTLS from the drop down listAuth MethodSelect Auth-Method 2 (non-EAP Inner Authentication)Auth ParamSelect Auth-Param 4 (MSCHAPv2)

The screenshot shows the Nomadix configuration interface for the 'Cellular' section. The left sidebar shows the navigation menu with 'Cellular' selected. The main content area displays the 'Cellular' settings. The 'Domain Name' is set to 'customer.cloud.global'. The 'Auth Type' is 'eap-ttls', 'Auth Method' is '2', and 'Auth Param' is '4'. The 'Cellular Network' section shows 'MCC' and 'MNC' fields. The 'Roaming Consortium' section shows 'Organization Identifier' and 'In-beacon' fields. The 'Other' section shows 'Type' and 'Chinese Name' fields.

- Scroll down to the end of the settings and click **Save**.

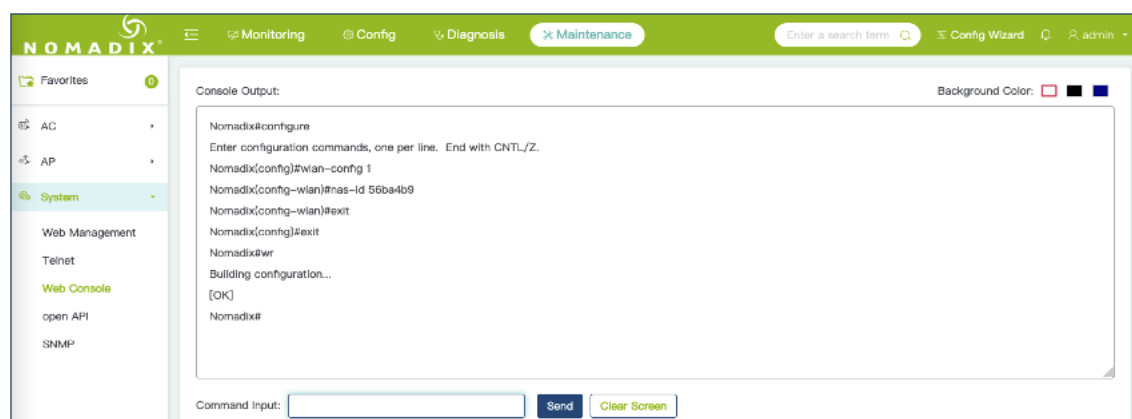
NAS ID

- Click on the **Maintenance** tab at in the top menu bar, expand the **System** section in the left-hand navigation menu and select **Web Console**:



- Using the command input box, send the sequence of commands shown below to configure the NAS-Identifier:

Commands:configure
wlan-config 1
nas-id <nas-identifier from TrustedWiFi settings>
exit
exit
wr



Hardware Configuration Guide for Ruckus One

Download this guide 

INTRODUCTION

Scope and Purpose

Thank you for purchasing the Trusted WiFi solution. This document is a hardware configuration guide describing how to setup Ruckus One for a Trusted WiFi Passpoint service.

- For more information on how to setup an end-to-end Trusted WiFi Passpoint service, please refer to the Trusted WiFi Passpoint Administration Guide.
- For complete information on how to setup Ruckus One, please refer to the vendor's original documentation.

Documentation Conventions

{{snippet.Documentation Conventions}}

Passpoint Overview

Passpoint – also known as Hotspot 2.0 – is an industry-wide next generation approach to public internet access driven by the Wi-Fi Alliance that brings the following benefits:

- Frictionless onboarding and roaming, thanks to a one-time registration followed by automatic access to interconnected hotspots
- More secure and private Wi-Fi connections, compared with general visitor networks

Passpoint is based on the IEEE 802.11u standard, which is a set of protocols enabling cellular-like roaming. Following the initial enrollment, frequent users such as visitors, guests or employees bypass repeated logins, forms and passwords, as their mobile devices automatically join the Wi-Fi subscriber service when they return to a venue or roam between inter-linked Passpoint enabled hotspots and providers, while being better protected against potential cyber threats.

If a device supports 802.11u and is enrolled to a service, it automatically communicates with the Wi-Fi infrastructure via the access points to discover the network SSID and connects securely to it by presenting its access credentials. Upon successful authentication, the device is provisioned with Passpoint standards-based management objects - known as Per-Provider Subscription Management Objects (PPS-MO).

GlobalReach - an ASSA ABLOY company - has been involved with Passpoint since its inception and even contributed to the creation and initial pilot testing of the standard. As a result, it is one of the few trusted worldwide experts on this topic, with a proven platform backed by real-world operational experiences at scale.

The best user experience is to offer Passpoint through a customer/brand mobile app integration, as it further simplifies the onboarding process, while incentivizing app downloads and customer loyalty, leading to further engagement and monetization opportunities. To this effect, Trusted WiFi offers a Software Development Kit (SDK) for easy app integration.

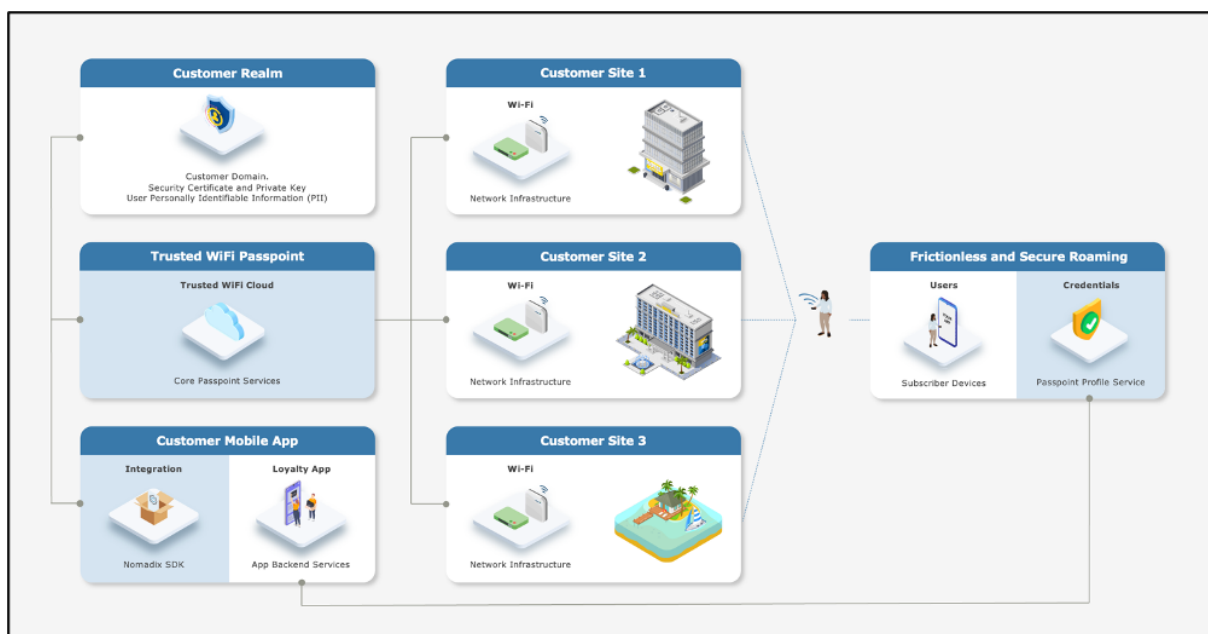
End-to-End Service Components

Implementing an end-to-end Trusted WiFi Passpoint service requires a combination of the following software and hardware components:

- **Trusted WiFi Passpoint:** the core services are offered and managed centrally via the **Trusted WiFi Passpoint Module** (hosted and operated by GlobalReach). Following an initial setup performed by the GlobalReach Operations team, Managed Service Providers (MSPs) can then add sites to customer realms and monitor the solution.
- **Customer Realm:** the service requires connectivity to a realm including domain, security certificate and private key, as well as the database holding the users' Personally Identifiable Information (PII). It is the responsibility of the Identity Provider – typically the customer/brand – to make this part available.
- **Mobile App:** the best user experience is to offer Passpoint through a customer/brand mobile app integration. Trusted WiFi offers a **Software Development Kit (SDK)** for easy implementation. It is the responsibility of the app owner – typically the customer/brand – to perform the integration.
- **Local Networks:** the local networks must be compliant with Passpoint (Hotspot 2.0). – Most recent Wi-Fi access points and controllers from major vendors support Passpoint today, however older models or products from more exotic manufacturers might not do. The configuration of the local infrastructure is typically handled by the Managed Service Providers (MSPs).
- **User Devices:** the subscribers' mobile devices (belonging to visitors, guests, employees, etc.) must be compliant with Passpoint (Hotspot 2.0) – All recent iOS and Android-based smartphones or tablets support Passpoint today, however older models or products running less popular operating systems might not do. Laptops compatibility is also more erratic. It is therefore essential to maintain a traditional onboarding service in parallel with Passpoint to handle non-compatible devices.

High-Level Topology

The diagram below illustrates the high-level topology for the end-to-end Trusted WiFi Passpoint service:



Glossary

The following is a glossary of the most common terms used regarding this solution.

Term	Abbreviation	Description
Deployment	-	Enabling of a Trusted WiFi product module for a property via the Trusted WiFi interface.
Module	-	Product or service purchased from Trusted WiFi that is managed through its own sub-section via the Trusted WiFi interface.
License	-	Legal agreement that grants users the right to use specific software, outlining terms and conditions for its usage, distribution, and potential modifications, while protecting the intellectual property of the software developer. GlobalReach licenses comprise of two different types: one-off licenses - typically to initially enable a software module. recurring licenses - typically including software updates and technical support, or more in the case of OpEx consumption commercial models based on price per month/quarter/year. Trusted WiFi is sold as a combination of one-off licenses to activate the service and recurring licenses based on a price per Wi-Fi access point per month.
Managed Service Provider	MSP	The third-party company that remotely manages and monitors a client's IT infrastructure and end-user systems, offering services like network and infrastructure management, security, and 24/7 technical support.
Organization	Org	A company account in Trusted WiFi.
Operator	-	An organization type account in Trusted WiFi that is used by MSPs to manage a property's Wi-Fi network.
Customer	-	An organization type account in Trusted WiFi typically used for customers/brands that allows grouping to view all properties belonging to the same company even if managed by several different MSPs.
Linked Organization	Linked Org	A link creating a relationship between an operator and a customer account, allowing a customer to view a property while allowing an operator to manage it.
User	-	An individual accessing a product, service or system. In the context of Trusted WiFi, a user is setup to access products and deployments information at Admin/Editor/Viewer levels, according to their relevant role permissions. In the context of a given product or service, a user is another term referring to a subscriber: the person at the end of the chain interacting with that product or service – usually through a device.
Property	-	Trusted WiFi concept representing an Individual site or location where products are deployed.
Linked Property	-	Site or location shared between operator and customer accounts.

Passpoint Software Development Kit	Passpoint SDK	The service that sits within the customer's mobile app and that is connected to the Trusted WiFi RADIUS infrastructure allowing Passpoint profiles to be created for a given Passpoint realm.
Passpoint Realm	-	The customer specific domain that is used to provision Passpoint profiles that are approved for connection to any associated network.
Passpoint Profile	-	The security certified profile that sits on a subscriber's device. If installed correctly it allows seamless authentication to the secure Passpoint SSID.
Secure Passpoint Service Set Identifier	Secure Passpoint SSID	The Wi-Fi network that is associated to the Passpoint realm configured to allow subscriber devices with a valid Passpoint profile to seamlessly connect to the Passpoint network at a property.
Subscriber	-	An individual person using a service.
Subscriber Device	-	The equipment – typically a smartphone, tablet or computer – the subscriber is using to connect to the service.
Collision-Resistant Unique Identifier	CUID	A unique identifier designed to be collision-resistant – meaning engineered to minimize the likelihood of generating duplicate IDs even in distributed systems – and more efficient in terms of space and database indexing performance due to its sequential nature. In the context of Passpoint, a CUID is delivered to a subscriber device when it requests a Passpoint profile.
Customer Loyalty Mobile Application	Mobile App	The iOS and/or Android digital application used by businesses to engage and reward their customers through loyalty programs. In the context of Passpoint, the best user experience is to offer Passpoint through a customer/brand mobile app integration using the Trusted WiFi SDK.

RUCKUS ONE CONFIGURATION

Supported Versions

This document is based on the Enterprise tier of Ruckus One. Other Ruckus One tiers – such as Hospitality – are supported, but the UI may differ.

Prerequisites

It is assumed the following prerequisites are met before configuring Ruckus One for a Trusted WiFi Passpoint service:

- A supported Ruckus One account is activated.
- A Ruckus One supported AP is running and licensed.
- A Trusted WiFi account with operator permissions.
- A core Trusted WiFi Passpoint service configured and tested.
- A property in Trusted WiFi with deployed Passpoint modules.

- A deployed and configured Wi-Fi network.

Notes

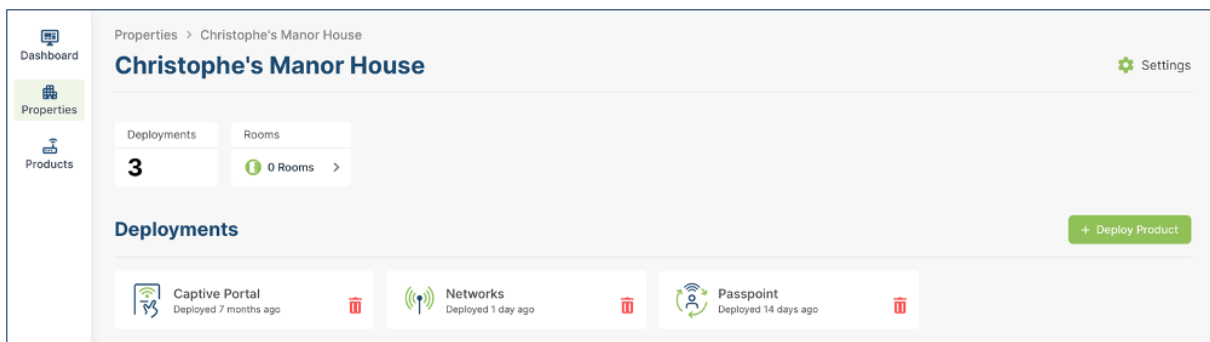
This document focuses on a specific part of the Ruckus One configuration only. Please refer to the Passpoint Administration Guide for instructions on how to configure the end-to-end Passpoint service. Please refer to the original Ruckus documentation for complete instructions on Ruckus One configuration.

Warning

The Ruckus configuration is focusing on Passpoint v1.0. All properties must share the same NAI realm, RADIUS IP / port settings and SSID name for the Passpoint service (CustomerPasspoint). Each property requires a separate RADIUS secret and NAS identifier, both generated when a configuration is activated within the Trusted WiFi management platform.

Core Passpoint Settings

- Log into Trusted WiFi, then click on the **Properties** icon in the left menu to view the properties list.
- Select or search for the property you wish to work on. This will open its deployments page:



- Click on the **Passpoint** tile and then click on the Configuration option in the left menu to display the Passpoint settings summary as per the example below:

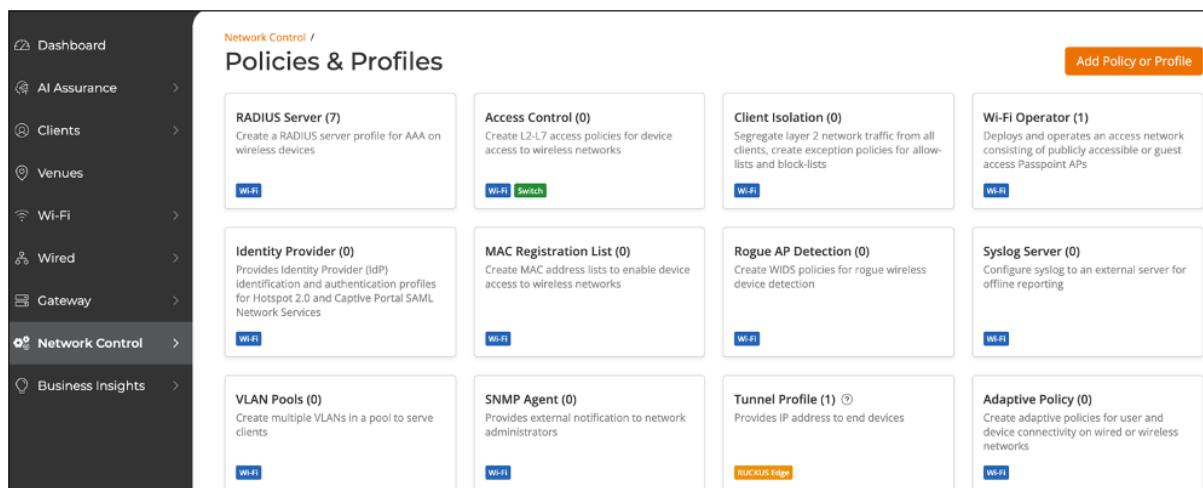
Configuration	
NAI Realm	customer.cloud.global
Friendly Name	customer.cloud.global
Primary RADIUS IP	116.214.120.1
Secondary RADIUS IP	116.214.121.1
RADIUS Authentication Port	1812
RADIUS Accounting Port	1813
NAS Identifier	1ft386a9
RADIUS Secret	5dc7b90b296789h08a6ef764b1e3d

- Take note of the details for your respective installation as they will be required at a later step.

Ruckus One Configuration

RADIUS Profile

- Log into the Ruckus One application using your login credential.
- Once logged in, proceed to the Dashboard screen by clicking on the **Manage my Account** button on the right menu.
- On the left-hand menu items click on **Network Control > Policies & Profiles** to display the page below:



- Click on the **RADIUS Server** option to display the page below:



- Click on **Add RADIUS Server** in the top right corner, to display the page below:

Network Control / Policies & Profiles / RADIUS Server /

Add RADIUS Server

Settings

Profile Name *

Type

☒ Authentication RADIUS Server

☐ Accounting RADIUS Server

Primary Server

IP Address * Port *

Shared Secret *

Add Secondary Server

Add
Cancel
Microsoft P

- Click **Add Secondary Sever** text and complete the fields as per Trusted WiFi RADIUS settings:
 FieldDescriptionProfile NameEnter the name of the profileTypeSelect Authentication RADIUS
 ServerPrimary ServerIP addressEnter the <Primary RADIUS IP> from Trusted WiFi.PortEnter
 the < RADIUS Authentication Port> from Trusted WiFi.Shared SecretEnter the < RADIUS
 Shared Secret> from Trusted WiFi.Secondary ServerIP addressEnter the <Secondary RADIUS
 IP> from Trusted WiFi.PortEnter the < RADIUS Authentication Port> from Trusted WiFi.Shared
 SecretEnter the < RADIUS Shared Secret> from Trusted WiFi.

Settings

Profile Name *

 ✓

Type

☒ Authentication RADIUS Server

☐ Accounting RADIUS Server

Primary Server

IP Address * Port *

Shared Secret *

Remove Secondary Server

Secondary Server

IP Address * Port *

Shared Secret *

- Click **Add** and the server will appear under your RADIUS Sever list.

- Click **Add RADIUS Server** again and repeat the same steps as above but select type as **Accounting RADIUS Server** and fill in the RADIUS Accounting information as per Trusted WiFi RADIUS settings.

Settings

Profile Name *

RADIUSPasspointAcct ✓

Type

☐ Authentication RADIUS Server

☒ Accounting RADIUS Server

Primary Server

IP Address * Port *

116.214.1 1813

Shared Secret *

[Remove Secondary Server](#)

Secondary Server

IP Address * Port *

116.214.1 1813

Shared Secret *

- Click **Add** and the server will appear under your RADIUS Sever list.

Passpoint (Hotspot 2.0)

- On the left-hand menu, click **Venues**.
- Select the desired **Venue** and select **Edit** as shown below.
- On the left-hand menu, click **Wi-Fi > Wi-Fi Networks List** to display the page below:

Wi-Fi /

Wi-Fi Networks

[Add Wi-Fi Network](#)

Network List (4) WLANs Report Applications Report Wireless Report

Search Name Type

Name ^	Description	Type	Venues	APs	Clients	VLAN	Security Protocol
RuckusCloudCP	Captive Portal - Clie...		0	0	0	VLAN-1	Open Captive Portal

- Click **Add Wi-Fi Network** to display the page below:
- Enter a **Network Name**.
- Under Network Type select **Hotspot 2.0 Access**.
- Select **Next**. The following page will appear:

Create New Network

Hotspot 2.0 Settings

Network Details
Hotspot 2.0 Settings
 Venues
 Summary

Security Protocol
 WPA2 (Recommended)

WPA2 is strong Wi-Fi security that is widely available on all mobile devices manufactured after 2006. WPA2 should be selected unless you have a specific reason to choose otherwise.
 6GHz radios are only supported with WPA3.

Wi-Fi Operator *
 Select... Add

Identity Provider *
 Select... Add

Select up to 6

Identity Group
 Select... View Details | Add

Show more settings

Cancel Back Next

- Security Protocol should be left as **WPA2 (Recommended)**.
- Under Wi-Fi Operator, click **Add**. The following pop up will display:

Wi-Fi / Wi-Fi Networks / Network List /

Create New Network

Hotspot 2.0 Settings

Network Details
Hotspot 2.0 Settings
 Venues
 Summary

Security Protocol
 WPA2 (Recommended)

WPA2 is strong Wi-Fi security that is widely available on all mobile devices manufactured after 2006. WPA2 should be selected unless you have a specific reason to choose otherwise.
 6GHz radios are only supported with WPA3.

Wi-Fi Operator *
 Select... Add

Identity Provider *
 Select... Add

Select up to 6

Identity Group
 Select... View Details | Add

Show more settings

Cancel Back Next

Add Wi-Fi Operator

Profile Name *

Domain *

One domain per line

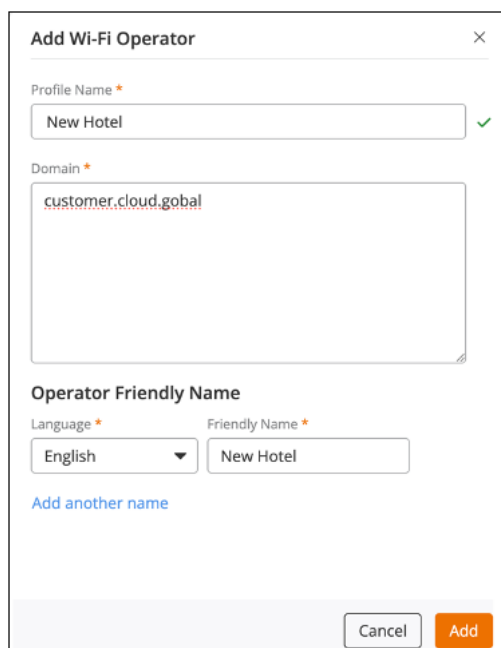
Operator Friendly Name

Language * Friendly Name *

Select... Add another name

Cancel Add

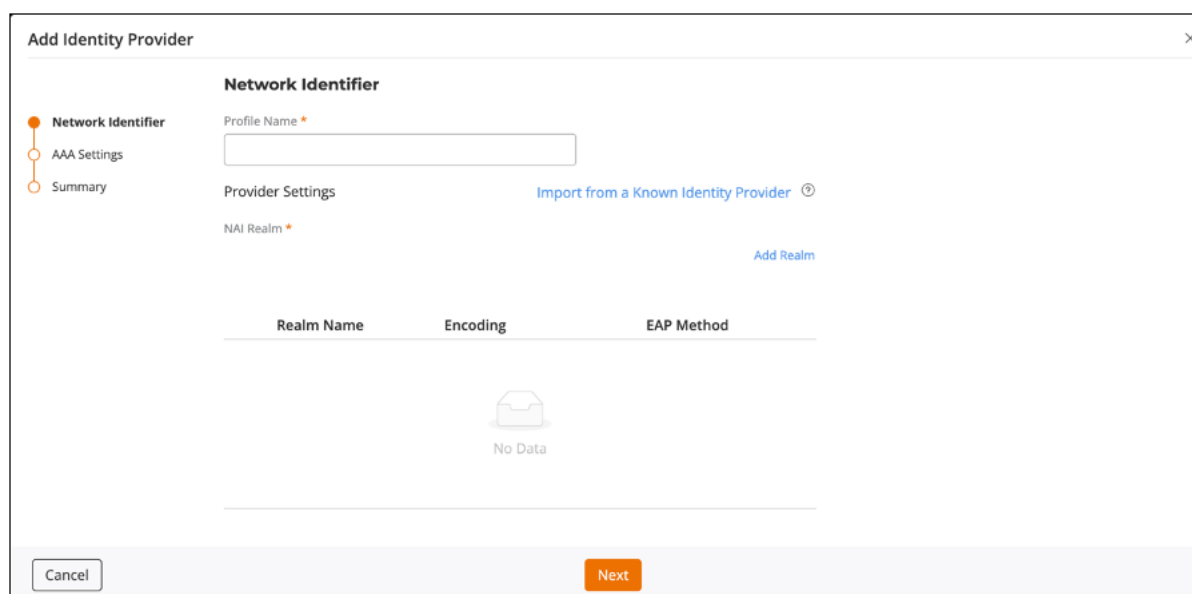
- Complete the fields as per the description below and click **Add** to save the Operator configuration:
 FieldDescriptionNameEnter the Profile Name.Domain NameEnter the Domain Name as per set up e.g. customer.cloud.globalLanguageSelect the appropriate language from the pull-down list and add a Friendly Name. Click on the +Add to save.



The 'Add Wi-Fi Operator' dialog box contains the following fields and controls:

- Profile Name ***: A text input field containing 'New Hotel' with a green checkmark to its right.
- Domain ***: A text input field containing 'customer.cloud.gobal'.
- Operator Friendly Name**:
 - Language ***: A dropdown menu showing 'English'.
 - Friendly Name ***: A text input field containing 'New Hotel'.
 - A link labeled 'Add another name' in blue text.
- At the bottom right are 'Cancel' and 'Add' buttons.

- On the same page, under Identify Provider, click **Add**. The follow pop up will appear:



The 'Add Identity Provider' dialog box features a sidebar with 'Network Identifier', 'AAA Settings', and 'Summary'. The main area is titled 'Network Identifier' and includes:

- Profile Name ***: An empty text input field.
- Provider Settings**: A section with a link 'Import from a Known Identity Provider ⓘ'.
- NAI Realm ***: A section with a link 'Add Realm'.
- A table with headers 'Realm Name', 'Encoding', and 'EAP Method'. Below the headers is a 'No Data' message with a folder icon.
- At the bottom are 'Cancel' and 'Next' buttons.

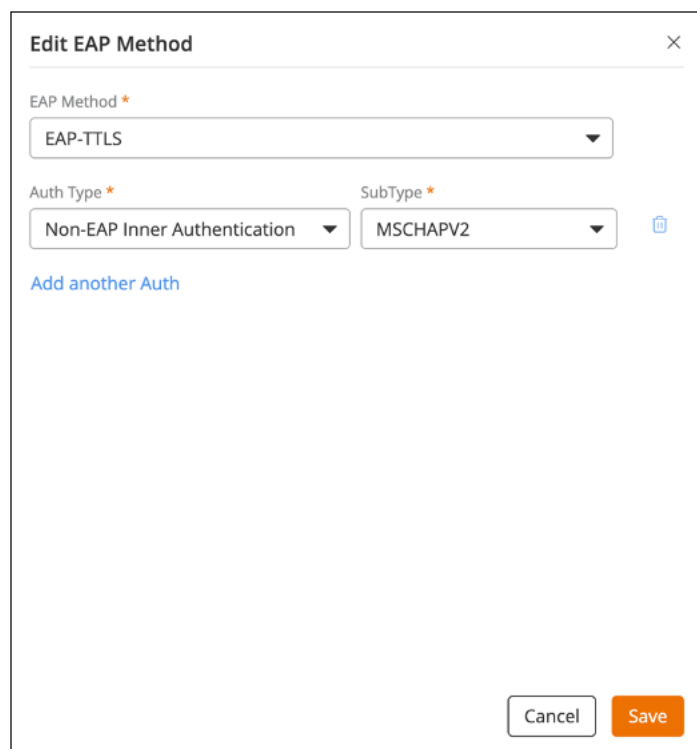
- Provide the name of the Identify provider.
- Click the **Add Realm** text, the following pop up will appear:

The screenshot displays the 'Add Realm' dialog box on the right and the 'Network Identifier' settings on the left. The 'Add Realm' dialog has a close button (X) in the top right corner. It contains a 'Realm Name' text field with an asterisk, an 'Encoding' dropdown menu set to 'RFC-4282', and an 'Add EAP Method' link. Below these is a table with columns 'EAP Method' and 'Auth', showing 'No Data'. At the bottom, there is a checkbox for 'Add another Realm', a 'Cancel' button, and an 'Add' button. The 'Network Identifier' settings on the left include a 'Profile Name' dropdown set to 'Identity Provider' with a green checkmark, a 'Provider Settings' link, and an 'NAI Realm' text field. A table below shows columns for 'Realm Name', 'Encoding', and 'EAP Method', also showing 'No Data'. A red error message at the bottom states 'NAI Realm list must contain at least one entry'. Navigation buttons 'Cancel', 'Next', and 'Add' are visible at the bottom of the interface.

- Complete the fields as per the description below and click **Add** to save the Realm configuration:
FieldDescriptionRealm NameEnter the name of realm as per settings. e.g. customer.cloud.globalEncodingSelect RFC-4282 from the drop- down.EAP MethodsClick on Add EAP Method.Select EAP-TTLS from the drop down.
- Click **Add** and select the **Add another Auth** as shown below:

The 'Edit EAP Method' dialog box has a close button (X) in the top right corner. It contains an 'EAP Method' dropdown menu set to 'EAP-TTLS' and an 'Add another Auth' link.

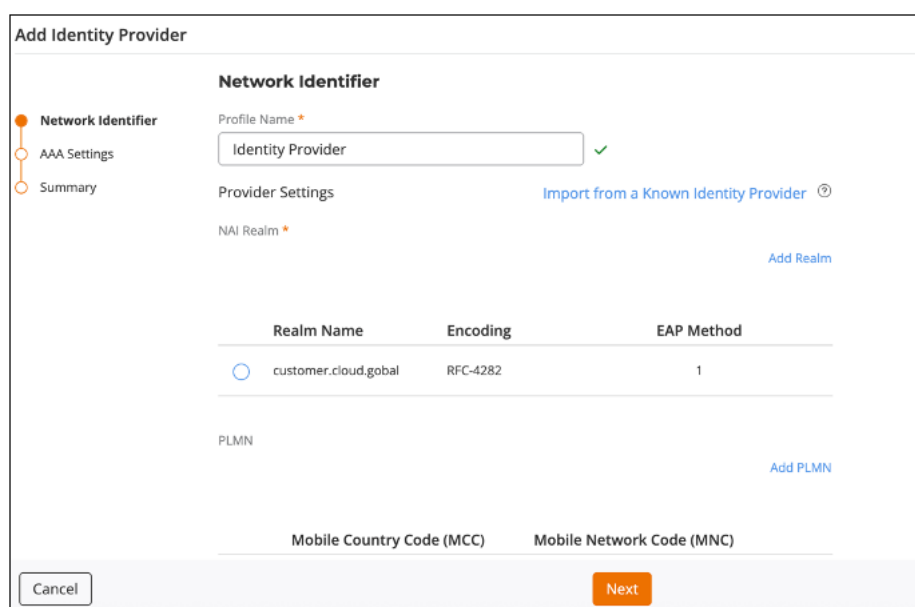
- Under Auth type, select **Non-EAP Inner Authentication**.
- Under SubType, select **MSCHAPV2**.



The 'Edit EAP Method' dialog box contains the following fields and controls:

- EAP Method ***: A dropdown menu with 'EAP-TTLS' selected.
- Auth Type ***: A dropdown menu with 'Non-EAP Inner Authentication' selected.
- SubType ***: A dropdown menu with 'MSCHAPV2' selected.
- Add another Auth**: A blue link.
- Buttons**: 'Cancel' and 'Save' buttons at the bottom right.

- Click **Save**.
- Click **Add** to save this Realm. The following page will display:



The 'Add Identity Provider' screen features a sidebar with 'Network Identifier', 'AAA Settings', and 'Summary'. The main content area is titled 'Network Identifier' and includes:

- Profile Name ***: A text field with 'Identity Provider' and a green checkmark.
- Provider Settings**: A section with a blue link 'Import from a Known Identity Provider ⓘ'.
- NAI Realm ***: A text field with a blue link 'Add Realm'.
- Table**: A table with columns 'Realm Name', 'Encoding', and 'EAP Method'.

Realm Name	Encoding	EAP Method
customer.cloud.gobal	RFC-4282	1
- PLMN**: A section with a blue link 'Add PLMN'.
- Mobile Country Code (MCC)** and **Mobile Network Code (MNC)**: Text fields.
- Buttons**: 'Cancel' and 'Next' buttons at the bottom.

- Once the Realm is added, click **Next** to display the **AAA Settings** tab, as shown below:

The screenshot shows the 'Add Identity Provider' configuration page with the 'AAA Settings' tab selected. On the left, a sidebar contains three items: 'Network Identifier', 'AAA Settings' (which is highlighted with an orange dot), and 'Summary'. The main content area is titled 'AAA Settings' and contains two sections. The 'Authentication Service' section has a dropdown menu for 'Authentication Server' currently set to 'Select RADIUS', with a blue 'Add Server' link to its right. The 'Accounting Service' section has a toggle switch that is currently turned off. At the bottom of the page, there are three buttons: 'Cancel', 'Back', and 'Next'.

- Select the **Authentication Service** you configured in the previous steps.
- Enable the **Accounting Service** toggle.
- Select the **Accounting Service** you configured in the previous steps.

This screenshot shows the same 'Add Identity Provider' configuration page, but with more details visible. The 'Authentication Server' dropdown is now set to 'RADIUSPasspointAuth', and the 'Accounting Service' toggle is now turned on. Below the 'Authentication Service' section, there are fields for 'Primary Server' (116.214.121) and 'Shared Secret' (masked with dots). Below the 'Accounting Service' section, there are fields for 'Primary Server' (116.214.121) and 'Shared Secret' (masked with dots). The 'Add Server' links are still present. The 'Cancel', 'Back', and 'Next' buttons remain at the bottom.

- Click **Next** to display the **Summary** tab.
- Once reviewed click **Add**. The following page will display again:

Wi-Fi / Wi-Fi Networks / Network List /

Create New Network

Network Details

Hotspot 2.0 Settings

Venues

Summary

Hotspot 2.0 Settings

Security Protocol

WPA2 (Recommended) ▼

WPA2 is strong Wi-Fi security that is widely available on all mobile devices manufactured after 2006. WPA2 should be selected unless you have a specific reason to choose otherwise. ⓘ 6GHz radios are only supported with WPA3.

Wi-Fi Operator *

New Hotel ▼ Add

Identity Provider *

Identity Provider × Add

Select up to 6

Identity Group

Select... ▼ View Details | Add

[Show more settings](#)

Cancel

Back

Next

- Click **Show more settings**, then click on the **Networking** tab as shown below.

[Show less settings](#)

VLAN

Hotspot 2.0

Network Control

Radio

Networking

Advanced

Enable Agile Multiband (AMB) ⓘ ☐

Enable 802.11k neighbor reports ☒

- Scroll down **RADIUS Options** and under **NAS ID** select **User-defined**.
- Enter the **< NAS ID >** from Trusted WiFi as shown below:

RADIUS Options

NAS ID ⓘ

Custom NAS ID *

NAS Request Timeout * Seconds
 NAS Max Retries * Retries
 NAS Reconnect Primary * Minutes

Called Station ID

Single Session ID Accounting ⓘ ☐

[Back](#) [Next](#)

- Click **Next**.
- Under **Venues**, select the desired venue and click **Activate** as shown below:

Create New Network

Venues
 Select venues to activate this network

1 selected [Activate](#) [Deactivate](#)

Venue	City	Country	Networks	Wi-Fi APs	Activated	APs	Radi
☐ GRT-HO	London	United Kingdom	1	0	☐		
☒ Tralee Lab	Tralee, County K...	Ireland	1	1	☐		

[Cancel](#) [Back](#) [Next](#)

- One **Activate** click **Next** to display the summary page.
- Once reviewed click **Add** to save this new Network. The following page will display with the network added:

Wi-Fi / Wi-Fi Networks

[Add Wi-Fi Network](#)

Network List (5) [WLANs Report](#) [Applications Report](#) [Wireless Report](#)

Q Search Name Type

Name ^	Description	Type	Venues	APs	Clients	VLAN	Security Proto...
☐ Passpoint	Hotspot 2.0 Access		1	1	0	VLAN-1	WPA2 Enterprise

Administration Guide

Download this guide 

Introduction

Scope and Purpose

Thank you for purchasing the Trusted WiFi solution. The aim of this document is to cover the configuration, monitoring and troubleshooting of an end-to-end Trusted WiFi Passpoint service, with a focus on the core services delivered by Trusted WiFi.

Separate complementary documents are available, covering a selection of network hardware equipment configuration, as well as integration with the Trusted WiFi Passpoint Software Development Kit (SDK).

Documentation Conventions

{{snippet.Documentation Conventions}}

Passpoint Overview

Passpoint – also known as Hotspot 2.0 – is an industry-wide next generation approach to public internet access driven by the Wi-Fi Alliance that brings the following benefits:

- Frictionless onboarding and roaming, thanks to a one-time registration followed by automatic access to interconnected hotspots
- More secure and private Wi-Fi connections, compared with general visitor networks

Passpoint is based on the IEEE 802.11u standard, which is a set of protocols enabling cellular-like roaming. Following the initial enrollment, frequent users such as visitors, guests or employees bypass repeated logins, forms and passwords, as their mobile devices automatically join the Wi-Fi subscriber service when they return to a venue or roam between inter-linked Passpoint enabled hotspots and providers, while being better protected against potential cyber threats.

If a device supports 802.11u and is enrolled to a service, it automatically communicates with the Wi-Fi infrastructure via the access points to discover the network SSID and connects securely to it by presenting its access credentials. Upon successful authentication, the device is provisioned with Passpoint standards-based management objects - known as Per-Provider Subscription Management Objects (PPS-MO).

GlobalReach - an ASSA ABLOY company - has been involved with Passpoint since its inception and even contributed to the creation and initial pilot testing of the standard. As a result, it is one of the few trusted worldwide experts on this topic, with a proven platform backed by real-world operational experiences at scale.

The best user experience is to offer Passpoint through a customer/brand mobile app integration, as it further simplifies the onboarding process, while incentivizing app downloads and customer loyalty, leading to further engagement and monetization opportunities. To this effect, Trusted WiFi offers a Software Development Kit (SDK) for easy app integration.

Seamless Onboarding and Roaming

Best user experience via brand mobile loyalty app integration:

- One-time setup in 3 "taps" via a mobile app
- Followed by automatic connection for returning visitors and roaming within any of the participating venues
- No manual SSID selection, no login/password, no reset after a given period
- No impact from MAC address randomization
- Incentive to download the app, leading to visitor engagement opportunities
- Non-compatible devices can still use the legacy onboarding process

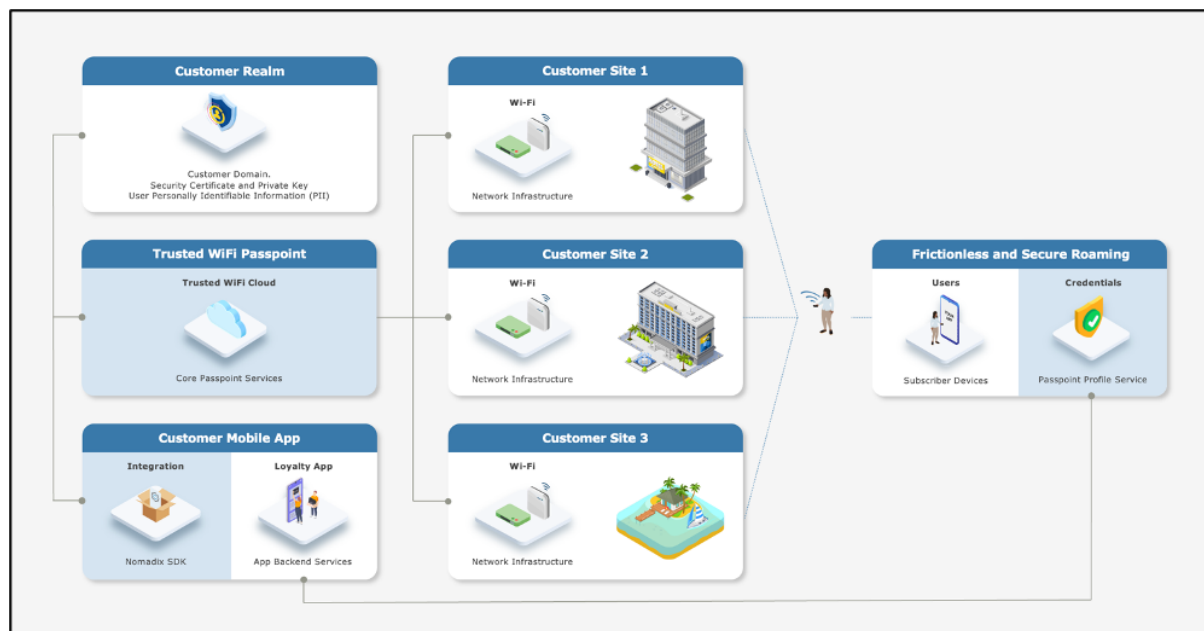
End-to-End Service Components

Implementing an end-to-end Trusted WiFi Passpoint service requires a combination of the following software and hardware components:

- **Trusted WiFi Passpoint:** the core services are offered and managed centrally via the Trusted WiFi Passpoint Module (hosted and operated by GlobalReach). Following an initial setup performed by the GlobalReach Operations team, Managed Service Providers (MSPs) can then add sites to customer realms and monitor the solution.
- **Customer Realm:** the service requires connectivity to a realm including domain, security certificate and private key, as well as the database holding the users' Personally Identifiable Information (PII). It is the responsibility of the Identity Provider – typically the customer/brand – to make this part available.
- **Mobile App:** the best user experience is to offer Passpoint through a customer/brand mobile app integration. Trusted WiFi offers a **Software Development Kit (SDK)** for easy implementation. It is the responsibility of the app owner – typically the customer/brand – to perform the integration.
- **Local Networks:** the local networks must be compliant with Passpoint (Hotspot 2.0). – Most recent Wi-Fi access points and controllers from major vendors support Passpoint today, however older models or products from more exotic manufacturers might not do. The configuration of the local infrastructure is typically handled by the Managed Service Providers (MSPs).
- **User Devices:** the subscribers' mobile devices (belonging to visitors, guests, employees, etc.) must be compliant with Passpoint (Hotspot 2.0) – All recent iOS and Android-based smartphones or tablets support Passpoint today, however older models or products running less popular operating systems might not do. Laptops compatibility is also more erratic. It is therefore essential to maintain a traditional onboarding service in parallel with Passpoint to handle non-compatible devices.

High-Level Topology

The diagram below illustrates the high-level topology for the end-to-end Trusted WiFi Passpoint service:



Glossary

The following is a glossary of the most common terms used regarding this solution.

Term	Abbreviation	Description
Deployment	-	Enabling of a Trusted WiFi product module for a property via the Trusted WiFi interface.
Module	-	Product or service purchased from Trusted WiFi that is managed through its own sub-section via the Trusted WiFi interface.
License	-	Legal agreement that grants users the right to use specific software, outlining terms and conditions for its usage, distribution, and potential modifications, while protecting the intellectual property of the software developer. GlobalReach licenses comprise of two different types: one-off licenses - typically to initially enable a software module. recurring licenses - typically including software updates and technical support, or more in the case of OpEx consumption commercial models based on price per month/quarter/year. Trusted WiFi is sold as a combination of one-off licenses to activate the service and recurring licenses based on a price per Wi-Fi access point per month.
Managed Service Provider	MSP	The third-party company that remotely manages and monitors a client's IT infrastructure and end-user systems, offering services like network and infrastructure management, security, and 24/7 technical support.
Organization	Org	A company account in Trusted WiFi.
Operator	-	An organization type account in Trusted WiFi that is used by MSPs to manage a property's Wi-Fi network.
Customer	-	An organization type account in Trusted WiFi typically used for customers/brands that allows grouping to view all properties belonging to the same company even if managed by several different MSPs.
Linked Organization	Linked Org	A link creating a relationship between an operator and a customer account, allowing a customer to view a property while allowing an operator to manage it.
User	-	An individual accessing a product, service or system. In the context of Trusted WiFi, a user is setup to access products and deployments information at Admin/Editor/Viewer levels, according to their relevant role permissions. In the context of a given product or service, a user is another term referring to a subscriber: the person at the end of the chain interacting with that product or service – usually through a device.
Property	-	Trusted WiFi concept representing an Individual site or location where products are deployed.
Linked Property	-	Site or location shared between operator and customer accounts.

Passpoint Software Development Kit	Passpoint SDK	The service that sits within the customer's mobile app and that is connected to the Trusted WiFi RADIUS infrastructure allowing Passpoint profiles to be created for a given Passpoint realm.
Passpoint Realm	-	The customer specific domain that is used to provision Passpoint profiles that are approved for connection to any associated network.
Passpoint Profile	-	The security certified profile that sits on a subscriber's device. If installed correctly it allows seamless authentication to the secure Passpoint SSID.
Secure Passpoint Service Set Identifier	Secure Passpoint SSID	The Wi-Fi network that is associated to the Passpoint realm configured to allow subscriber devices with a valid Passpoint profile to seamlessly connect to the Passpoint network at a property.
Subscriber	-	An individual person using a service.
Subscriber Device	-	The equipment – typically a smartphone, tablet or computer – the subscriber is using to connect to the service.
Collision-Resistant Unique Identifier	CUID	A unique identifier designed to be collision-resistant – meaning engineered to minimize the likelihood of generating duplicate IDs even in distributed systems – and more efficient in terms of space and database indexing performance due to its sequential nature. In the context of Passpoint, a CUID is delivered to a subscriber device when it requests a Passpoint profile.
Customer Loyalty Mobile Application	Mobile App	The iOS and/or Android digital application used by businesses to engage and reward their customers through loyalty programs. In the context of Passpoint, the best user experience is to offer Passpoint through a customer/brand mobile app integration using the Trusted WiFi SDK.

TRUSTED WIFI PASSPOINT CORE SERVICES

Prerequisites

Trusted WiFi Components

The following Trusted WiFi products, licenses and services must be purchased:

- Trusted WiFi Passpoint core services module
- A one-time Passpoint activation fee per customer realm
- A recurring yearly Passpoint hosting fee per customer realm
- A recurring license per month per AP
- Trusted WiFi Passpoint Software Development Kit (SDK)
- A one-time SDK activation fee per customer realm, available either standalone or in combination with the one-time Passpoint activation fee

Warning

A purchase order is required to be sent to support@globalreachtech.com including: Operator details – company responsible for installing and managing the network at the site Customer name User contact details (name and email address)

Third-Party Components

The following third-party services must be available:

- A customer realm including domain, security certificate and private key, as well as the database holding the users' Personally Identifiable Information (PII). It is the responsibility of the Identity Provider – typically the customer/brand – to make this part available. This information must be emailed to support@globalreachtech.com for the GlobalReach's team to provision the realm on Trusted WiFi.
- The best user experience is to offer Passpoint through a customer/brand mobile app integration. Once they have access to the Trusted WiFi Passpoint SDK, it is the responsibility of the app owner – typically the customer/brand – to perform the integration.
- Local networks must be Passpoint compliant. Most recent Wi-Fi access points and controllers from major vendors support Passpoint today, however older models or products from smaller manufacturers might not do. The configuration of the local infrastructure is typically handled by the Managed Service Providers (MSPs).
- The subscribers' mobile devices (belonging to visitors, guests, employees, etc.) must be Passpoint compliant. All recent iOS and Android-based smartphones or tablets support Passpoint today, however older models or products running less popular operating systems might not do. Laptops compatibility is also more erratic. It is therefore essential to maintain a traditional onboarding service in parallel with Passpoint to handle non-compatible devices.

Assumptions

The instructions provided in this document assume the following:

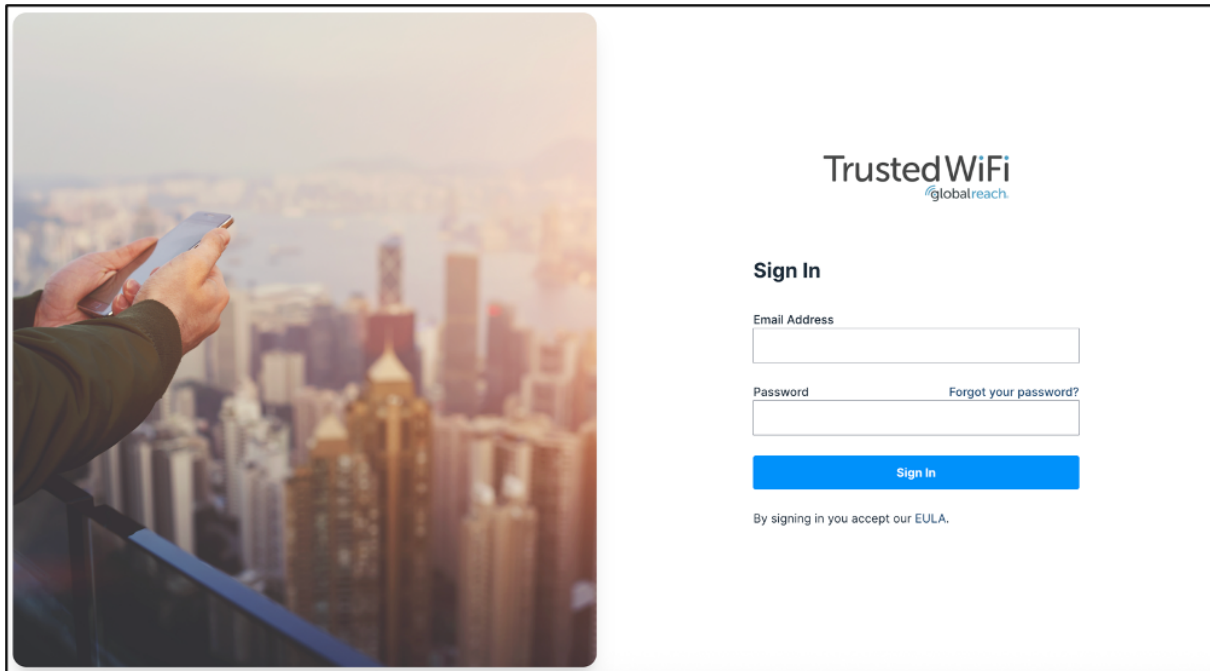
Warning

The Trusted WiFi Passpoint module is activated and licensed. The Trusted WiFi Passpoint SDK is activated and licensed. Wi-Fi networks are Passpoint compliant, installed and ready on site. Subscriber devices are Passpoint compliant. The Managed Service Provider (MSP) has Trusted WiFi account with operator permissions. All properties must share the same NAI realm, RADIUS IP / port settings and SSID name for the Passpoint service (CustomerPasspoint). Each property requires a separate RADIUS secret and NAS identifier, both generated when a configuration is activated within the Trusted WiFi management platform.

Trusted WiFi Configuration

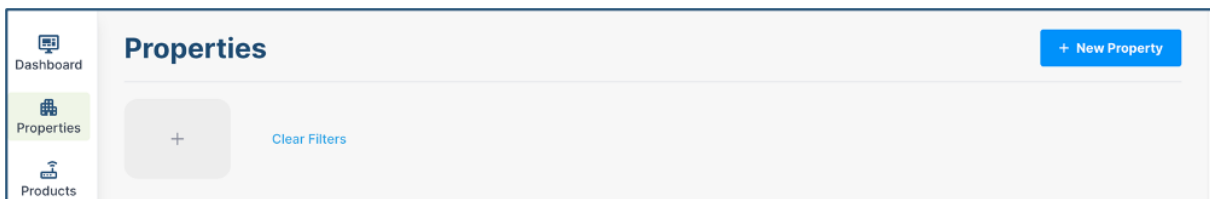
Trusted WiFi Login

- In a web browser, navigate to: <https://trustedwifi.cloud.global/> and enter your email and password.



Property Creation

- Select **Properties** from the left navigation menu and click on the **+ New Property** button at the top of the page to display the New Property form:



Properties

New Property

Name *

Site Code *

Customer Code

Customer

Sub brand

Address

Address 1

Address 2

City

State

Postcode

Country code *

Latitude

Longitude

Contact Details

Primary Contact Number

Primary Support Number

Localization Preferences

Timezone *

Locales

Default Locale *

Temperature Unit *

Distance Unit *

Currency *

- Complete the mandatory fields as per the description below:

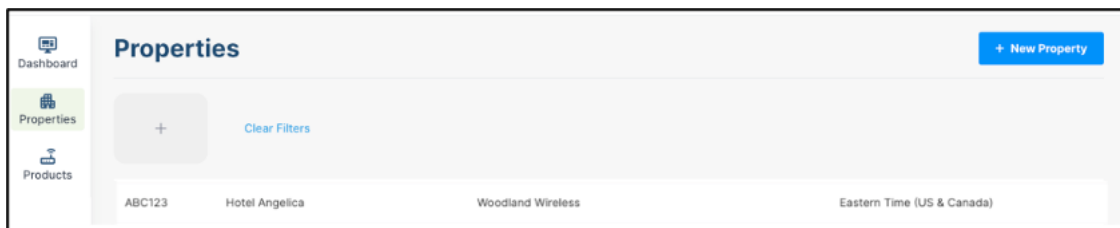
Field	Description
Name	The hotel or site name.
Site Code	This can be any unique alpha-numeric identifier. It is best practice to use the same ID assigned to the property from the hotel brand. If the Gateway is going into a lab, then this can be a random number.
Country Code	Select the appropriate country from the pull-down list.
Timezone	The time zone where the hotel is located. This is important as it affects operations scheduled to take place at a specific time.
Default Locale	Refers to the local language. Currently only English (en) is supported.
Temperature Unit	Whether the administrator prefers to display temperature information using Fahrenheit or Celsius.
Distance Unit	Whether the administrator prefers to display distance measurements in Imperial or Metric.
Currency	The currency of the property.
- Select the **Customer** from the drop-down list. This will link the property to the correct account where the Passpoint realm is then shared with the property.
- Click **Create Property** to save the details.

Warning

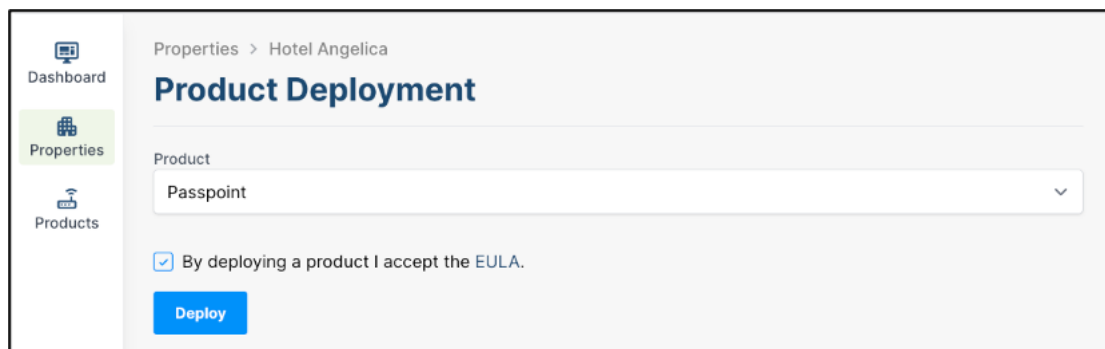
All properties from a given customer must share the same realm, which is pre-provisioned to their account by the GlobalReach Support team – as explained further in this document. Please check with GlobalReach what is the correct customer name to use and contact support@globalreachtech.com to link your operator account if the customer name doesn't appear in the drop-down list.

Passpoint Module Deployment

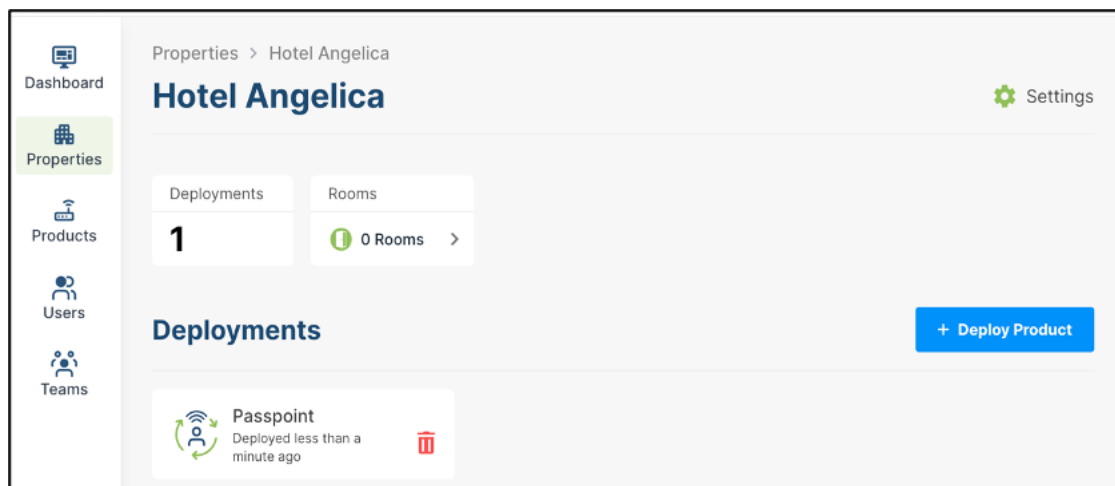
- Scroll down the list or use the search field at the top to find the desired property:



- Select the property and click the **+ Deploy Product** button. Then Select the **Passpoint** module from the drop-down list and accept the End User License Agreement (EULA), then click the **Deploy** button:



- The Passpoint deployment is now featured on the property page and the service can be configured:



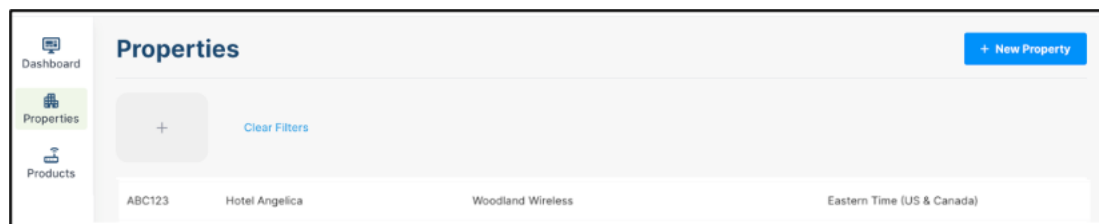
Warning

Once a property features deployed products, it cannot be deleted until the individual product deployments are first deleted to avoid deletion of live implementations.

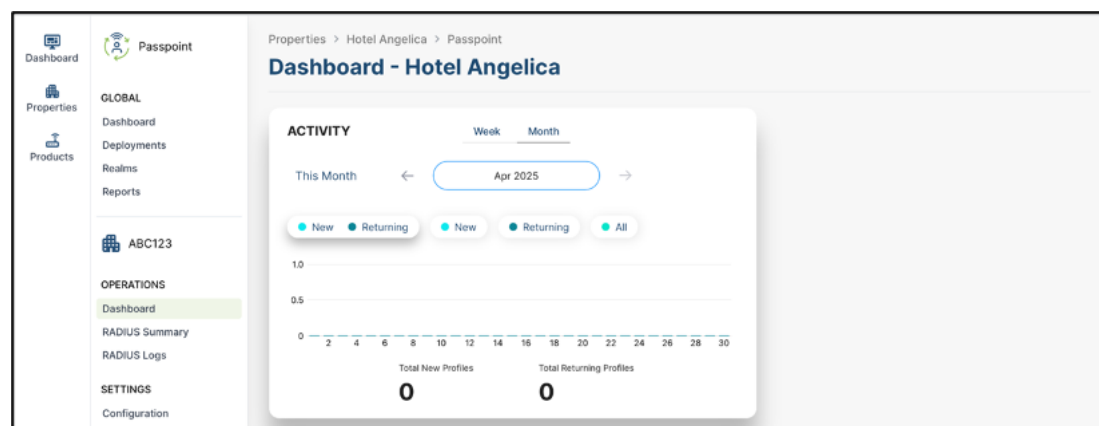
Property Realm Details

Once a Passpoint module is linked to a property, the realm details can be accessed.

- Scroll down the list or use the search field at the top to find the desired property:



- Select the property and click the **Passpoint** tile to display the dashboard page below:



Warning

If the Passpoint Unlicensed message is displayed, it means this deployment is not showing valid licenses. Please contact the GlobalReach Support team urgently to resolve.

- Select the **Configuration** option in the left-hand navigation menu and click on **Edit**:

Properties > Hotel Angelica > Passpoint

Configuration

NAI Realm	
Friendly Name	
Primary RADIUS IP	116.214.120.1
Secondary RADIUS IP	116.214.121.1
RADIUS Authentication Port	1812
RADIUS Accounting Port	1813
NAS Identifier	1f386a9
RADIUS Secret	5dc7b90b296789h08a6ef764b1e3d

[Edit](#)

- Select the relevant customer **NAI Realm** from the drop-down list and click **Apply** to save the changes.

Properties > Hotel Angelica > Passpoint

Edit Configuration

RADIUS Configuration
Default

NAI Realm
customer.cloud.global

[Cancel](#) [Apply](#)

- Take note of the details for your respective installation as they will be required at a later step to configure the network hardware.

Configuration

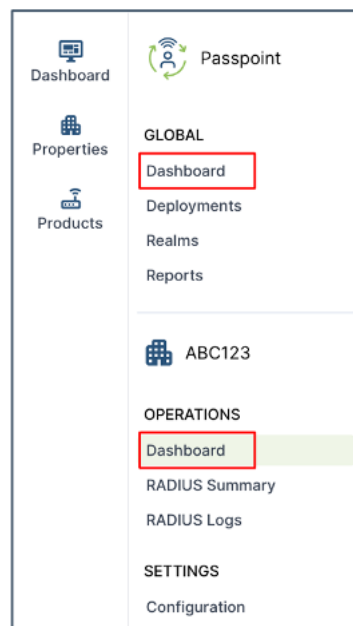
NAI Realm	customer.cloud.global
Friendly Name	customer.cloud.global
Primary RADIUS IP	116.214.120.1
Secondary RADIUS IP	116.214.121.1
RADIUS Authentication Port	1812
RADIUS Accounting Port	1813
NAS Identifier	1f386a9
RADIUS Secret	5dc7b90b296789h08a6ef764b1e3d

Warning

The NAS Identifier and the RADIUS Secret are always unique per property.

Analytics Dashboards

Trusted WiFi provides analytics dashboards containing anonymous information on usage of the Passpoint service, available both at global and individual property levels.



Warning

Some of the analytics might not be available depending on the operator account permissions.

Global Profiles View

The global profiles view is available to customers/brands accounts only, as it offers an estate-wide summary showing the number of profiles being requested via the loyalty app, as well as the number of subscriber profiles being used across the estate. This information includes all sites, irrespective of the different Managed Service Providers.

Operator accounts will see profiles requests only across the properties that manage.

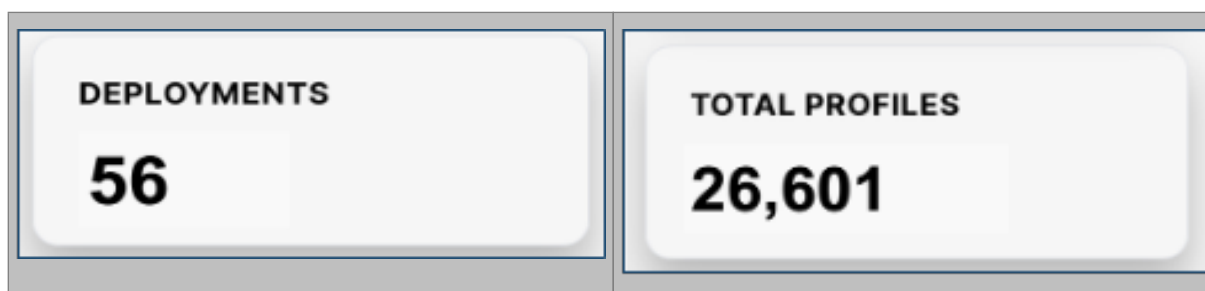
Profiles Widget:

The Profiles Widget shows all profiles requests, broken down by the selected period. It can also be further filtered by device type (toggles these on/off as required):



Deployments and Profiles Widget

The deployments widget shows how many instances of Passpoint have been setup across all customer properties, and associated to the customer's Passpoint realm. The profiles widget shows the total number of Passpoint profiles requested to date, across all properties.



Global Activity View

The global activity view is available to customers/brands accounts only, as it offers an estate-wide summary showing the profiles activity across the estate. This information includes all sites, irrespective of the different Managed Service Providers.

Operator accounts will see profile activity only across the properties that manage.

New vs Returning Filter

The new vs returning filter shows how many unique profiles were seen as new connections vs returning connections across all the customer's Passpoint networks during the selected period.

- **New filter:** total unique profiles seen as connecting for the first time (anywhere) during the selected period.
- **Returning filter:** total unique profiles seen as returning (from anywhere) connections during the selected period.
- **All filter:** total of all unique profiles seen connecting.



Property Level Activity View

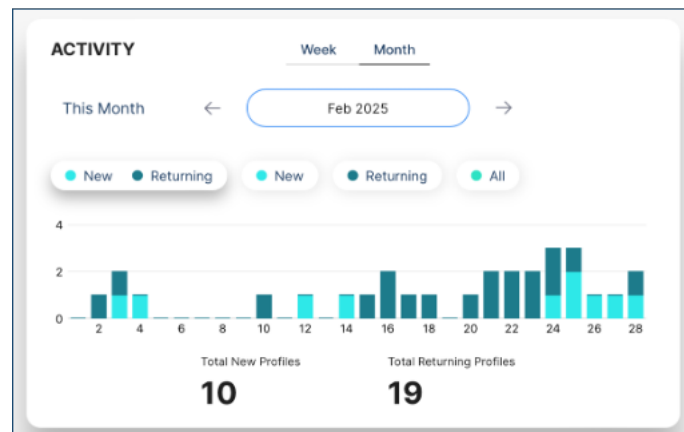
This view is available to all users who have access to property information within the Trusted WiFi management platform.

New Vs Returning Filter

The new vs returning filter shows how many unique profiles were seen as new connections vs returning connections at a specific property during the selected period.

- **New filter:** total unique profiles seen as connecting for the first time (first time ever using the customer's realm) during the selected period.
- **Returning filter:** total unique profiles seen as returning connections (previously seen using the customer's realm at this property or another property using the same realm) during the selected period.

- **All filter:** total of all unique profiles seen connecting to this specific property during the selected period.

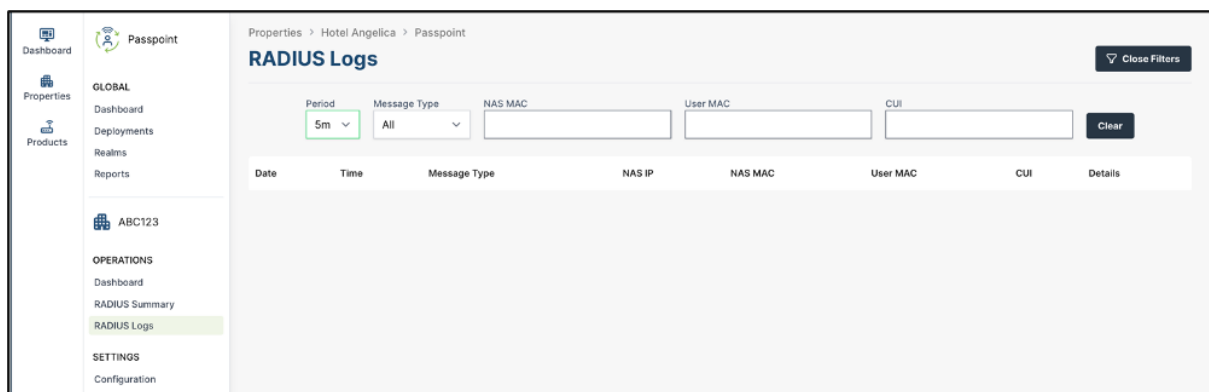


Troubleshooting

RADIUS Logs

Network Device Issues

- RADIUS logs are accessed from the left-hand navigation menu for an individual property's Passpoint deployment:



- Indicate the **NAS MAC** to find an individual device.
- Use the **Period Filter** to narrow-down the search if you know the timeframe you are looking for.
- Use the **Message Type** to narrow-down by Rejects or Accepts seen on RADIUS from that device.

Warning

If the RADIUS configuration is not setup correctly, all Passpoint RADIUS requests will be seen to be coming from unidentified subscriber devices, meaning they will get dropped silently.

The table in the RADIUS requests details section below provides a description of the error messages.

Subscriber Access Issues

- From the same RADIUS logs page as above, indicate the **CUI** to find an individual device.
- Use the **Period Filter** to narrow-down the search if you know the timeframe you are looking for.

- Use the **Message Type** to narrow-down by Rejects or Accepts seen on RADIUS from that device.

The table in the RADIUS requests details section below provides a description of the error messages.

RADIUS Summary

- The RADIUS summary is accessed from the left-hand navigation menu for an individual property's Passpoint deployment:



- This page presents a quick overview of the RADIUS accepts and rejects for allow an operator to check at a glance what is happening across the property network devices.

RADIUS Requests Details

The following sequence is what is expected to occur when a network device (AP) gets a request from a subscriber device to use the Passpoint service.

Warning

Logs showing these messages out of sequence could be a case of a step being skipped or having an issue when making that particular call. This is typically seen when there is a network issue or if the realm is not connected.

RADIUS Requests Sequence

Details	Description	Message Type
Network/TTLS Authentication	Pre-starting of the authentication process. The NAS IP is the public IP address at this stage.	Client
Profile Authentication	RADIUS request sent.	Accept
MSCHAPv2 [Success]	Authentication was successful and the subscriber device should have access to the Passpoint service.	Accept

RADIUS Rejects Descriptions

Details	Description	Message Type
Missing username/realm	No profile or Corrupt profile.	Reject
MS-CHAP2-Response is incorrect	Corrupt profile reject.	Reject
CUI not found	No profile or Corrupt profile.	Reject

CUSTOMER REALM

Prerequisites

Operating a Passpoint service requires connectivity to a customer realm including domain, security certificate and private key, as well as the database holding the users' Personally Identifiable Information (PII).

Warning

It is the responsibility of the Identity Provider – typically the customer/brand – to make these components available. This information must be emailed to support@globalreachtech.com for the GlobalReach Support team to provision the realm on Trusted WiFi.

Custom Domain

Any fully qualified custom domain name (FQDN) can be used, but must be present within the TLS certificate – for example: `passpoint.customer.com`

Warning

The domain must be under administrative control such that certificate Extended Validation (EV) can be completed successfully by the certificate authority.

Custom Certificates

The Passpoint service requires that the RADIUS server has a TLS certificate installed that matches the domain for the secure SSID through which Passpoint will operate. The certificate will be used to sign Passpoint profiles installed on subscriber devices, so the domain and certificate details should reflect desired branding.

The RADIUS certificate must meet the following requirements:

- CommonName (CN) must match the domain.
- DNS subjectAltName must match the domain.
- Must include Extended Validation (*EV).

For example, if the custom domain is `passpoint.customer.com`, the RADIUS certificate must have both the CN and subjectAltName fields set to `passpoint.customer.com`.

*EV SSL certificate validation: the CA will perform a rigorous verification of the site's ownership as well as the legitimacy of the company before issuing your certificate. The vetting process is more intensive than any other validation level and thus commands higher trust with end-users. Website owners will need to provide acceptable documents for the company during the validation process. These verifying documents attest that they have rights to the specific domain and that the business itself is legitimate.

The EV process ensures a thorough investigation is performed on the organization, and this information is displayed on the certificate itself. Historically, it's also been the only certificate that enables the organization's name to appear on the browser to indicate your website's identity.

Trusted WiFi Realm Configuration

The following is required for the setup of the customer realm on the Trusted WiFi platform:

- Chosen Realm (Domain) name
- Security Certificates
- Private key

Warning

These parameters are not self-configurable, so they must be emailed to support@globalreachtech.com to be provisioned on the Trusted WiFi platform. The certificates must be kept up to date for the service to work as expected. If the certificate expires, the subscriber will get a security warning in their browser when they try to connect to the Passpoint service at a supporting property. If your certificate or key is about to expire, please provide an updated version to support@globalreachtech.com so it can be updated in the Trusted WiFi platform.

Once the customer realm is setup by the GlobalReach's Operations team, it will appear as the NIA Realm in the Passpoint configuration section in Trusted WiFi – as described in an earlier chapter of this document.

Configuration	
NAI Realm	customer.cloud.global
Friendly Name	customer.cloud.global
Primary RADIUS IP	116.214.120.1
Secondary RADIUS IP	116.214.121.1
RADIUS Authentication Port	1812
RADIUS Accounting Port	1813
NAS Identifier	1ft386a9
RADIUS Secret	5dc7b90b296789h08a6ef764b1e3d

MOBILE APP INTEGRATION

Trusted WiFi Passpoint SDK

The best user experience is delivered through a loyalty app integration, offering the following benefits:

- No additional user information to collect - guest or visitor is already known from the app.

- One time download of the Passpoint profile in 3 “taps”.
- Incentive to increase the app uptake.
- Additional services and engagement possibilities once users have the app.
- Non-compatible devices can still use the legacy onboarding process.

Warning

Access to the GlobalReach Passpoint SDK and its documentation are subject to the purchase of the one-time activation fee per customer realm.

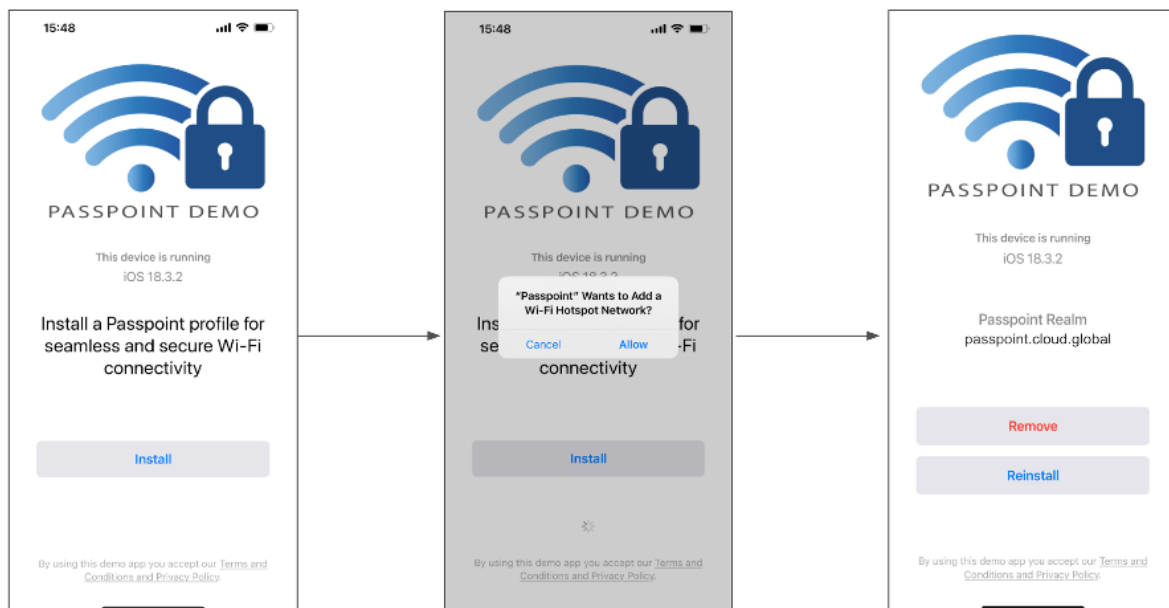
The Trusted WiFi Passpoint SDK offers a customer/brand the ability to insert an option for their app members to pre-download a secure Passpoint profile to their subscriber device before they travel to a property connected to the customer realm.

It also allows for actions such as removing a Passpoint profile or requesting a new one. This is helpful for support, in case subscriber profiles become corrupted or broken.

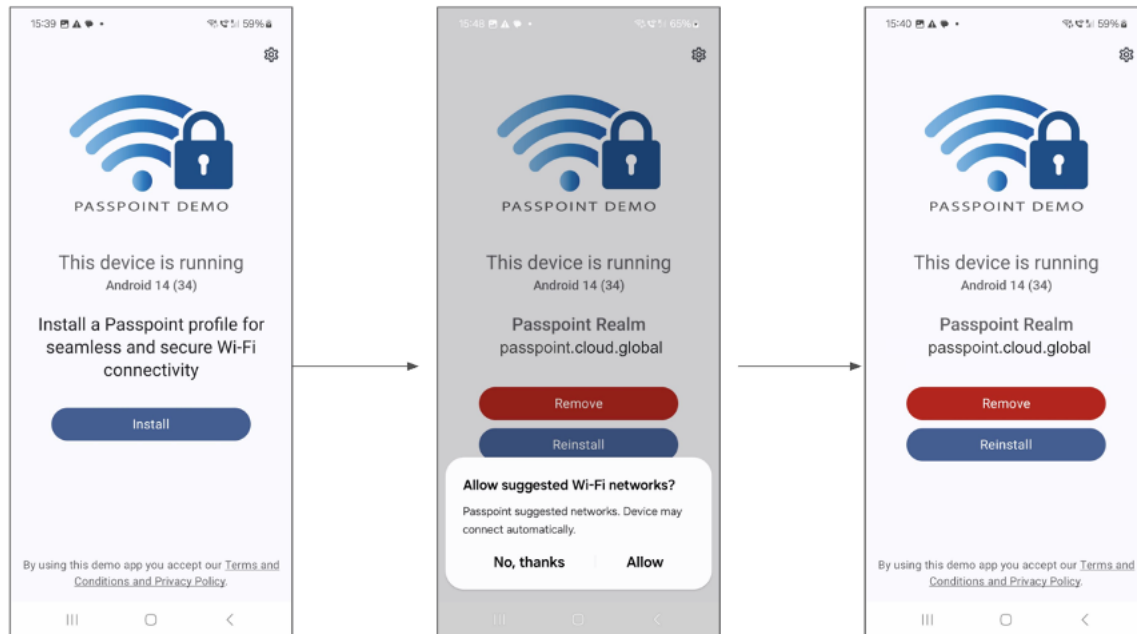
Following the purchase of the one-time activation fee per customer realm, the GlobalReach Operations team will require the name and email address of the developers in need of access to the SDK GitHub repository containing the technical information required to perform the integration.

The following screenshots illustrate Passpoint profile download via a demo app:

iOS Profile Download



Android profile download



WI-FI NETWORKS CONFIGURATION

Prerequisites

The local networks must be compliant with Passpoint (Hotspot 2.0). – Most recent Wi-Fi access points and controllers from major vendors support Passpoint today, however older models or products from smaller manufacturers might not do. The configuration of the local infrastructure is typically handled by the Managed Service Providers (MSPs).

Warning

Wi-Fi networks must be within their vendor's support lifecycle and running valid licenses. Wi-Fi networks must be Passpoint compliant.

Configuration

Hardware configuration is addressed via separate guides. At the time of writing, the following documents are available:

- Hardware Configuration Guide for Aruba Central
- Hardware Configuration Guide for Aruba Mobility Controllers
- • Hardware Configuration Guide for Cisco Meraki
- Hardware Configuration Guide for Nomadix WLAN Controllers
- Hardware Configuration Guide for Ruckus SmartZone
- Hardware Configuration Guide for Ruckus ZoneDirector

Please refer to these documents for further configuration information.

SUBSCRIBER DEVICES

Prerequisites

The subscribers' mobile devices (belonging to visitors, guests, employees, etc.) must be compliant with Passpoint (Hotspot 2.0) – All recent iOS and Android-based smartphones or tablets support Passpoint today, however older models or products running less popular operating systems might not do. Laptops compatibility is also more erratic. It is therefore essential to maintain a traditional onboarding service in parallel with Passpoint to handle non-compatible devices.

Operation

If a Passpoint profile is installed correctly, the subscriber device should connect automatically to the customer's private Passpoint SSID on the supported property network.

Warning

The Passpoint service does not use captive portal or CNA pop-up, the subscriber devices automatically join the relevant Passpoint Wi-Fi SSID. Do not manually connect to the SSID, as this will not work and will cause the subscriber device to prompt for a username and password.

Troubleshooting

If a subscriber device is prompted for user name/password, the following may help to further assess identifying Passpoint profile related issues:

- If username/password are blank, this usually means there is no Passpoint profile found on the subscriber device
- If username is filled in, but password is blank, this usually means the Passpoint profile is corrupted and will not work.

If the Passpoint profile is not installed, use the customer app to download it to the subscriber device.

Aruba Central

Download this guide 

Introduction

Scope and Purpose

Thank you for purchasing the Trusted WiFi solution. This document is a hardware configuration guide describing how to setup Aruba Central for a Trusted WiFi Passpoint service.

- For more information on how to setup an end-to-end Trusted WiFi Passpoint service, please refer to the Trusted WiFi Passpoint Administration Guide.
- For complete information on how to setup Aruba Central, please refer to the vendor's original documentation.

Documentation Conventions

{{snippet.Documentation Conventions}}

Passpoint Overview

Passpoint – also known as Hotspot 2.0 – is an industry-wide next generation approach to public internet access driven by the Wi-Fi Alliance that brings the following benefits:

- Frictionless onboarding and roaming, thanks to a one-time registration followed by automatic access to interconnected hotspots
- More secure and private Wi-Fi connections, compared with general visitor networks

Passpoint is based on the IEEE 802.11u standard, which is a set of protocols enabling cellular-like roaming. Following the initial enrollment, frequent users such as visitors, guests or employees bypass repeated logins, forms and passwords, as their mobile devices automatically join the Wi-Fi subscriber service when they return to a venue or roam between inter-linked Passpoint enabled hotspots and providers, while being better protected against potential cyber threats.

If a device supports 802.11u and is enrolled to a service, it automatically communicates with the Wi-Fi infrastructure via the access points to discover the network SSID and connects securely to it by presenting its access credentials. Upon successful authentication, the device is provisioned with Passpoint standards-based management objects - known as Per-Provider Subscription Management Objects (PPS-MO).

GlobalReach - an ASSA ABLOY company - has been involved with Passpoint since its inception and even contributed to the creation and initial pilot testing of the standard. As a result, it is one of the few trusted worldwide experts on this topic, with a proven platform backed by real-world operational experiences at scale.

The best user experience is to offer Passpoint through a customer/brand mobile app integration, as it further simplifies the onboarding process, while incentivizing app downloads and customer loyalty, leading to further engagement and monetization opportunities. To this effect, Trusted WiFi offers a Software Development Kit (SDK) for easy app integration.

Seamless Onboarding and Roaming

Best user experience via brand mobile loyalty app integration:

- One-time setup in 3 "taps" via a mobile app
- Followed by automatic connection for returning visitors and roaming within any of the participating venues
- No manual SSID selection, no login/password, no reset after a given period
- No impact from MAC address randomization
- Incentive to download the app, leading to visitor engagement opportunities
- Non-compatible devices can still use the legacy onboarding process

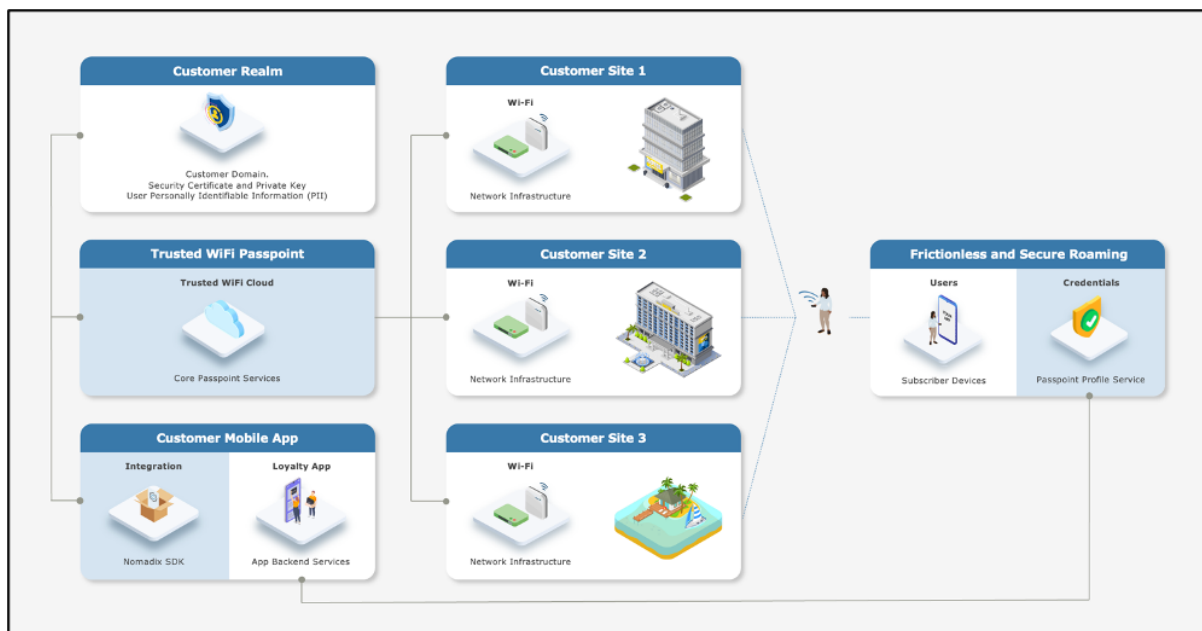
End-to-End Service Components

Implementing an end-to-end Trusted WiFi Passpoint service requires a combination of the following software and hardware components:

- **Trusted WiFi Passpoint:** the core services are offered and managed centrally via the **Trusted WiFi Passpoint Module** (hosted and operated by GlobalReach). Following an initial setup performed by the GlobalReach Operations team, Managed Service Providers (MSPs) can then add sites to customer realms and monitor the solution.
- **Customer Realm:** the service requires connectivity to a realm including domain, security certificate and private key, as well as the database holding the users' Personally Identifiable Information (PII). It is the responsibility of the Identity Provider – typically the customer/brand – to make this part available.
- **Mobile App:** the best user experience is to offer Passpoint through a customer/brand mobile app integration. Trusted WiFi offers a **Software Development Kit (SDK)** for easy implementation. It is the responsibility of the app owner – typically the customer/brand – to perform the integration.
- **Local Networks:** the local networks must be compliant with Passpoint (Hotspot 2.0). – Most recent Wi-Fi access points and controllers from major vendors support Passpoint today, however older models or products from more exotic manufacturers might not do. The configuration of the local infrastructure is typically handled by the Managed Service Providers (MSPs).
- **User Devices:** the subscribers' mobile devices (belonging to visitors, guests, employees, etc.) must be compliant with Passpoint (Hotspot 2.0) – All recent iOS and Android-based smartphones or tablets support Passpoint today, however older models or products running less popular operating systems might not do. Laptops compatibility is also more erratic. It is therefore essential to maintain a traditional onboarding service in parallel with Passpoint to handle non-compatible devices.

High-Level Topology

The diagram below illustrates the high-level topology for the end-to-end Trusted WiFi Passpoint service:



Glossary

The following is a glossary of the most common terms used regarding this solution.

Term	Abbreviation	Description
Deployment	-	Enabling of a Trusted WiFi product module for a property via the Trusted WiFi interface.
Module	-	Product or service purchased from Trusted WiFi that is managed through its own sub-section via the Trusted WiFi interface.
License	-	Legal agreement that grants users the right to use specific software, outlining terms and conditions for its usage, distribution, and potential modifications, while protecting the intellectual property of the software developer. GlobalReach licenses comprise of two different types: one-off licenses - typically to initially enable a software module. recurring licenses - typically including software updates and technical support, or more in the case of OpEx consumption commercial models based on price per month/quarter/year. Trusted WiFi is sold as a combination of one-off licenses to activate the service and recurring licenses based on a price per Wi-Fi access point per month.
Managed Service Provider	MSP	The third-party company that remotely manages and monitors a client's IT infrastructure and end-user systems, offering services like network and infrastructure management, security, and 24/7 technical support.
Organization	Org	A company account in Trusted WiFi.
Operator	-	An organization type account in Trusted WiFi that is used by MSPs to manage a property's Wi-Fi network.
Customer	-	An organization type account in Trusted WiFi typically used for customers/brands that allows grouping to view all properties belonging to the same company even if managed by several different MSPs.
Linked Organization	Linked Org	A link creating a relationship between an operator and a customer account, allowing a customer to view a property while allowing an operator to manage it.
User	-	An individual accessing a product, service or system. In the context of Trusted WiFi, a user is setup to access products and deployments information at Admin/Editor/Viewer levels, according to their relevant role permissions. In the context of a given product or service, a user is another term referring to a subscriber: the person at the end of the chain interacting with that product or service – usually through a device.
Property	-	Trusted WiFi concept representing an Individual site or location where products are deployed.
Linked Property	-	Site or location shared between operator and customer accounts.

Passpoint Software Development Kit	Passpoint SDK	The service that sits within the customer's mobile app and that is connected to the Trusted WiFi RADIUS infrastructure allowing Passpoint profiles to be created for a given Passpoint realm.
Passpoint Realm	-	The customer specific domain that is used to provision Passpoint profiles that are approved for connection to any associated network.
Passpoint Profile	-	The security certified profile that sits on a subscriber's device. If installed correctly it allows seamless authentication to the secure Passpoint SSID.
Secure Passpoint Service Set Identifier	Secure Passpoint SSID	The Wi-Fi network that is associated to the Passpoint realm configured to allow subscriber devices with a valid Passpoint profile to seamlessly connect to the Passpoint network at a property.
Subscriber	-	An individual person using a service.
Subscriber Device	-	The equipment – typically a smartphone, tablet or computer – the subscriber is using to connect to the service.
Collision-Resistant Unique Identifier	CUID	A unique identifier designed to be collision-resistant – meaning engineered to minimize the likelihood of generating duplicate IDs even in distributed systems – and more efficient in terms of space and database indexing performance due to its sequential nature. In the context of Passpoint, a CUID is delivered to a subscriber device when it requests a Passpoint profile.
Customer Loyalty Mobile Application	Mobile App	The iOS and/or Android digital application used by businesses to engage and reward their customers through loyalty programs. In the context of Passpoint, the best user experience is to offer Passpoint through a customer/brand mobile app integration using the Trusted WiFi SDK.

ARUBA CENTRAL CONFIGURATION

Supported Versions

Aruba Central supports Passpoint on AOS v10. Aruba recommends upgrading to the latest version of AOS.

Warning

The configuration of the RADIUS is documented using the WLAN configuration option. The alternative method, under Configuration > Authentication, is not covered in this document.

Prerequisites

It is assumed the following prerequisites are met before configuring Aruba Central for a Trusted WiFi Passpoint service:

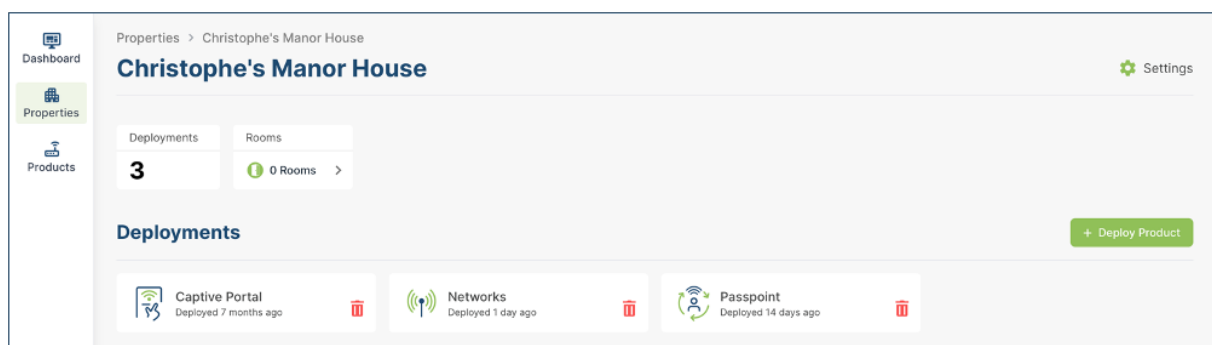
- A supported Aruba Central running version 10.0 or greater, activated and licensed.
- A Trusted WiFi account with operator permissions.
- A core Trusted WiFi Passpoint service configured and tested.
- A property in Trusted WiFi with deployed Passpoint modules.
- A deployed and configured Wi-Fi network.

Warning

The Aruba configuration is focusing on Passpoint v1.0. All properties must share the same NAI realm, RADIUS IP / port settings and SSID name for the Passpoint service (CustomerPasspoint). Each property requires a separate RADIUS secret and NAS identifier, both generated when a configuration is activated within the Trusted WiFi management platform.

Core Passpoint Settings

- Log into Trusted WiFi, then click on the **Properties** icon in the left menu to view the properties list.
- Select or search for the property you wish to work on. This will open its deployments page:



- Click on the **Passpoint** tile and then click on the **Configuration** option in the left menu to display the Passpoint settings summary as per the example below:

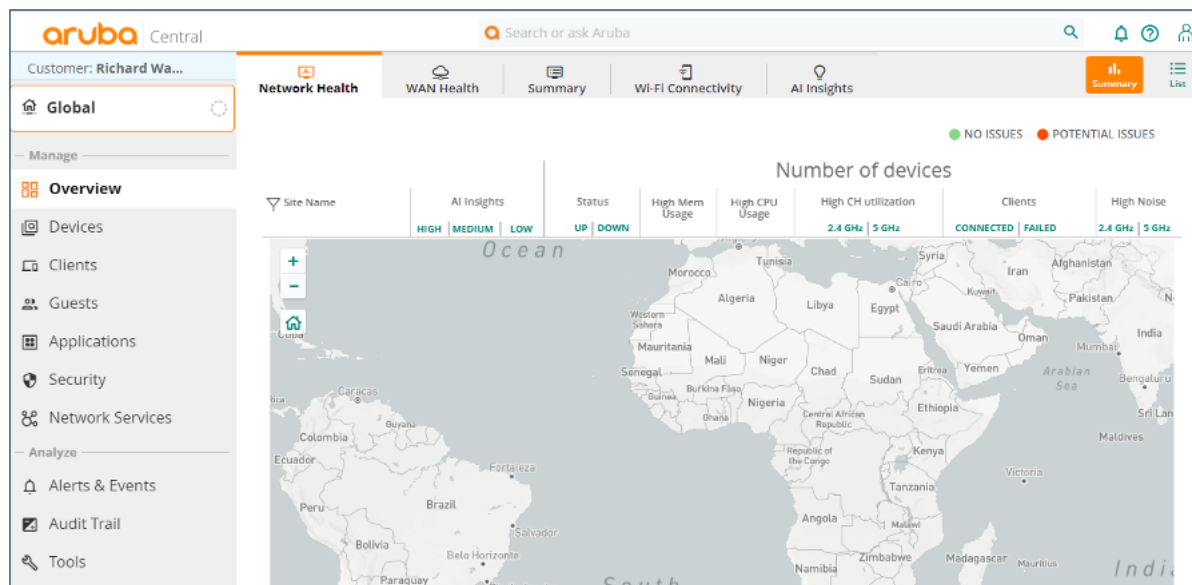
Configuration	
NAI Realm	customer.cloud.global
Friendly Name	customer.cloud.global
Primary RADIUS IP	116.214.120.1
Secondary RADIUS IP	116.214.121.1
RADIUS Authentication Port	1812
RADIUS Accounting Port	1813
NAS Identifier	1f386a9
RADIUS Secret	5dc7b90b296789h08a6ef764b1e3d

- Take note of the details for your respective installation as they will be required at a later step.

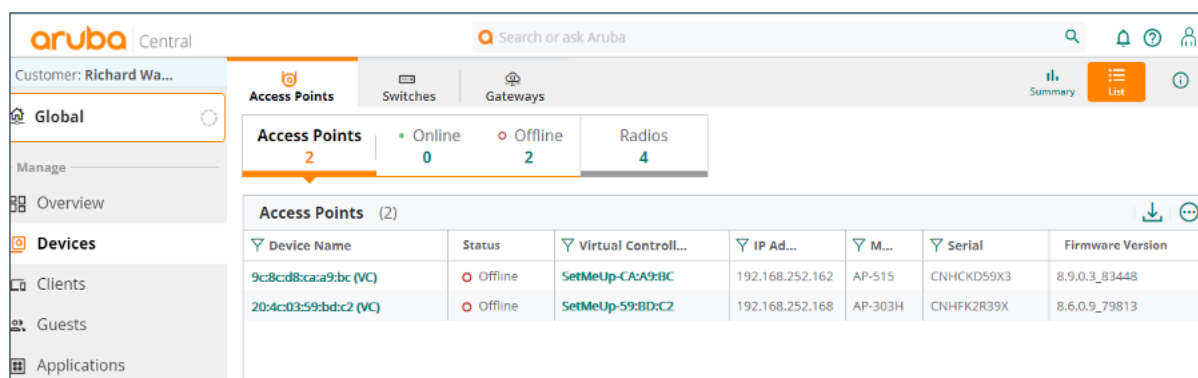
Aruba Central Configuration

SSID

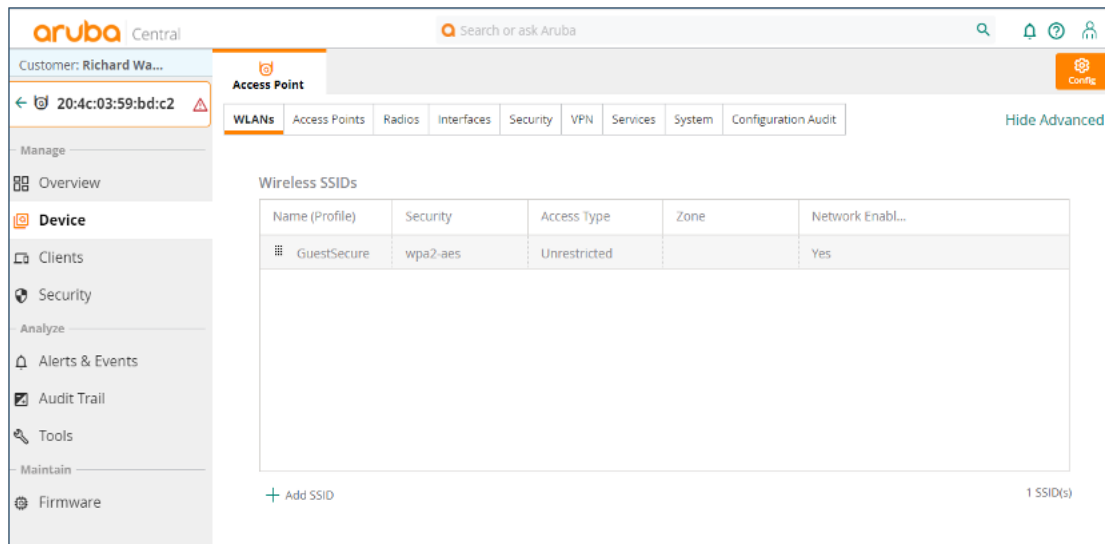
- Log into Aruba Central with your credentials to display the below page:



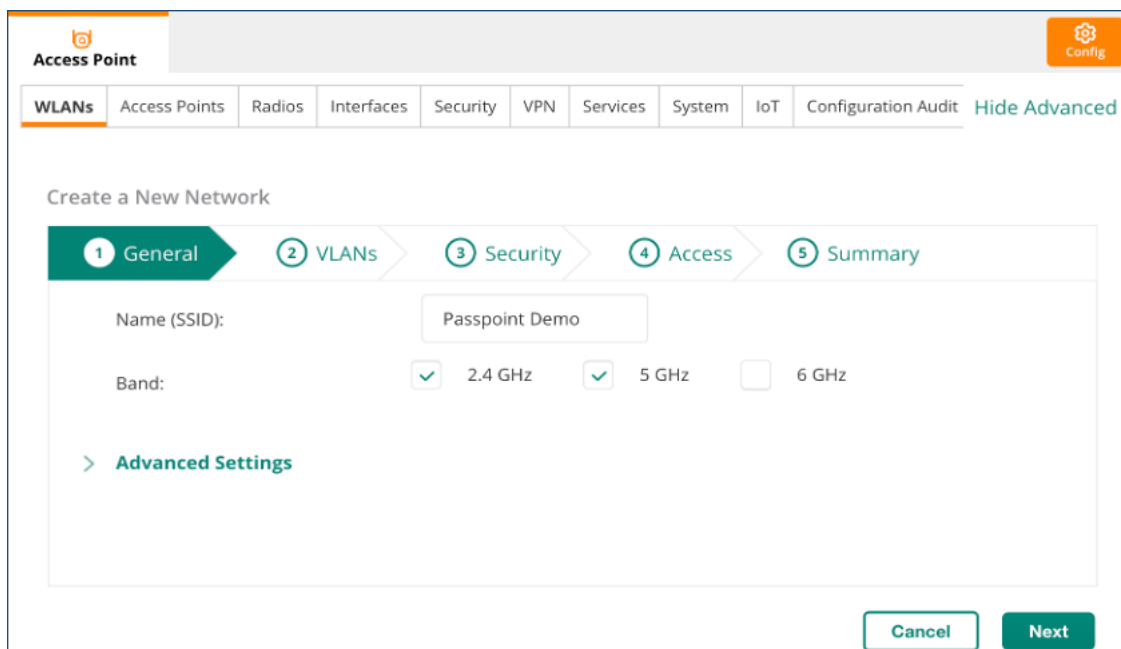
- Click on **Devices** in the left-hand navigation menu to display the below page:



- Click on one of the **Access Points** to display a detailed summary, which also changes the left navigation menu. Click on the **Device** tab on the left and select the **WLANS** tab:



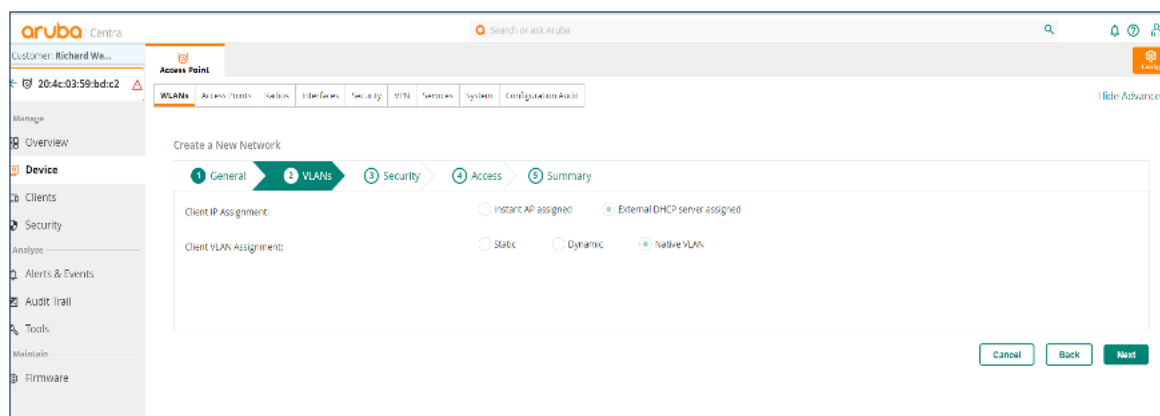
- Click on **+ Add SSID** on the bottom of the screen to display the below page:



- Enter the **Name** of the new Passpoint SSID, (for example: CustomerPasspoint) and click **Next**.

Client IP and VLAN Assignment

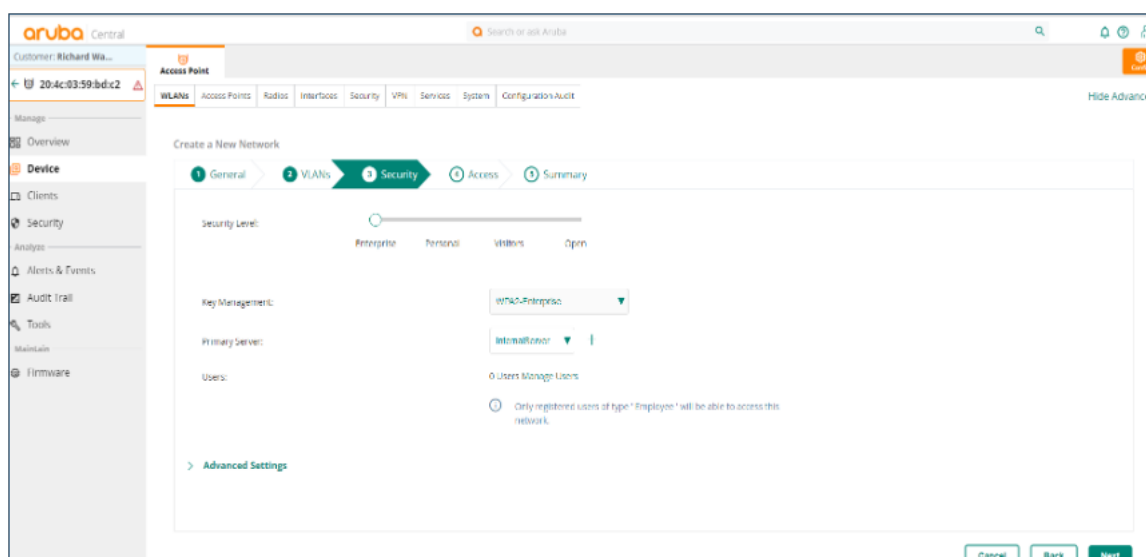
- In the **WLANs tab>VLANs** tab below:



- Set the **Client IP Assignment** to **External DHCP server assigned** and the **Client VLAN Assignment** to **Native VLAN**, then click **Next**.

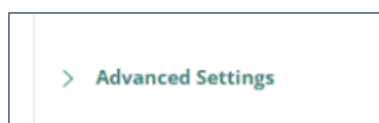
Security and RADIUS

- In the **Security** tab below:



- Move the **Security Level** slider to **Enterprise** and set the Key Management to **WPA Enterprise** from the drop-down list.
- Click on the **+** to the right of the **Primary Server** drop-down list to display the page below:

- Complete the fields as per the description below:
 FieldDescriptionServer
 TypeRADIUS.NameEnter a name.IP Addresses/FQDNEnter the <Primary RADIUS IP> from Trusted WiFi.NAS IdentifierEnter the <NAS Identifier> from Trusted WiFi.Auth PortEnter the < RADIUS Authentication Port> from Trusted WiFi.Accounting PortEnter the < RADIUS Accounting Port> from Trusted WiFi.Shared SecretEnter the < RADIUS Shared Secret> from Trusted WiFi.Confirm Shared SecretRe-enter the < RADIUS Shared Secret> from Trusted WiFi.
- Click **OK** to save, repeat this operation for the **Secondary Server** and click **OK** to save.
- Back to the **Security** tab, ensure the RADIUS server you just configured is selected as the default and click on the **Advanced Settings** option at the bottom of the screen:



- Scroll down to the **Passpoint Server Profile** option:

Access Point

WLANs | Access Points | Radios | Interfaces | Security | VPN | Services | System | IoT | Configuration Audit

Max Authentication Failures: 0

Enforce DHCP: ☐

Use IP for Calling Station ID: ☐

Called Station ID Type: MAC Address ▼

Called Station ID Include SSID: ☐

Authentication Survivability: ☐

Passpoint Service Profile: None ▼ [Manage Passpoint Services](#)

Auth Packet Delimiter Character:

Auth Packet Uppercase Support: ☐

Delete PMK Cache For Inactive Clients: ☐ ⚠

[+ Accounting](#)

[+ Fast Roaming](#)

- Click on the **Manage Passpoint Services** option, then click on **+ Add Profile** at the bottom of the screen to display page below:

Add Profile

Name: This field is required

[+ Access Network](#)

[+ Identity Provider](#)

[+ Roaming Consortium](#)

[Cancel](#) [Save](#)

- Enter the **Name** of the Passpoint profile, then click on **Access Network** to display the options below:

Add Profile

Name:

☒ **Access Network**

Domain Name: Internet: ☒

Radius Location Data: ☐ Radius Chargeable User Identity: ☐

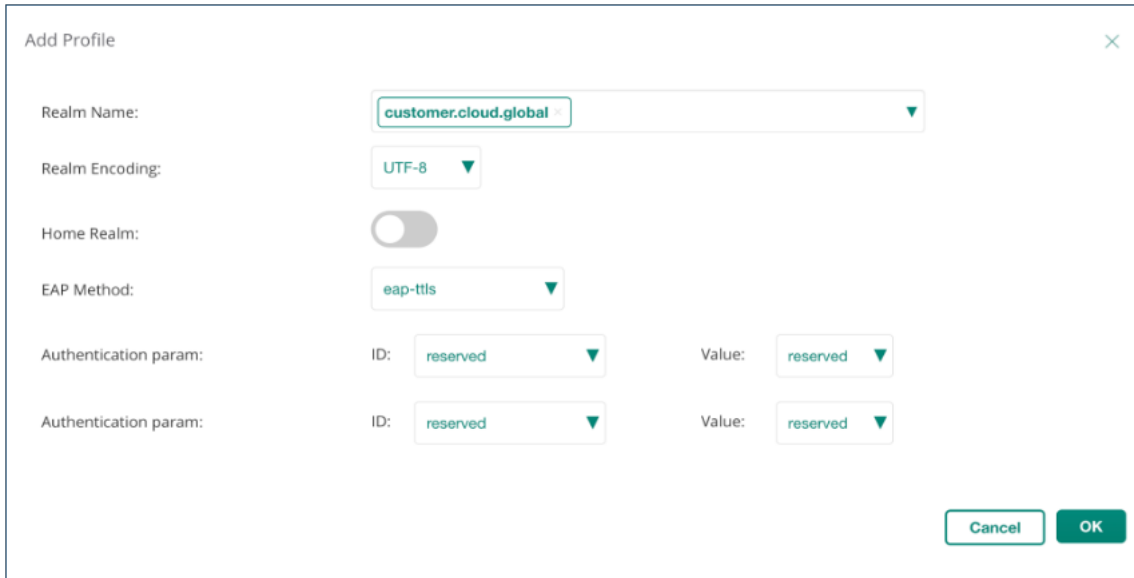
Operator Friendly Name: Venue/Operator Language Code: eng ▼

Venue Name: Venue Group: business ▼ Venue Type: research-and-dev-facility ▼

Network Type: free-public ▼ IPv4: None ▼ IPv6: None ▼

[Cancel](#) [Save](#)

- Complete the fields as per the description below:FieldDescriptionNameName of the Passpoint profile.Domain NameEnter the domain name.Network TypeSelect free-public from the drop down list.IPv4Select free-public from the drop down list. Select public from the drop down list.
- The other fields can be left as they are.
- Click **Identity Provider** and then **+Add** to display the page below:

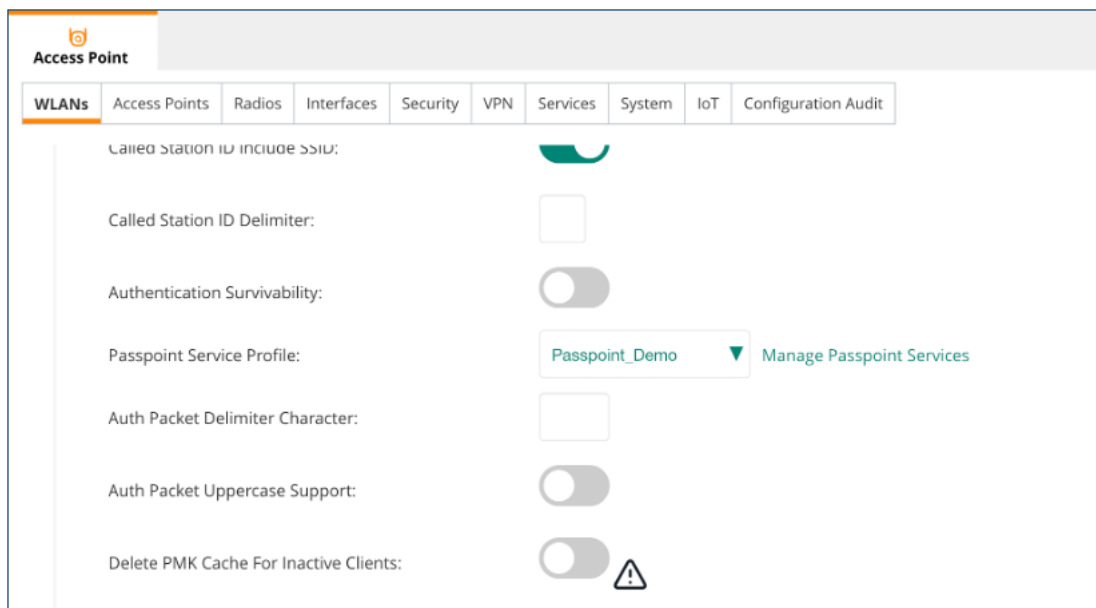


The 'Add Profile' dialog box contains the following fields:

- Realm Name:** A dropdown menu with 'customer.cloud.global' selected.
- Realm Encoding:** A dropdown menu with 'UTF-8' selected.
- Home Realm:** A toggle switch currently turned off.
- EAP Method:** A dropdown menu with 'eap-ttls' selected.
- Authentication param:** Two sets of fields, each with an 'ID' dropdown (both set to 'reserved') and a 'Value' dropdown (both set to 'reserved').

At the bottom right are 'Cancel' and 'OK' buttons.

- Complete the fields as per the description below:FieldDescriptionRealm NameSelect the realm name.Realm EncodingSelect UTF-8 from the Realm Encoding drop-down list.EAP MethodSelect eap-ttls from the drop-down list.
- The other fields can be left as they are.
- Click on **OK** to return to the Add Profile screen and then click **Save** and **X** to the **Manage Passpoint Service** screen pop-up.
- Ensure you select the newly created **Passpoint Service Profile**, as shown below:



The 'Access Point' configuration screen shows a tabbed interface with 'WLANs' selected. The configuration for the selected WLAN is as follows:

- Called Station ID Include SSID:** A toggle switch currently turned on.
- Called Station ID Delimiter:** A text input field.
- Authentication Survivability:** A toggle switch currently turned off.
- Passpoint Service Profile:** A dropdown menu with 'Passpoint_Demo' selected. A link 'Manage Passpoint Services' is visible to the right.
- Auth Packet Delimiter Character:** A text input field.
- Auth Packet Uppercase Support:** A toggle switch currently turned off.
- Delete PMK Cache For Inactive Clients:** A toggle switch currently turned off, with a warning icon (triangle with exclamation mark) to its right.

- Click **Next** to the Access tab and click **Next** again to the Summary tab:

Access Point

Config

WLANs

Access Points

Radios

Interfaces

Security

VPN

Services

System

IoT

Configuration Audit

Hide Advanced

Create a New Network

1 General

2 VLANs

3 Security

4 Access

5 Summary

Network Summary

General

ESSID

Passpoint Demo

Multicast Optimization

disabled

Band

2.4, 5

DTIM Interval

1 beacons

Primary Usage

employee

Inactivity Timeout

1000 secs

Dynamic Multicast OPT

disabled

Content Filtering

disabled

Airtime

unlimited

Hide SSID

disabled

Broadcast filtering

arp

Security

Security Level

Enterprise

Auth Server 1

ServerSettings

Auth Server 2

ServerSettings2

Key Management

WPA2-Enterprise

MAC Authentication

disabled

VLANs

Client IP Assignment

External DHCP Server Assigned

Client VLAN Assignment

Native VLAN

VLAN

Access

ROLE ASSIGNMENTS FOR AUTHENTICATED USERS

disabled

- Click **Finish**. This WLAN configuration can now be applied to any predefined group of APs.